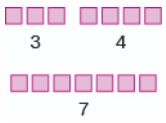
Answer Key

Chapter 1

Be Prepared

1.1 215

1.2 342,006

1.3 

1.4 910

1.5 2,162

1.6 284

1.7 81

1.8 17

1.9 5,394

Try It

1.1 (a) 2, 9, 241, 376
(b) 0, 2, 9, 241, 376

1.2 (a) 7, 13, 201
(b) 0, 7, 13, 201

1.3 176

1.4 237

1.5 (a) ten millions (b) tens
(c) hundred thousands
(d) millions (e) ones

1.6 (a) billions
(b) ten thousands
(c) tens
(d) hundred thousands
(e) hundred millions

1.7 nine trillion, two hundred fifty-eight billion, one hundred thirty-seven million, nine hundred four thousand, sixty-one

1.8 seventeen trillion, eight hundred sixty-four billion, three hundred twenty-five million, six hundred nineteen thousand, four

1.9 three hundred sixteen million, one hundred twenty-eight thousand, eight hundred thirty-nine

1.10 thirty-one million, five hundred thirty-six thousand

1.11 53,809,051

1.12 2,022,714,466

1.13 34,000,000 miles

1.14 204,000,000 pounds

1.15 160

1.16 880

1.17 17,900

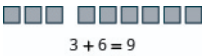
1.18 5,000

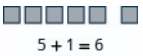
1.19 64,000

1.20 156,000

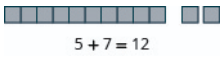
1.21 (a) eight plus four; the sum of eight and four
(b) eighteen plus eleven; the sum of eighteen and eleven

1.22 (a) twenty-one plus sixteen; the sum of twenty-one and sixteen
(b) one hundred plus two hundred; the sum of one hundred and two hundred

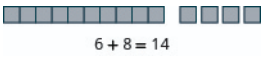
1.23 

1.24 

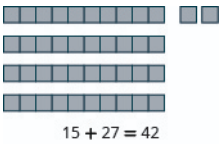
1.25



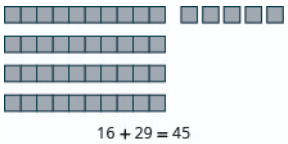
1.26



1.27



1.28



1.29 (a) $0 + 19 = 19$
(b) $39 + 0 = 39$

1.30 (a) $0 + 24 = 24$
(b) $57 + 0 = 57$

1.31 $9 + 7 = 16; 7 + 9 = 16$

1.32 $8 + 6 = 14; 6 + 8 = 14$

1.33 $32 + 54 = 86$

1.34 $25 + 74 = 99$

1.35 $35 + 98 = 133$

1.36 $72 + 89 = 161$

1.37 $456 + 376 = 832$

1.38 $269 + 578 = 847$

1.39 $4{,}597 + 685 = 5{,}282$

1.40 $5{,}837 + 695 = 6{,}532$

1.41 $46{,}195 + 397 + 6{,}281 = 52{,}873$

1.42 $53{,}762 + 196 + 7{,}458 = 61{,}416$

1.43 Translate: $17 + 26$;
Simplify: 43

1.44 Translate: $28 + 14$;
Simplify: 42

1.45 Translate: $29 + 76$;
Simplify 105

1.46 Translate 37 + 69; Simplify 106

1.47 He rode 140 miles.

1.48 The total number is 720 students.

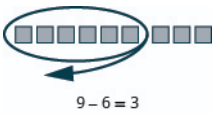
1.49 The perimeter is 30 inches.

1.50 The perimeter is 36 inches.

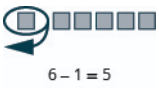
1.51 (a) twelve minus four; the difference of twelve and four
(b) twenty-nine minus eleven; the difference of twenty-nine and eleven

1.52 (a) eleven minus two; the difference of eleven and two
(b) twenty-nine minus twelve; the difference of twenty-nine and twelve

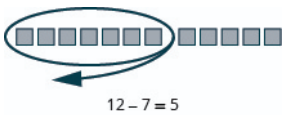
1.53



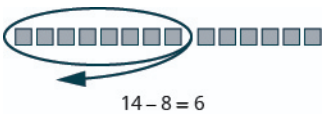
1.54



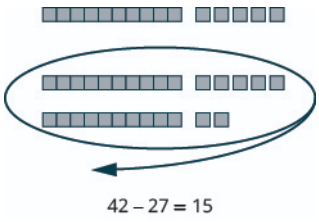
1.55



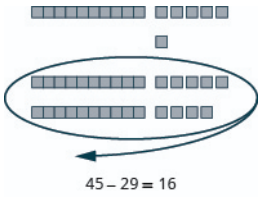
1.56



1.57



1.58



1.59 7 − 0 = 7; 7 + 0 = 7

1.60 6 − 2 = 4; 2 + 4 = 6

1.61 86 − 54 = 32 because 54 + 32 = 86

1.62 99 − 74 = 25 because 74 + 25 = 99

1.63 93 − 58 = 35 because 58 + 35 = 93

1.64 81 − 39 = 42 because 42 + 39 = 81

1.65 439 − 52 = 387 because 387 + 52 = 439

1.66 318 − 75 = 243 because 243 + 75 = 318

1.67 832 − 376 = 456 because 456 + 376 = 832

1.68 847 − 578 = 269 because 269 + 578 = 847

1.69 4,585 − 697 = 3,888 because 3,888 + 697 = 4,585

1.70 5,637 − 899 = 4,738 because 4,738 + 899 = 5,637

1.71 (a) 14 − 9 = 5
(b) 37 − 21 = 16

1.72 (a) 11 − 6 = 5
(b) 67 − 18 = 49

1.73 The difference is 19 degrees Fahrenheit.

1.74 The difference is 17 degrees Fahrenheit.

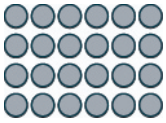
1.75 The difference is $149.

1.76 The difference is $136.

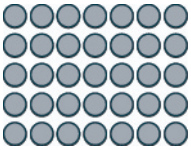
1.77 (a) eight times seven ; the product of eight and seven
(b) eighteen times eleven ; the product of eighteen and eleven

1.78 (a) thirteen times seven ; the product of thirteen and seven
(b) five times sixteen; the product of five and sixteen

1.79



1.80



1.81 (a) 0 (b) 0

1.82 (a) 0 (b) 0

1.83 (a) 19 (b) 39

1.84 (a) 24 (b) 57

1.85 54 and 54; both are the same.

1.86 48 and 48; both are the same.

1.87 512

1.88 342

1.89 1,735

1.90 3,234

1.91 3,354

1.92 3,776

1.93 (a) 540 (b) 5,400

1.94 (a) 750 (b) 7,500

1.95 127,995

1.96 653,462

1.97 365,462

1.98 504,108

1.99 13 · 28; 364

1.100 47 · 14; 658

1.101 2(167); 334

1.102 2(258); 516

1.103 Valia donated 144 water bottles.

1.104 Vanessa brought 80 hot dogs.

1.105 Erin needs 28 dahlias.

1.106 There are 36 women in the choir.

1.107 Jane needs 320 tiles.

1.108 Yousef needs 1,080 tiles.

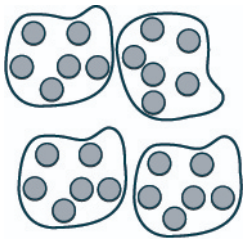
1.109 The area of the rug is 40 square feet.

1.110 The area of the driveway is 900 square feet

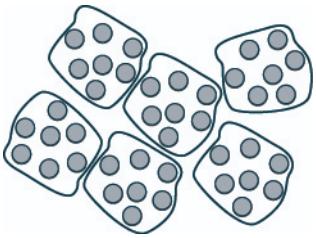
1.111 (a) eighty-four divided by seven; the quotient of eighty-four and seven
(b) eighteen divided by six; the quotient of eighteen and six.
(c) twenty-four divided by eight; the quotient of twenty-four and eight

1.112 (a) seventy-two divided by nine; the quotient of seventy-two and nine
(b) twenty-one divided by three; the quotient of twenty-one and three
(c) fifty-four divided by six; the quotient of fifty-four and six

1.113



1.114



1.115 (a) 9 (b) 3

1.116 (a) 4 (b) 5

1.117 (a) 1 (b) 27

1.118 (a) 16 (b) 4

1.119 (a) 0 (b) undefined

1.120 (a) 0 (b) undefined

1.121 659

1.122 679

1.123 861

1.124 651

1.125 704

1.126 809

1.127 476 with a remainder of 4

1.128 539 with a remainder of 7

1.129 114 R11

1.130 121 R9

1.131 307 R49

1.132 308 R77

1.133 91 ÷ 13; 7

1.134 52 ÷ 13; 4

1.135 Marcus can fill 15 cups.

1.136 Andrea can make 9 bows.

Section 1.1 Exercises

1. (a) 5, 125 (b) 0, 5, 125

3. (a) 50, 221 (b) 0, 50, 221

5. 561

7. 407

9. (a) thousands
(b) hundreds (c) tens
(d) ten thousands
(e) hundred thousands

11. (a) hundred thousands
(b) millions
(c) thousands (d) tens
(e) ten thousands

13. One thousand, seventy-eight

15. Three hundred sixty-four thousand, five hundred ten

17. Five million, eight hundred forty-six thousand, one hundred three

19. Thirty seven million, eight hundred eighty-nine thousand, five

21. Fourteen thousand, four hundred ten

23. Six hundred thirteen thousand, two hundred

25. Two million, six hundred seventeen thousand, one hundred seventy-six

27. Twenty three million, eight hundred sixty-seven thousand

29. One billion, three hundred seventy-seven million, five hundred eighty-three thousand, one hundred fifty-six

31. 412

33. 35,975

35. 11,044,167

37. 3,226,512,017

39. 7,173,000,000

41. 39,000,000,000,000

43. (a) 390 (b) 2,930

45. (a) 13,700 (b) 391,800

47. (a) 1,490 (b) 1,500

49. (a) 64,000 (b) 63,900

51. Twenty four thousand, four hundred ninety-three dollars

53. (a) \$24,490 (b) \$24,500 (c) \$24,000 (d) \$20,000

55. (a) 1,000,000,000 (b) 1,400,000,000 (c) 1,356,000,000

57. Answers may vary. The whole numbers are the counting numbers with the inclusion of zero.

Section 1.2 Exercises

59. five plus two; the sum of 5 and 2.

61. thirteen plus eighteen; the sum of 13 and 18.

63. two hundred fourteen plus six hundred forty-two; the sum of 214 and 642

65.



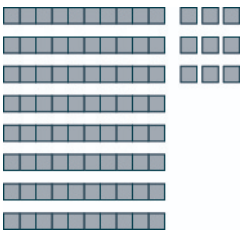
$2 + 4 = 6$

67.



$8 + 4 = 12$

69.



$14 + 75 = 89$

71.



$16 + 25 = 41$

73.

+	0	1	2	3	4	5	6	7	8	9
0	0	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9	10
2	2	3	4	5	6	7	8	9	10	11
3	3	4	5	6	7	8	9	10	11	12
4	4	5	6	7	8	9	10	11	12	13
5	5	6	7	8	9	10	11	12	13	14
6	6	7	8	9	10	11	12	13	14	15
7	7	8	9	10	11	12	13	14	15	16
8	8	9	10	11	12	13	14	15	16	17
9	9	10	11	12	13	14	15	16	17	18

75.

+	3	4	5	6	7	8	9
6	9	10	11	12	13	14	15
7	10	11	12	13	14	15	16
8	11	12	13	14	15	16	17
9	12	13	14	15	16	17	18

77.

+	5	6	7	8	9
5	10	11	12	13	14
6	11	12	13	14	15
7	12	13	14	15	16
8	13	14	15	16	17
9	14	15	16	17	18

79. ⓐ 13 ⓑ 13

81. ⓐ 11 ⓑ 11

83. 78

85. 99

87. 85

89. 142

91. 493

93. 861

95. 1,031

97. 6,850

99. 22,333

101. 1,104,860

103. 29,191

105. 101,531

107. $13 + 18 = 31$

109. $90 + 65 = 155$

111. $33 + 49 = 82$

113. $599 + 250 = 849$

115. $628 + 77 = 705$

117. $915 + 1{,}482 = 2{,}397$

119. The total cost was $1,875.

121. Ethan rode 138 miles.

123. The total square footage in the rooms is 1,167 square feet.

125. Natalie's total salary is $237,186.

127. The perimeter of the figure is 44 inches.

129. The perimeter of the figure is 56 meters.

131. The perimeter of the figure is 71 yards.

133. The perimeter of the figure is 62 feet.

135. The total number of calories was 640.

137. Yes, he scored 406 points.

138. Yes, the total weight is 1091 pounds.

139. Answers will vary.

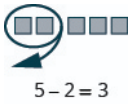
Section 1.3 Exercises

141. fifteen minus nine; the difference of fifteen and nine

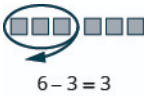
143. forty-two minus thirty-five; the difference of forty-two and thirty-five

145. six hundred seventy-five minus three hundred fifty; the difference of six hundred seventy-five and three hundred fifty

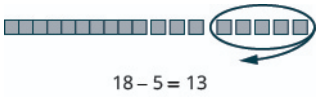
147.



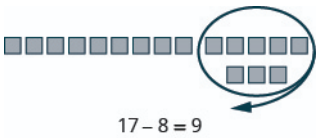
149.



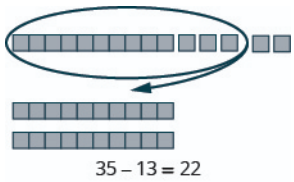
151.



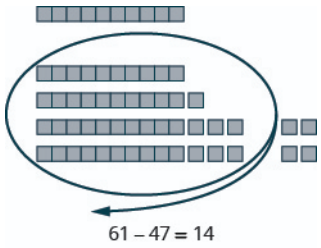
153.



155.



157.



159. 5

161. 8

163. 22

165. 33

167. 123

169. 1,321

171. 28

173. 222

175. 346

177. 3,519

179. 1,186

181. 34,668

183. 10 − 3; 7

185. 15 − 4; 11

187. 9 − 6; 3

189. 75 − 28; 47

191. 45 − 20; 25

193. 92 − 67; 25

195. 16 − 12; 4

197. 61 − 38; 23

199. 29

201. 72

203. 1,060

205. 22

207. 75 + 35; 110

209. 41 − 13; 28

211. 100 − 76; 24

213. The difference between the high and low temperature was 17 degrees

215. The difference between the third grade and second grade was 13 children.

217. The difference between the regular price and sale price is $251.

219. John needs to save $155 more.

221. 157 miles

223. Answers may vary.

Section 1.4 Exercises

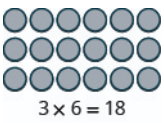
225. four times seven; the product of four and seven

227. five times twelve; the product of five and twelve

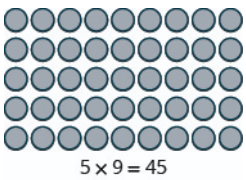
229. ten times twenty-five; the product of ten and twenty-five

231. forty-two times thirty-three; the product of forty-two and thirty-three

233.



235.



237.

×	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7	8	9
2	0	2	4	6	8	10	12	14	16	18
3	0	3	6	9	12	15	18	21	24	27
4	0	4	8	12	16	20	24	28	32	36
5	0	5	10	15	20	25	30	35	40	45
6	0	6	12	18	24	30	36	42	48	54
7	0	7	14	21	28	35	42	49	56	63
8	0	8	16	24	32	40	48	56	64	72
9	0	9	18	27	36	45	54	63	72	81

239.

×	3	4	5	6	7	8	9
4	12	16	20	24	28	32	36
5	15	20	25	30	35	40	45
6	18	24	30	36	42	48	54
7	21	28	35	42	49	56	63
8	24	32	40	48	56	64	72
9	27	36	45	54	63	72	81

241.

×	3	4	5	6	7	8	9
6	18	24	30	36	42	48	54
7	21	28	35	42	49	56	63
8	24	32	40	48	56	64	72
9	27	36	45	54	63	72	81

243.

×	5	6	7	8	9
5	25	30	35	40	45
6	30	36	42	48	54
7	35	42	49	56	63
8	40	48	56	64	72
9	45	54	63	72	81

245. 0

247. 0

249. 43

251. 28

253. 240,055

255. (a) 42 (b) 42

257. 395

259. 1,650

261. 23,947

263. 1,976

265. 7,008

267. 2,295

269. 230

271. 3,600

273. 88,000

275. 50,000,000

277. 34,333

279. 422,506

281. 804,285

283. 26,624

285. 245,340

287. 1,476,417

289. $18 \cdot 33$; 594

291. 51(67); 3,417

293. 2(249); 498

295. 10(375); 3,750

297. 1,406

299. 148

301. 11,000

303. 11,424

305. 0

307. 15,383

309. 50 − 18; 32

311. 2(35); 70

313. 20 + 980; 1,000

315. 12(875); 10,500

317. 89 − 74; 15

319. 3,075 + 95; 3,170

321. 814 − 366; 448

323. Tim brought 54 cans of soda to the party.

325. There were 308 students.

327. Rey donated 180 t-shirts.

329. Stephanie should use 20 cups of fruit juice.

331. There are 100 senators in the U.S. senate.

333. The area of the wall is 117 square feet.

335. The area of the room is 1,428 square feet.

337. The area of the court is 4,700 square feet.

339. Javier's portfolio gained $3,600.

341. Answers will vary.

Section 1.5 Exercises

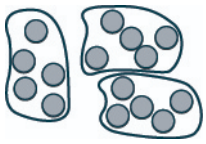
343. fifty-four divided by nine; the quotient of fifty-four and nine

345. thirty-two divided by eight; the quotient of thirty-two and eight

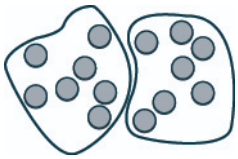
347. forty-eight divided by six; the quotient of forty-eight and six

349. sixty-three divided by seven; the quotient of sixty-three and seven

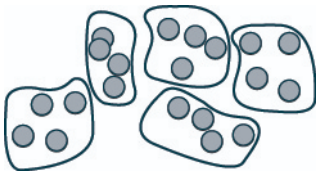
351.



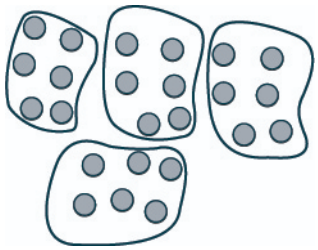
353.



355.



357.



359. 9

361. 9

363. 7

365. 9

367. 9

369. 5

371. 1

373. 1

375. 23

377. 19

379. 0

381. undefined

383. undefined

385. 0

387. 24

389. 12

391. 93

393. 132

395. 871

397. 7,831

399. 2,403

401. 901

403. 704

405. 10,209

407. 352 R6

409. 6,913 R1

411. 86,234 R4

413. 43,338 R2

415. 382 R5

417. 849

419. 96

421. 1,986 R17

423. 3,060

425. 72

427. 1,060

429. 35

431. 45 ÷ 15; 3

433. 288 ÷ 24; 12

435. Ric can fill 32 bags.

437. There are 25 groups.

439. Marta can wrap 16 cakes from 1 roll.

441. The difference is 28 miles per gallon.

443. They will need 5 vans for the field trip

445. Bill hiked 50 miles

447. LaVonne treated 48 patients last week.

449. Jenna uses 26 pairs of contact lenses, but there is 1 day left over, so she needs 27 pairs for 365 days.

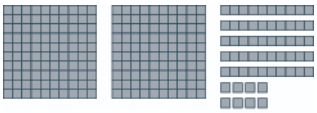
451. Answers may vary. Using multiplication facts can help you check your answers once you've finished division.

Review Exercises

453. (a) 2, 99 (b) 0, 2, 99

455. (a) 4, 90 (b) 0, 4, 90

457.



Place Value	Digit	Total Value
hundreds	2	200
tens	5	50
ones	8	8
		258

459. (a) tens
(b) hundred thousands
(c) ones
(d) ten thousands
(e) thousands

461. Five thousand, two hundred eighty

463. Five million, twelve thousand, five hundred eighty-two

466. 15,253

467. 340,912,061

469. 410

471. 3,560

473. 39,000

475. 81,500

477. four plus three; the sum of four and three

479. five hundred seventy-one plus six hundred twenty-nine; the sum of five hundred seventy-one and six hundred twenty-nine

481.



483.

+	0	1	2	3	4	5	6	7	8	9
0	0	1	2	3	4	5	6	7	8	9
1	1	2	3	4	5	6	7	8	9	10
2	2	3	4	5	6	7	8	9	10	11
3	3	4	5	6	7	8	9	10	11	12
4	4	5	6	7	8	9	10	11	12	13
5	5	6	7	8	9	10	11	12	13	14
6	6	7	8	9	10	11	12	13	14	15
7	7	8	9	10	11	12	13	14	15	16
8	8	9	10	11	12	13	14	15	16	17
9	9	10	11	12	13	14	15	16	17	18

485. (a) 19 (b) 19

487. (a) 13 (b) 13

489. 79

491. 154

493. 19,827

495. 30 + 12; 42

497. 39 + 25; 64

499. $76

501. 46 feet

503. fourteen minus five; the difference of fourteen and five

505. three hundred fifty-one minus two hundred forty-nine; the difference between three hundred fifty-one and two hundred forty-nine

507.



509. 3

511. 14

513. 23

515. 322

517. 380

519. 9,985

521. 19 − 13; 6

523. 74 − 8; 66

525. 58 degrees Fahrenheit

527. eight times five; the product of eight and five

529. ten times ninety-five; the product of ten and ninety-five

531.



533.

×	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6	7	8	9
2	0	2	4	6	8	10	12	14	16	18
3	0	3	6	9	12	15	18	21	24	27
4	0	4	8	12	16	20	24	28	32	36
5	0	5	10	15	20	25	30	35	40	45
6	0	6	12	18	24	30	36	42	48	54
7	0	7	14	21	28	35	42	49	56	63
8	0	8	16	24	32	40	48	56	64	72
9	0	9	18	27	36	45	54	63	72	81

535. 0

537. 99

539. (a) 28 (b) 28

541. 27,783

543. 640

545. 79,866

547. 1,873,608

549. 15(28); 420

551. 2(575); 1,150

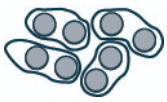
553. 48 marigolds

555. 1,800 seats

557. fifty-four divided by nine; the quotient of fifty-four and nine

559. seventy-two divided by eight; the quotient of seventy-two and eight

561.



563. 7

565. 13

567. 97

569. undefined

571. 638

573. 300 R5

575. 64 ÷ 16; 4

577. 9 baskets

Practice Test

579. (a) 4, 87 (b) 0, 4, 87

581. (a) 613 (b) 55,208

583. 68

585. 17

587. 32

589. 0

591. 0

593. 66

595. 3,325

597. 490

599. 442

601. 11

603. 16 + 58; 74

605. 32 − 18; 14

607. 2(524); 1,048

609. 300 − 50; 250

611. Stan had $344 left.

613. Clayton walked 30 blocks.

Chapter 2

Be Prepared

2.1 112

2.2 180,096

2.3 807

2.4 expression

2.5 1,024

2.6 73

2.7 19

2.8 42

2.9 $x - 8$

2.10 4 and 215

2.11 15

2.12 2, 3, 5, 6, 10

2.13 prime

2.13 2^4

Try It

2.1 (a) 18 plus 11; the sum of eighteen and eleven
(b) 27 times 9; the product of twenty-seven and nine
(c) 84 divided by 7; the quotient of eighty-four and seven
(d) *p* minus *q*; the difference of *p* and *q*

2.2 (a) 47 minus 19; the difference of forty-seven and nineteen
(b) 72 divided by 9; the quotient of seventy-two and nine
(c) *m* plus *n*; the sum of *m* and *n*
(d) 13 times 7; the product of thirteen and seven

2.3 (a) fourteen is less than or equal to twenty-seven
(b) nineteen minus two is not equal to eight
(c) twelve is greater than four divided by two
(d) *x* minus seven is less than one

2.4 (a) nineteen is greater than or equal to fifteen
(b) seven is equal to twelve minus five
(c) fifteen divided by three is less than eight
(d) *y* minus three is greater than six

2.5 (a) > (b) <

2.6 (a) < (b) >

2.7 (a) equation
(b) expression

2.8 (a) expression
(b) equation

2.9 41^5

2.10 7^9

2.11 (a) $4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4$
(b) $a \cdot a \cdot a \cdot a \cdot a \cdot a \cdot a$

2.12 (a) $8 \cdot 8 \cdot 8 \cdot 8 \cdot 8 \cdot 8 \cdot 8 \cdot 8$
(b) $b \cdot b \cdot b \cdot b \cdot b \cdot b$

2.13 (a) 125 (b) 1

2.14 (a) 49 (b) 0

2.15 (a) 2 (b) 14

2.16 (a) 35 (b) 99

2.17 18

2.18 9

2.19 16

2.20 23

2.21 86

2.22 1

2.23 81

2.24 75

2.25 (a) 10 (b) 19

2.26 (a) 4 (b) 12

2.27 (a) 13 (b) 5

2.28 (a) 8 (b) 16

2.29 64

2.30 216

2.31 64

2.32 81

2.33 33

2.34 10

2.35 40

2.36 9

2.37 The terms are $4x$, $3b$, and 2. The coefficients are 4, 3, and 2.

2.38 The terms are $9a$, $13a^2$, and a^3, The coefficients are 9, 13, and 1.

2.39 9 and 15; $2x^3$ and $8x^3$; y^2 and $11y^2$

2.40 $4x^3$ and $6x^3$; $8x^2$ and $3x^2$; 19 and 24

2.41 $16x + 17$

2.42 $17y + 7$

2.43 $4x^2 + 14x$

2.44 $12y^2 + 15y$

2.45 (a) $47 - 41$ (b) $5x \div 2$

2.46 (a) $17 + 19$ (b) $7x$

2.47 (a) $x + 11$ (b) $11a - 14$

2.48 (a) $j + 19$ (b) $2x - 21$

2.49 (a) $4(p + q)$ (a) $4p + q$

2.50 (a) $2x - 8$ (b) $2(x - 8)$

2.51 $w - 5$

2.52 $l + 2$

2.53 $6q - 7$

2.54 $4n + 8$

2.55 no

2.56 yes

2.57 yes

2.58 yes

2.59 $x + 1 = 7$; $x = 6$

2.60 $x + 3 = 4$; $x = 1$

2.61 $x = 13$

2.62 $x = 5$

2.63 $y = 28$

2.64 $y = 46$

2.65 $x = 22$

2.66 $y = 4$

2.67 $a = 37$

2.68 $n = 41$

2.69 $7 + 6 = 13$

2.70 $8 + 6 = 14$

2.71 $6 \cdot 9 = 54$

2.72 $21 \cdot 3 = 63$

2.73 $2(x - 5) = 30$

2.74 $2(y - 4) = 16$

2.75 $x + 7 = 37$; $x = 30$

2.76 $y + 11 = 28$; $y = 17$

2.77 $z - 17 = 37$; $z = 54$

2.78 $x - 19 = 45$; $x = 64$

2.79 (a) yes (b) no

2.80 (a) no (b) yes

2.81 (a) yes (b) no

2.82 (a) no (b) yes

2.83 (a) no (b) yes

2.84 (a) yes (b) no

2.85 (a) yes (b) no

2.86 (a) no (b) yes

2.87 Divisible by 2, 3, 5, and 10

2.88 Divisible by 2 and 3, not 5 or 10.

2.89 Divisible by 2 and 3, not 5 or 10.

2.90 Divisible by 3 and 5.

2.91 1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 48, 96

2.92 1, 2, 4, 5, 8, 10, 16, 20, 40, 80

2.93 composite

2.94 prime

2.95 $2 \cdot 2 \cdot 2 \cdot 2 \cdot 5$, or $2^4 \cdot 5$

2.96 $2 \cdot 2 \cdot 3 \cdot 5$, or $2^2 \cdot 3 \cdot 5$

2.97 $2 \cdot 3 \cdot 3 \cdot 7$, or $2 \cdot 3^2 \cdot 7$

2.98 $2 \cdot 3 \cdot 7 \cdot 7$, or $2 \cdot 3 \cdot 7^2$

2.99 $2 \cdot 2 \cdot 2 \cdot 2 \cdot 5$, or $2^4 \cdot 5$

2.100 $2 \cdot 2 \cdot 3 \cdot 5$, or $2^2 \cdot 3 \cdot 5$

2.101 $2 \cdot 3 \cdot 3 \cdot 7$, or $2 \cdot 3^2 \cdot 7$

2.102 $2 \cdot 3 \cdot 7 \cdot 7$, or $2 \cdot 3 \cdot 7^2$

2.103 36

2.104 72

2.105 60

2.106 105

2.107 440

2.108 360

Section 2.1 Exercises

1. 16 minus 9, the difference of sixteen and nine

3. 5 times 6, the product of five and six

5. 28 divided by 4, the quotient of twenty-eight and four

7. *x* plus 8, the sum of *x* and eight

9. 2 times 7, the product of two and seven

11. fourteen is less than twenty-one

13. thirty-six is greater than or equal to nineteen

15. 3 times n equals 24, the product of three and n equals twenty-four

17. *y* minus 1 is greater than 6, the difference of *y* and one is greater than six

19. 2 is less than or equal to 18 divided by 6; 2 is less than or equal to the quotient of eighteen and six

21. *a* is not equal to 7 times 4, *a* is not equal to the product of seven and four

23. equation

25. expression

27. expression

29. equation

31. 3^7

33. x^5

35. $5 \cdot 5 \cdot 5$

37. $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$

39. (a) 43 (b) 55

41. 5

43. 34

45. 58

47. 6

49. 13

51. 4

53. 35

55. 10

57. 41

59. 81

61. 149

63. 50

Section 2.2 Exercises

69. 22

71. 26

73. 144

75. 32

77. 27

79. 21

81. 41

83. 9

84. 225

85. 73

87. 54

89. $15x^2$, $6x$, 2

91. $10y^3$, y, 2

93. 8

95. 5

97. x^3 and $8x^3$; 14 and 5

99. $16ab$ and $4ab$; $16b^2$ and $9b^2$

101. $13x$

103. $26a$

105. $7c$

107. $12x + 8$

109. $10u + 3$

111. $12p + 10$

113. $22a + 1$

115. $17x^2 + 20x + 16$

117. $8 + 12$

119. $14 - 9$

121. $9 \cdot 7$

123. $36 \div 9$

125. $x - 4$

127. $6y$

129. $8x + 3x$

131. $y \div 3$

133. $8\,(y - 9)$

135. $5\,(x + y)$

137. $b + 15$

139. $b - 4$

141. $2n - 7$

143. He will pay $750. His insurance company will pay $1350.

Section 2.3 Exercises

147. (a) yes (b) no

149. (a) no (b) yes

151. (a) yes (b) no

153. (a) no (b) yes

155. (a) no (b) yes

157. (a) no (b) yes

159. $x + 2 = 5$; $x = 3$

161. $x + 3 = 6$; $x = 3$

163. $a = 16$

165. $p = 5$

167. $r = 24$

169. $x = 7$

171. $p = 69$

173. $d = 67$

175. $y = 22$

177. $u = 30$

179. $f = 178$

181. $n = 32$

183. $p = 48$

185. $y = 467$

187. $8 + 9 = 17$

189. $23 - 19 = 4$

191. $3 \cdot 9 = 27$

193. $54 \div 6 = 9$

195. $2(n - 10) = 52$

197. $3y + 10 = 100$

199. $p + 5 = 21$; $p = 16$

201. $r + 18 = 73$; $r = 55$

203. $d - 30 = 52$; $d = 82$

205. $u - 12 = 89$; $u = 101$

207. $c - 325 = 799$; $c = 1124$

209. $1300

211. $460

Section 2.4 Exercises

215. 2, 4, 6, 8, 10 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 38, 40, 42, 44, 46, 48

217. 4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48

219. 6, 12, 18, 24, 30, 36, 42, 48

221. 8, 16, 24, 32, 40, 48

223. 10, 20, 30, 40

225. Divisible by 2, 3, 4, 6

227. Divisible by 3, 5

229. Divisible by 2, 3, 4, 6

231. Divisible by 2, 3, 4, 5, 6, 10

233. Divisible by 2, 4

235. Divisible by 3, 5

237. Divisible by 2, 5, 10

239. Divisible by 2, 5, 10

241. Divisible by 3, 5

243. 1, 2, 3, 4, 6, 9, 12, 18, 36

245. 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60

247. 1, 2, 3, 4, 6, 8, 9, 12, 16, 18, 24, 36, 48, 72,144

249. 1, 2, 3, 4, 6, 7, 12, 14, 21, 28, 42, 49, 84, 98, 147, 196, 294, 588

251. prime

253. composite

255. prime

257. composite

259. composite

261. composite

263.

Weeks after graduation	Total number of dollars Frank put in the account	Simplified Total
0	100	100
1	100 + 15	115
2	100 + 15 • 2	130
3	100 + 15 • 3	145
4	100 + 15 • 4	160
5	100 + 15 • 5	175
6	100 + 15 • 6	190
20	100 + 15 • 20	400
x	100 + 15 • x	$100 + 15x$

Section 2.5 Exercises

267. $2 \cdot 43$

269. $2 \cdot 2 \cdot 3 \cdot 11$

271. $3 \cdot 3 \cdot 7 \cdot 11$

273. $5 \cdot 23$

275. $3 \cdot 3 \cdot 5 \cdot 5 \cdot 11$

277. $2 \cdot 2 \cdot 2 \cdot 7$

279. $2 \cdot 2 \cdot 2 \cdot 3 \cdot 7$

281. $17 \cdot 23$

283. $2 \cdot 2 \cdot 2 \cdot 2 \cdot 3 \cdot 3 \cdot 3$

285. $2 \cdot 2 \cdot 2 \cdot 2 \cdot 3 \cdot 3 \cdot 3 \cdot 5$

287. $2 \cdot 3 \cdot 5 \cdot 5$

289. $3 \cdot 5 \cdot 5 \cdot 7$

291. $2 \cdot 2 \cdot 3 \cdot 3$

293. $2 \cdot 5 \cdot 5 \cdot 7$

295. 24

297. 30

299. 120

301. 300

303. 24

305. 120

307. 420

309. 42

311. 120

313. 40

Review Exercises

317. 3 times 8, the product of three and eight.

319. 24 divided by 6, the quotient of twenty-four and six.

321. 50 is greater than or equal to 47

323. The sum of n and 4 is equal to 13

325. equation

327. expression

329. 2^3

331. x^6

333. $8 \cdot 8 \cdot 8 \cdot 8$

335. $y \cdot y \cdot y \cdot y \cdot y$

337. 81

339. 128

341. 20

343. 18

345. 74

347. 31

349. 58

351. 26

353. $12n^2, 3n, 1$

355. 6

357. 3 and 4; $3x$ and x

359. $24a$

361. $14x$

363. $12n + 11$

365. $10y^2 + 2y + 3$

367. $x - 6$

369. $3n \cdot 9$

371. $5(y + 1)$

373. $c + 3$

375. (a) yes (b) no

377. (a) yes (b) no

379. (a) no (b) yes

381. $x + 3 = 5$; $x = 2$

383. $c = 6$

385. $x = 11$

387. $y = 23$

389. $p = 34$

391. $7 + 33 = 40$

393. $4 \cdot 8 = 32$

395. $2(n - 3) = 76$

397. $x + 8 = 35$; $x = 27$

399. $q - 18 = 57$; $q = 75$

401. $h = 42$

403. $z = 33$

405. $q = 8$

407. $v = 56$

409. 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39, 42, 45, 48

411. 8, 16, 24, 32, 40, 48

413. 2, 3, 6

415. 2, 3, 5, 6, 10

417. 1, 2, 3, 5, 6, 10, 15, 30

419. 1, 2, 3, 4, 5, 6, 9, 10, 12, 15, 18, 20, 30, 36, 45, 60, 90, 180

421. prime

423. composite

425. $2 \cdot 2 \cdot 3 \cdot 7$

427. $2 \cdot 5 \cdot 5 \cdot 7$

429. 45

431. 175

433. Answers will vary

Practice Test

435. 15 minus x, the difference of fifteen and x.

437. equation

439. (a) n^6
(b) $3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 = 243$

441. 36

443. 5

445. 45

447. 125

449. 36

451. $x + 5$

453. $3(a - b)$

455. $n = 31$

457. $y - 15 = 32$; $y = 47$

459. 4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48

461. $2^3 \cdot 3^3 \cdot 5$

Chapter 3

Be Prepared

3.1 

3.2 $<$

3.3 14

3.3 20

3.3 $3 + (-7)$

3.4 5

3.5 $20 - (-15)$

3.6 -11

3.7 $20 \div 13$

3.8 -15

3.9 -3

3.10 0

3.11 16

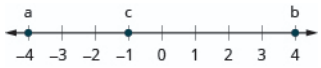
3.12 $x - 5$

Try It

3.1



3.2



3.3 (a) > (b) < (c) > (d) >

3.4 (a) < (b) > (c) < (d) >

3.5 (a) −4 (b) 3

3.6 (a) −8 (b) 5

3.7 1

3.8 5

3.9 (a) −4 (b) 4

3.10 (a) −11 (b) 11

3.11 (a) 12 (b) −28

3.12 (a) 9 (b) −37

3.13 (a) 17 (b) 39 (c) −22 (d) −11

3.14 (a) 23 (b) 21 (c) −37 (d) −49

3.15 (a) > (b) > (c) < (d) =

3.16 (a) > (b) = (c) > (d) <

3.17 (a) 3 (b) 18

3.18 (a) 11 (b) 63

3.19 2

3.20 3

3.21 16

3.22 9

3.23 (a) −9 (b) 15 (c) −20 (d) 11−(−4)

3.24 (a) 19 (b) −22 (c) −9 (d) −8−(−5)

3.25 5 yards

3.26 −30 feet

3.27



6

3.28



7

3.29



−6

3.30



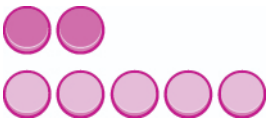
−7

3.31



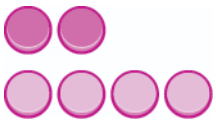
−2

3.32



−3

3.33



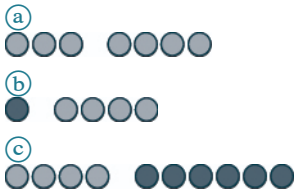
2

3.34



3

3.35



3.36



3.37 ⓐ −17 ⓑ 57

3.38 ⓐ −46 ⓑ 26

3.39 −50

3.40 −70

3.41 13

3.42 0

3.43 ⓐ 2 ⓑ −12

3.44 ⓐ 2 ⓑ −1

3.45 ⓐ −6 ⓑ 10

3.46 ⓐ −1 ⓑ 17

3.47 9

3.48 13

3.49 196

3.50 8

3.51 −7 + 4 = −3

3.52 −8 + (−6) = −14

3.53 [9 + (−16)] + 4 = −3

3.54 [−8 + (−12)] + 7 = −13

3.55 4 degrees Celsius

3.56 −33 feet

3.57 14-yard line

3.58 37-yard line

3.59



2

3.60



3

3.61



−2

3.62



−3

3.63



−10

3.64



−11

3.65



10

3.66



11

3.67 (a)



(b)



(c)



(d)



3.68 (a)



(b)



(c)



(d)



3.69 (a)



−2

(b)



4

3.70 (a)



−3

(b)



3

3.71 (a) 8, 8 (b) −18, −18

3.72 (a) 8, 8 (b) −22, −22

3.73 (a) 19, 19 (b) −4, −4

3.74 (a) 23, 23 (b) 3, 3

3.75 −29

3.76 −26

3.77 3

3.78 13

3.79 −69

3.80 −47

3.81 (a) −2 (b) −15

3.82 (a) −2 (b) −7

3.83 (a) −2 (b) 36

3.84 (a) −19 (b) 9

3.85 (a) 14 − (−23) = 37
(b) −17 − 21 = −38

3.86 (a) 11 − (−19) = 30
(b) −11 − 18 = −29

3.87 45 degrees Fahrenheit

3.88 9 degrees Fahrenheit

3.89 10,103 feet

3.90 233 feet

3.91 (a) $48 (b) −$2 (c) $18

3.92 (a) −$54 (b) No, −$5

3.93 (a) −48 (b) 28 (c) −63
(d) 60

3.94 (a) −56 (b) 54 (c) −28
(d) 39

3.95 (a) −9 (b) 17

3.96 (a) −8 (b) 16

3.97 (a) −7 (b) 39

3.98 (a) −9 (b) 23

3.99 (a) −6 (b) 36

3.100 (a) −28 (b) 52

3.101 −63

3.102 −84

3.103 (a) 81 (b) −81

3.104 (a) 49 (b) −49

3.105 29

3.106 52

3.107 4

3.108 9

3.109 21

3.110 6

3.111 39

3.112 13

3.113 −8

3.114 19

3.115 −5 (12) = −60

3.116 8 (−13) = −104

3.117 −63 ÷ −9 = 7

3.118 −72 ÷ −9 = 8

3.119 (a) no (b) no (c) yes

3.120 (a) no (b) no (c) yes

3.121 −4

3.122 −19

3.123 −6

3.124 −4

3.125 $4x = 12$; $x = 3$

3.126 $3x = 6$; $x = 2$

3.127 7

3.128 11

3.129 −12

3.130 −9

3.131 $x + 7 = -2$; $x = -9$

3.132 $y + 11 = 2$; $y = -9$

3.133 $p - 2 = -4$; $p = -2$

3.134 $q - 7 = -3$; $q = 4$

3.135 $132 = -12y$; $y = -11$

3.136 $117 = -13z$; $z = -9$

Section 3.1 Exercises

1.



3.



5. (a) > (b) < (c) < (d) >

7. (a) < (b) > (c) < (d) >

9. (a) −2 (b) 6

11. (a) 8 (b) −1

13. 4

15. 15

17. (a) −3 (b) 3

19. (a) −12; (b) 12

21. (a) 7 (b) 25 (c) 0

23. (a) 32 (b) 18 (c) 16

25. (a) 28 (b) 15

27. (a) −19 (b) −33

29. (a) < (b) =

31. (a) > (b) >

33. 4

35. 56

37. 0

39. 8

41. 80

43. (a) −8 (b) −(−6), or 6
(c) −3 (d) 4−(−3)

45. (a) −20 (b) −(−5), or 5
(c) −12 (d) 18−(−7)

47. −6 degrees

49. −40 feet

51. −12 yards

53. $3

55. +1

57. (a) 20,320 feet
(b) −282 feet

59. (a) $540 million
(b) −$27 billion

61. Sample answer: I have experienced negative temperatures.

Section 3.2 Exercises

63.



11

65.



−9

67.



−2

69.



1

71. −80

73. 32

75. −135

77. 0

79. −22

81. 108

83. −4

85. 29

87. ⓐ −18 ⓑ −87

89. ⓐ −47 ⓑ 16

91. ⓐ −4 ⓑ 10

93. ⓐ −13 ⓑ 5

95. −8

97. 10

99. 64

101. 121

103. −14 + 5 = −9

105. −2 + 8 = 6

107. −15 + (−10) = −25

109. [−1 + (−12)] + 6 = −7

111. [10 + (−19)] + 4 = −5

113. 7°F

115. −$118

117. −8 yards

119. 25-yard line

121. 20 feet

123. −32

125. Sample answer: In the first case, there are more negatives so the sum is negative. In the second case, there are more positives so the sum is positive.

Section 3.3 Exercises

127.



6

129.



−4

131.



−9

133.



12

135. (a) 9 (b) 9

137. (a) 16 (b) 16

139. (a) 17 (b) 17

141. (a) 45 (b) 45

143. 27

145. 29

147. −39

149. −48

151. −42

153. −59

155. −51

157. 9

159. −2

161. −2

163. 22

165. 53

167. −20

169. 0

171. 4

173. 6

175. −8

177. −11

179. (a) −3 (b) −9

181. (a) 3 (b) 7

183. −8

185. −192

187. (a) 3 − (−10) = 13
(b) 45 − (−20) = 65

189. (a) −6 − 9 = −15
(b) −16 − (−12) = −4

191. (a) −17 − 8 = −25
(b) −24 − 37 = −61

193. (a) 6 − 21 = −15
(b) −19 − 31 = −50

195. −10°

197. 96°

199. 21-yard line

201. $65

203. −$40

205. $26

207. 13°

209. Sample answer: On a number line, 9 is 15 units away from −6.

Section 3.4 Exercises

211. −32

213. −35

215. 36

217. −63

219. −6

221. 14

223. −4

225. −8

227. 13

229. −12

231. −49

233. −47

235. 43

237. −125

239. 64

241. −16

243. 90

245. −88

247. 9

249. 41

251. −5

253. −9

255. −29

257. 5

259. (a) 1 (b) 33

261. (a) −5 (b) 25

263. 11

265. 21

267. 38

269. −56

271. −3·15 = −45

273. −60 ÷ (−20) = 3

275. $\frac{-6}{a+b}$

277. −10 (p − q)

279. −$3,600

281. Sample answer: Multiplying two integers with the same sign results in a positive product. Multiplying two integers with different signs results in a negative product.

283. Sample answer: In the first expression the base is positive and after you raise it to the power you should take the opposite. Then in the second expression the base is negative so you simply raise it to the power.

Section 3.5 Exercises

285. (a) no (b) no (c) yes

287. (a) no (b) no (c) yes

289. $n = -7$

291. $p = -17$

293. $u = -4$

295. $h = 6$

297. $x = -16$

299. $r = -14$

301. $3x = 6$; $x = 2$

303. $2x = 8$; $x = 4$

305. $x = 9$

307. $c = -8$

309. $p = 3$

311. $q = -12$

313. $x = 20$

315. $z = 0$

317. $n + 4 = 1$; $n = -3$

319. $8 + p = -3$; $p = -11$

321. $a - 3 = -14$; $a = -11$

323. $-42 = -7x$; $x = 6$

325. $-15f = 75$; $f = -5$

327. $-6 + c = 4$; $c = 10$

329. $m - 9 = -4$; $m = 5$

331. (a) $x = 8$ (b) $x = 5$

333. (a) $p = -9$ (b) $p = 30$

335. $a = 20$

337. $m = 7$

339. $u = -52$

341. $r = -9$

343. $d = 5$

345. $x = -42$

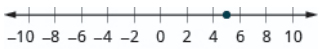
347. 17 cookies

349. Sample answer: It is helpful because it shows how the counters can be divided among the envelopes.

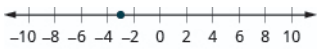
351. Sample answer: The operation used in the equation is multiplication. The inverse of multiplication is division, not addition.

Review Exercises

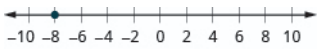
353.



355.



357.



359. <

361. >

363. >

365. −6

367. 4

369. (a) −8 (b) 8

371. (a) −32 (b) 32

373. 21

375. 36

377. 0

379. 14

381. −33

383. <

385. =

387. 55; −55

389. 7

391. 54

393. −1

395. −16

397. −3

399. −10°

401. 10

403. 1

405. 96

407. −50

409. −1

411. 21

413. (a) 3 (b) −16

415. −27

417. −8 + 2 = −6

419. 10 + [−5 + (−6)] = −1

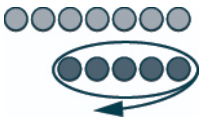
421. 16 degrees

423.



5

425.



7

427. 8

429. −38

431. −58

433. −1

435. (a) −2 (b) −11

437. 41

439. −12 − 5 = −17

441. −2 degrees

443. −36

445. 121

447. −7

449. −8

451. −45

453. −9

455. −81

457. 54

459. 4

461. −66

463. −58

465. −12(6) = −72

467. ⓐ no ⓑ yes ⓒ no

469. −12

471. −7

473. $3x = 9$; $x = 3$

475. 9

477. 4

479. $-6y = -42$; $y = 7$

481. $m + 4 = -48$; $m = -52$

483. Answers will vary.

Practice Test

485. ⓐ < ⓑ >

487. ⓐ 7 ⓑ −8

489. 5

491. −27

493. 11

495. 54

497. −8

499. 22

501. 39

503. 34

505. −7 − (−4) = −3

507. 4°F

509. $n = -1$

511. $r = 6$

513. $y - 8 = -32$; $y = -24$

Chapter 4

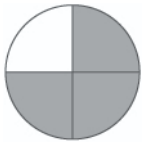
Be Prepared

4.1 11

4.2 $-2 > -5$

4.3 $2 \cdot 2 \cdot 2 \cdot 2 \cdot 3$

4.4



4.5 Answers may vary. Acceptable answers include $\frac{10}{12}$, $\frac{15}{18}$, $\frac{50}{60}$, etc.

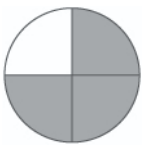
4.6 3

4.7 $\frac{1}{12}$

4.8 $\frac{13}{5}$

4.9 $5x + 5$

4.10

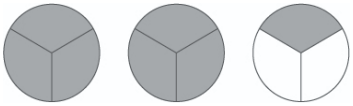


4.11 $\frac{5}{6}$

4.12 $\frac{10}{12}, \frac{15}{18}$

4.13 2

4.14



4.15 $2\frac{3}{4}$

4.16 $\frac{7}{2}$

4.17 1

4.18 $y = 6$

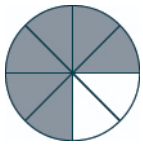
4.19 -6

Try It

4.1 ⓐ $\frac{3}{8}$ ⓑ $\frac{4}{9}$

4.2 ⓐ $\frac{3}{5}$ ⓑ $\frac{3}{4}$

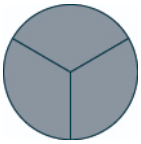
4.3



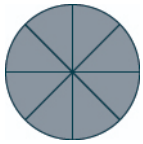
4.4



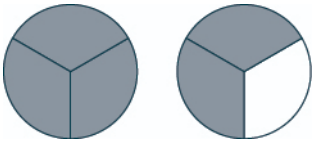
4.5



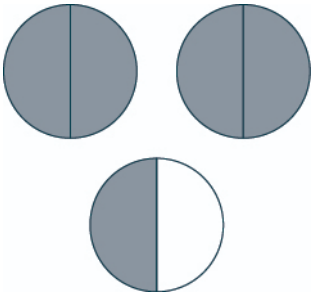
4.6



4.7



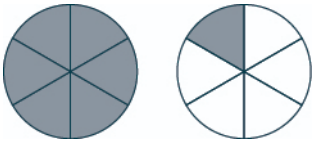
4.8



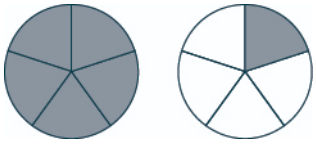
4.9 $\frac{5}{3} = 1\frac{2}{3}$

4.10 $\frac{13}{8} = 1\frac{5}{8}$

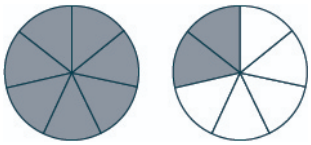
4.11



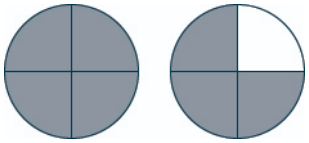
4.12



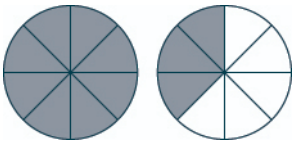
4.13 $1\frac{2}{7}$



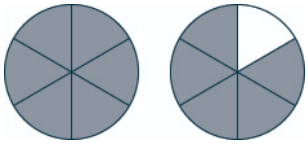
4.14 $1\frac{3}{4}$



4.15 $\frac{11}{8}$



4.16 $\frac{11}{6}$



4.17 $1\frac{6}{7}$.

4.18 $1\frac{5}{9}$

4.19 $3\frac{2}{7}$

4.20 $4\frac{4}{11}$

4.21 $\frac{26}{7}$

4.22 $\frac{23}{8}$

4.23 $\frac{50}{11}$

4.24 $\frac{34}{3}$

4.25 2

4.26 3

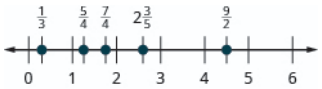
4.27 Correct answers include $\frac{6}{10}$, $\frac{9}{15}$, and $\frac{12}{20}$.

4.28 Correct answers include $\frac{8}{10}$, $\frac{12}{15}$, and $\frac{16}{20}$.

4.29 $\frac{18}{21}$

4.30 $\frac{30}{100}$

4.31



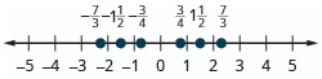
4.32



4.33



4.34



4.35 (a) > (b) > (c) < (d) <

4.36 (a) > (b) < (c) > (d) <

4.37 $\frac{2}{3}$

4.38 $\frac{3}{4}$

4.39 $-\frac{3}{4}$

4.40 $-\frac{2}{3}$

4.41 $-\frac{9}{7}$

4.42 $-\frac{9}{5}$

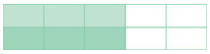
4.43 $\frac{23}{40}$

4.44 $\frac{5}{8}$

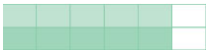
4.45 $\frac{x}{y}$

4.46 $\frac{a}{b}$

4.47 $\frac{3}{10}$



4.48 $\frac{5}{12}$



4.49 $\frac{2}{15}$

4.50 $\frac{21}{40}$

4.51 $\frac{5}{14}$

4.52 $\frac{14}{27}$

4.53 $-\frac{4}{21}$

4.54 $-\frac{3}{16}$

4.55 (a) 9 (b) $-33a$

4.56 (a) 24 (b) $\frac{44x}{3}$

4.57 (a) $\frac{7}{5}$ (b) -8 (c) $-\frac{4}{11}$ (d) $\frac{1}{14}$

4.58 (a) $\frac{7}{3}$ (b) -12 (c) $-\frac{9}{14}$ (d) $\frac{1}{21}$

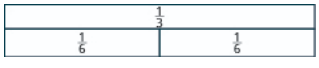
4.59

Number	Opposite	Absolute Value	Reciprocal
$-\frac{5}{8}$	$\frac{5}{8}$	$\frac{5}{8}$	$-\frac{8}{5}$
$\frac{1}{4}$	$-\frac{1}{4}$	$\frac{1}{4}$	4
$\frac{8}{3}$	$-\frac{8}{3}$	$\frac{8}{3}$	$\frac{3}{8}$
-8	8	8	$-\frac{1}{8}$

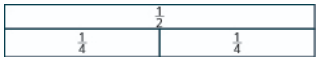
4.60

Number	Opposite	Absolute Value	Reciprocal
$-\frac{4}{7}$	$\frac{4}{7}$	$\frac{4}{7}$	$-\frac{7}{4}$
$\frac{1}{8}$	$-\frac{1}{8}$	$\frac{1}{8}$	8
$\frac{9}{4}$	$-\frac{9}{4}$	$\frac{9}{4}$	$\frac{4}{9}$
-1	1	1	-1

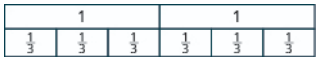
4.61 2



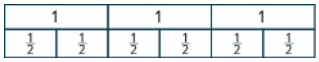
4.62 2



4.63 6



4.64 6



4.65 $-\frac{9}{14}$

4.66 $-\frac{10}{21}$

4.67 $\frac{21}{5p}$

4.68 $\frac{15}{8q}$

4.69 $\frac{4}{5}$

4.70 $\frac{5}{4}$

4.71 $\frac{4}{15}$

4.72 $\frac{2}{3}$

4.73 2

4.74 $\frac{9}{4}$

4.75 −15

4.76 $-\frac{85}{6}$

4.77 $\frac{5}{8}$

4.78 $\frac{7}{8}$

4.79 2

4.80 $\frac{5}{2}$

4.81 $\frac{9s}{14}$

4.82 $\frac{5y}{6}$

4.83 $\frac{a-b}{cd}$

4.84 $\frac{p+q}{r}$

4.85 $\frac{4}{5}$

4.86 $\frac{11}{14}$

4.87 $-\frac{2}{7}$

4.88 $-\frac{10}{3}$

4.89 $\frac{3}{4b}$

4.90 $\frac{4}{q}$

4.91 $\frac{25}{49}$.

4.92 $\frac{1}{2}$

4.93 $-\frac{3}{5}, \frac{3}{-5}$

4.94 $\frac{-2}{7}, \frac{2}{-7}$

4.95 $\frac{10}{9}$

4.96 $\frac{1}{2}$

4.97 $\frac{-3}{4}$

4.98 $-\frac{2}{3}$

4.99 $\frac{2}{9}$

4.100 $\frac{8}{5}$

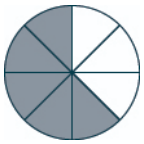
4.101 4

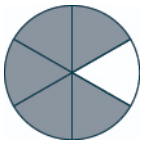
4.102 2

4.103 $\frac{5}{8}$

4.104 $\frac{5}{6}$

4.105 $\frac{5}{6}$





4.106 1

4.107 $\frac{x+3}{4}$

4.108 $\frac{y+5}{8}$

4.109 $\frac{1}{d}$

4.110 $\frac{3}{m}$

4.111 $\frac{9p}{8}$

4.112 $\frac{9q}{5}$

4.113 $-\frac{2}{3}$

4.114 $-\frac{2}{3}$

4.115 $\frac{3}{8}$, models may differ.

4.116 $\frac{1}{6}$, models may differ

4.117 $\frac{3}{7}$

4.118 $\frac{1}{2}$

4.119 $\frac{x-2}{7}$

4.120 $\frac{y-13}{14}$

4.121 $-\frac{16}{x}$

4.122 $-\frac{22}{a}$

4.123 −1

4.124 $-\frac{2}{3}$

4.125 60

4.126 15

4.127 96

4.128 224

4.129 $\frac{9}{12}, \frac{10}{12}$

4.130 $-\frac{35}{60}, \frac{44}{60}$

4.131 $\frac{52}{96}, \frac{51}{96}$

4.132 $\frac{72}{224}, \frac{189}{224}$

4.133 $\frac{7}{12}$

4.134 $\frac{7}{10}$

4.135 $\frac{5}{8}$

4.136 $\frac{1}{2}$

4.137 $\frac{79}{60}$

4.138 $\frac{103}{60}$

4.139 $\frac{1}{96}$

4.140 $\frac{75}{224}$

4.141 $\frac{37}{210}$

4.142 $-\frac{25}{96}$

4.143 $\frac{3y+14}{18}$

4.144 $\frac{5x+14}{30}$

4.145 (a) $-\frac{11}{12}$ (b) $-\frac{1}{8}$

4.146 (a) $-\frac{10}{3}$ (b) $\frac{13}{12}$

4.147 (a) $\frac{(27a-32)}{36}$ (b) $\frac{2a}{3}$

4.148 (a) $\frac{(24k+25)}{30}$ (b) $\frac{24k}{5}$

4.149 $\frac{1}{90}$

4.150 272

4.151 2

4.152 $\frac{2}{7}$

4.153 (a) -1 (b) $-\frac{1}{2}$

4.154 (a) $\frac{7}{6}$ (b) $-\frac{1}{4}$

4.155 $-\frac{3}{4}$

4.156 $-\frac{23}{8}$

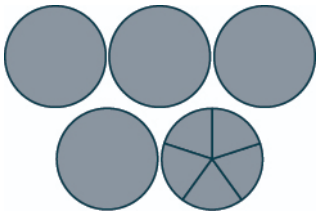
4.157 $-\frac{1}{2}$

4.158 $\frac{2}{3}$

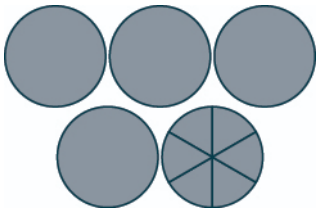
4.159 $-\frac{5}{2}$

4.160 $\frac{3}{2}$

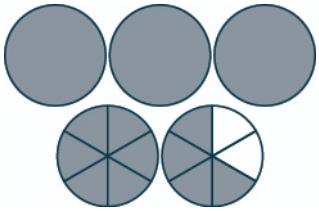
4.161 5



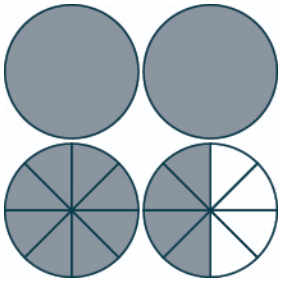
4.162 5



4.163 $4\frac{2}{3}$



4.164 $3\frac{1}{2}$



4.165 $5\frac{6}{7}$

4.166 $7\frac{9}{11}$

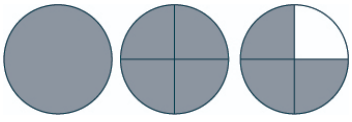
4.167 $16\frac{1}{2}$

4.168 $15\frac{1}{3}$

4.169 $9\frac{1}{3}$

4.170 $6\frac{3}{5}$

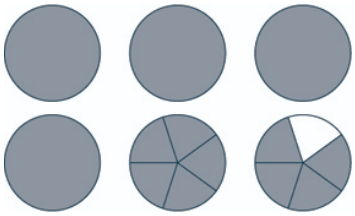
4.171 $\frac{3}{4}$



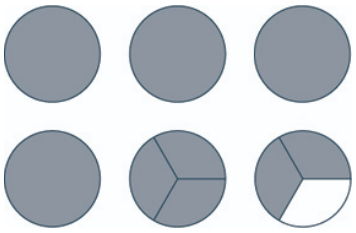
4.172 $\frac{4}{5}$



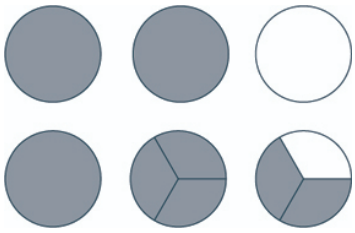
4.173 $\frac{9}{5}$



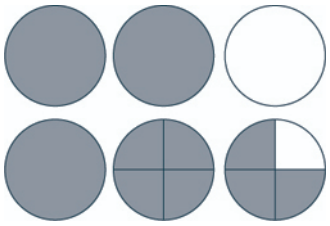
4.174 $\frac{5}{3}$



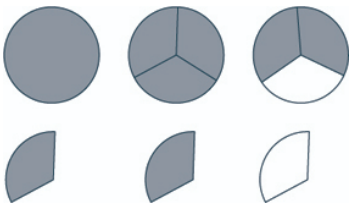
4.175 $\frac{2}{3}$



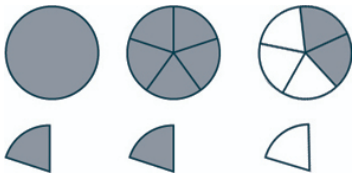
4.176 $\frac{3}{4}$



4.177 $\frac{2}{3}$



4.178 $\frac{2}{5}$



4.179 $2\frac{2}{3}$

4.180 $1\frac{5}{7}$

4.181 $2\frac{2}{3}$

4.182 $1\frac{5}{7}$

4.183 $6\frac{7}{12}$

4.184 $12\frac{3}{10}$

4.185 $4\frac{7}{10}$

4.186 $2\frac{11}{12}$

4.187 $-\frac{41}{8}$

4.188 $-\frac{757}{63}$

4.189 (a) no (b) yes (c) no

4.190 (a) no (b) no (c) yes

4.191 $-\frac{1}{2}$

4.192 $-\frac{4}{15}$

4.193 −1

4.194 $-\frac{6}{7}$

4.195 $-\frac{19}{3}$

4.196 $\frac{23}{2}$

4.197 −125

4.198 −243

4.199 245

4.200 132

4.201 −48

4.202 23

4.203 35

4.204 18

4.205 -91

4.206 -108

4.207 $\frac{n}{7} = -21; n = -147$

4.208 $\frac{n}{8} = -56; n = -448$

4.209 $\frac{q}{-8} = 72; q = -576$

4.210 $\frac{p}{-9} = 81; p = -729$

4.211 $\frac{2}{5}f = 16; f = 40$

4.212 $\frac{3}{4}f = 21; f = 28$

4.213 $\frac{n}{\frac{2}{3}} = \frac{5}{12}; n = \frac{5}{18}$

4.214 $\frac{c}{\frac{3}{8}} = \frac{4}{9}; c = \frac{1}{6}$

4.215 $\frac{5}{8} + x = \frac{1}{4}; x = -\frac{3}{8}$

4.216 $1\frac{3}{4} - x = \frac{5}{6};\ x = \frac{11}{12}$

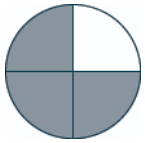
Section 4.1 Exercises

1. ⓐ $\frac{1}{4}$ ⓑ $\frac{3}{4}$ ⓒ $\frac{3}{8}$ ⓓ $\frac{5}{8}$

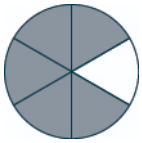
3.



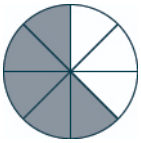
5.



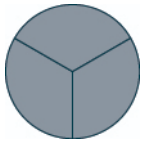
7.



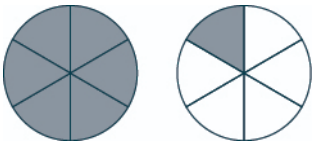
9.



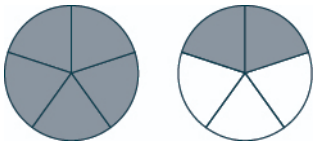
11.



13.



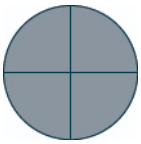
15.



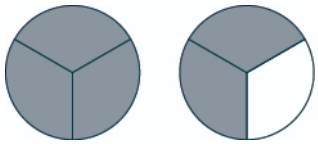
17. ⓐ $\frac{5}{4} = 1\frac{1}{4}$ ⓑ $\frac{7}{4} = 1\frac{3}{4}$ ⓒ $\frac{11}{8} = 1\frac{3}{8}$

19. ⓐ $\frac{11}{4} = 2\frac{3}{4}$ ⓑ $\frac{19}{8} = 2\frac{3}{8}$

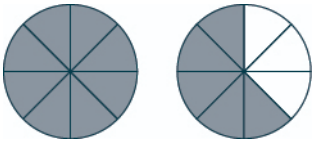
21.



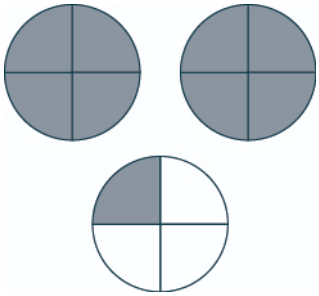
23.



25.



27.



29. $1\frac{2}{3}$

31. $2\frac{3}{5}$

33. $3\frac{1}{9}$

35. $3\frac{2}{15}$

37. $\frac{7}{5}$

39. $\frac{17}{6}$

41. $\frac{19}{7}$

43. $\frac{32}{9}$

45. 4

47. 9

49. 9

51. Answers may vary. Correct answers include $\frac{2}{6}, \frac{3}{9}, \frac{4}{12}$.

53. Answers may vary. Correct answers include $\frac{10}{12}, \frac{15}{18}, \frac{20}{24}$.

55. Answers may vary. Correct answers include $\frac{10}{18}, \frac{15}{27}, \frac{20}{36}$.

57.



59.



61.



63.



65. <

67. >

69. <

71. <

73. (a) 2 (b) $6\frac{1}{4}$

75. Answers will vary.

Section 4.2 Exercises

77. $\frac{1}{3}$

79. $\frac{3}{4}$

81. $-\frac{5}{11}$

83. $-\frac{12}{7}$

85. $\frac{10}{21}$

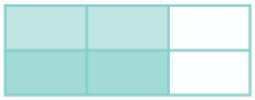
87. $-\frac{7}{8}$

89. $\frac{x}{y}$

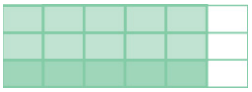
91. $-\frac{x}{4y}$

93. $\frac{2x^2}{3y}$

95. $\frac{1}{3}$



97. $\frac{5}{18}$



99. $\frac{2}{15}$

101. $\frac{27}{40}$

103. $\frac{1}{4}$

105. $-\frac{1}{6}$

107. $-\frac{2}{9}$

109. $-\frac{21}{50}$

111. $\frac{11}{30}$

113. $\frac{20}{11}$

115. $9n$

117. $7p$

119. −34

121. $\frac{3}{8}$

123. $\frac{8}{27}$

125. $\frac{1296}{625}$

127. $\frac{4}{3}$

129. $-\frac{17}{5}$

131. $\frac{8}{11}$

133. $-\frac{1}{19}$

135. 1

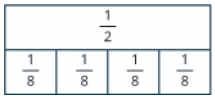
136.

Number	Opposite	Absolute Value	Reciprocal
$-\frac{4}{7}$	$\frac{4}{7}$	$\frac{4}{7}$	$-\frac{7}{4}$
$\frac{1}{8}$	$-\frac{1}{8}$	$\frac{1}{8}$	8
$\frac{9}{4}$	$-\frac{9}{4}$	$\frac{9}{4}$	$\frac{4}{9}$
-1	1	1	-1

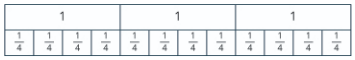
137.



139. 4



141. 12



143. 4

145. $\frac{16}{15}$

147. $-\frac{5}{4}$

149. 1

151. $\frac{18}{5y}$

153. $\frac{25}{2c}$

155. $-\frac{3}{4}$

157. $\frac{2}{9}$

159. $\frac{14r}{15s}$

161. -12

163. $-\frac{1}{25}$

165. 9

167. $\frac{8}{7}$

169. (a) $4 \cdot \frac{2}{3}$ cups$= \frac{8}{3}$ cups $= 2\frac{2}{3}$ cups
(b) Answers will vary.

171. $\frac{1}{8}$ yard

173. Answers will vary.

175. Answers will vary.

Section 4.3 Exercises

177. $\frac{44}{21}$

179. $\frac{35}{8}$

181. $-\frac{16}{3}$

183. $-\frac{63}{16}$

185. $\frac{3}{2}$

187. $-\frac{4}{3}$

189. 2

191. 5

193. $\frac{7v}{13}$

195. $\frac{a}{b}$

197. $\frac{A}{3-B}$

199. $\frac{3}{2}$

201. $-\frac{15}{22}$

203. $-\frac{3}{10}$

205. $\frac{1}{6}$

207. $\frac{3r}{5s}$

209. $\frac{9}{2y}$

211. 28

213. $-\frac{1}{18}$

215. $\frac{-4}{9}, -\frac{4}{9}$

217. $\frac{13}{-6}, \frac{-13}{6}$

219. $\frac{12}{7}$

221. $\frac{5}{2}$

223. $\frac{23}{4}$

225. $-\frac{1}{3}$

227. $\frac{4}{5}$

229. $-\frac{1}{2}$

231. 2

233. $\frac{5}{6}$

235. $\frac{26}{25}$

237. $\frac{11}{6}$

239. $\frac{5}{2}$

241. -10

243. −2

245. $\frac{51}{20}$

247. $\frac{18}{7}$

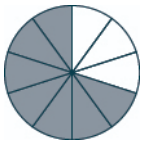
249. (a) $\frac{4}{3} = 1\frac{1}{3}$ cups
(b) $\frac{80}{3} = 26\frac{2}{3}$ cups

251. Answers will vary.

253. Answers will vary.

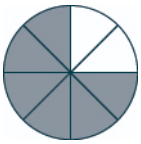
Section 4.4 Exercises

255.



$\frac{7}{10}$

257.



$\frac{3}{4}$

259. $\frac{7}{9}$

261. $\frac{16}{15}$

263. $\frac{y+2}{3}$

265. $\frac{14}{q}$

267. $\frac{9a}{7}$

269. $\frac{-4x}{5}$

271. $-\frac{3}{4}$

273. $-\frac{7}{8}$

275. $\frac{8}{19}$

277. $-\frac{13}{12}$

279.



$\frac{1}{2}$

281. $\frac{1}{5}$

283. $\frac{5}{13}$

285. $\frac{1}{6}$

287. $-\frac{8}{3}$

289. $\frac{x-8}{19}$

291. $\frac{11z-8}{13}$

293. $-\frac{14}{c}$

295. $-\frac{55}{v}$

297. $\frac{3d}{11}$

299. $-\frac{14s}{3}$

301. $\frac{2}{7}$

303. $-\frac{3}{11}$

305. $-\frac{1}{8}$

307. $\frac{6-s}{11}$

309. $-\frac{2}{9}$

311. $\frac{49}{27}$

313. $\frac{1}{4}$ cup

315. No, adding up the number of pieces gives $\frac{13}{12}$, which is greater than 1. (Answers may vary.)

Section 4.5 Exercises

317. 20

319. 48

321. 240

323. 245

325. 60

327. $\frac{5}{20}, \frac{4}{20}$

329. $\frac{14}{24}, \frac{15}{24}$

331. $\frac{33}{48}, -\frac{20}{48}$

333. $\frac{20}{60}, \frac{45}{60}, \frac{36}{60}$

335. $\frac{9}{20}$

337. $\frac{11}{24}$

339. $\frac{3}{8}$

341. $\frac{2}{3}$

343. $\frac{23}{20}$

345. $\frac{19}{24}$

347. $\frac{1}{48}$

349. $\frac{1}{24}$

351. $\frac{1}{12}$

353. $\frac{7}{60}$

355. $-\frac{53}{80}$

357. $-\frac{291}{245}$

359. $\frac{1}{20}$

361. $\frac{11}{40}$

363. $\frac{11}{6}$

365. $\frac{7}{10}$

367. $\frac{3y+4}{6}$

369. $\frac{4x-5}{20}$

371. (a) $\frac{5}{6}$ (b) 4

373. (a) $-\frac{37}{40}$ (b) $-\frac{1}{10}$

375. (a) $\frac{9a}{14}$ (b) $\frac{9a-14}{24}$

377. (a) $-\frac{4q}{27}$ (b) $\frac{12-25q}{45}$

379. $\frac{3}{4}$

381. $\frac{11}{24}$

383. $\frac{7}{18}$

385. $-\frac{2}{15}$

387. $\frac{-33-8x}{88}$

389. $\frac{16}{39}$

391. $\frac{1}{81}$

393. 32

395. $\frac{36}{25}$

397. $\frac{60}{7}$

399. $\frac{15}{2}$

401. $\frac{3}{13}$

403. $\frac{19}{30}$

405. −9

407. $\frac{91}{60}$

409. $\frac{11}{40}$

411. $\frac{5}{6}$

413. 1

415. $\frac{22}{7}$

417. (a) $\frac{1}{2}$ (b) -1

419. (a) 0 (b) $-\frac{1}{6}$

421. (a) $\frac{1}{3}$ (b) -1

423. (a) $\frac{1}{6}$ (b) $\frac{2}{3}$

425. $\frac{4}{15}$

427. $-\frac{9}{16}$

429. −2

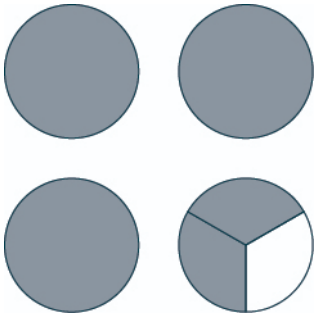
431. 3

433. She needs $2\frac{3}{8}$ cups

435. Answers will vary.

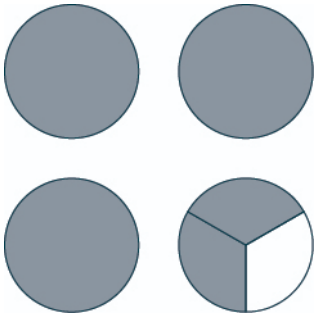
Section 4.6 Exercises

437.



$3\frac{2}{3}$

439.



$3\frac{2}{3}$

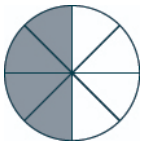
441. $7\frac{5}{9}$

443. 11

445. $11\frac{1}{3}$

447. $11\frac{1}{3}$

449.



$\frac{1}{2}$

451. $1\frac{1}{6}$

453. $6\frac{2}{5}$

455. $1\frac{7}{9}$

457. $\frac{5}{6}$

459. $7\frac{11}{12}$

461. $16\frac{1}{6}$

463. $5\frac{11}{20}$

465. $-1\frac{11}{24}$

467. $6\frac{17}{18}$

469. $\frac{7}{9}$

471. 23

473. $8\frac{3}{4}$

475. $\frac{4}{15}$

477. $5\frac{3}{5}$

479. $\frac{1}{16}$

481. $13\frac{1}{13}$

483. $7\frac{1}{30}$

485. $\frac{10}{27}$

487. $2\frac{7}{45}$

489. $\frac{17}{24}$

491. $\frac{7}{12}$ yards

493. $13\frac{1}{4}$ inches

495. Answers will vary.

497. Answers will vary.

Section 4.7 Exercises

499. (a) no (b) yes (c) no

501. (a) no (b) yes (c) no

503. $m = \frac{1}{2}$

505. $h = -\frac{2}{3}$

507. $c = -1$

509. $z = -1$

511. $p = \frac{37}{40}$

513. $k = -\frac{7}{15}$

515. $k = \frac{18}{7}$

517. $v = -\frac{11}{3}$

519. $b = -27$

521. $x = -256$

523. $q = 160$

525. $s = 45$

527. $y = -42$

529. $k = \frac{17}{20}$

531. $p = 100$

533. $m = -16$

535. $b = -21$

537. $v = 36$

539. $y = 0$

541. $g = \frac{1}{9}$

543. $q = -\frac{3}{4}$

545. $n = \frac{14}{5}$

547. $y = -\frac{25}{24}$

549. $d = -\frac{8}{15}$

551. $\frac{n}{6} = -24; n = -144$

553. $\frac{m}{-7} = -8; m = 56$

555. $\frac{f}{-4} = -20; f = 80$

557. $\frac{g}{9} = 14; g = 126$

559. $\frac{2}{5}q = 20; q = 50$

561. $\frac{4}{9}p = -28; p = -63$

563. $\frac{h}{2} = 43; h = 86$

565. $\frac{\frac{a}{2}}{3} = \frac{3}{4};\ a = \frac{1}{2}$

567. $\frac{3}{4} + x = \frac{1}{8}; x = -\frac{5}{8}$

569. $y - \frac{1}{3} = -\frac{1}{6}; y = \frac{1}{6}$

571. 30 inches

573. Answers will vary.

Review Exercises

575. $\frac{5}{9}$

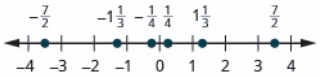
577. $\frac{3}{2} = 1\frac{1}{2}$

579. $5\frac{8}{11}$

581. $\frac{49}{5}$

583. Answers may vary.

585.



587. >

589. $-\frac{3}{4}$

591. $-\frac{x}{y}$

593. $-\frac{4}{7}$

595. $\frac{24m}{11}$

597. 6

599. $\frac{4}{15}$

601. −4

603. 4

605. $\frac{4}{15}$

607. $-\frac{5}{2b}$

609. $-\frac{268}{11}$

611. 8

613. $\frac{V}{h-6}$

615. $-\frac{2}{9}$

617. 22

619. $\frac{7}{36}$

621. $\frac{5}{8}$

623. $\frac{3}{5}$

625. $\frac{x+7}{10}$

627. $\frac{1}{2}$

629. $-\frac{19}{15}$

631. $\frac{8}{15}$

633. 15

635. 60

637. $\frac{9}{24}$ and $\frac{20}{24}$

639. $\frac{20}{60}$, $\frac{45}{60}$ and $\frac{48}{60}$

641. $\frac{1}{4}$

643. $-\frac{77}{90}$

645. $\frac{3y-10}{30}$

647. $\frac{14d}{11}$

649. $\frac{256}{225}$

651. (a) $-\frac{8}{5}$ (b) $-\frac{11}{20}$

653. $13\frac{2}{3}$

655. $8\frac{1}{11}$

657. $5\frac{1}{10}$

659. $1\frac{1}{3}$

661. (a) no (b) yes (c) no

663. $n = -\frac{5}{11}$

665. $h = -\frac{51}{40}$

667. $z = -23$

669. $q - \frac{1}{10} = \frac{1}{2}; q = \frac{3}{5}$

671. $\frac{3}{8}y = 24; y = 64$

Practice Test

673. $\frac{23}{7}$

675. $\frac{1}{4}$

677. $\frac{1}{4}$

679. $16u$

681. -2

683. $\frac{12}{25}$

685. 5

687. $\frac{5p}{2q}$

689. 13

691. $-\frac{7}{13}$

693. -1

695. $-\frac{9}{4x}$

697. 3

699. $y = \frac{4}{5}$

701. $f = \frac{13}{12}$

703. $c = -27$

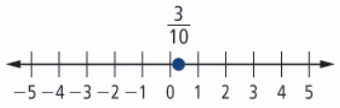
Chapter 5

Be Prepared

5.1 Four million, nine hundred twenty-six thousand, fifteen

5.2 750

5.3



5.4 $\frac{7}{10}$

5.5 $\frac{27}{100}$

5.6 4

5.7 0.03

5.8 >

5.9 <

5.10 $\frac{5}{12}$

5.11 20

5.12 −294

5.13 5

5.14 50

5.15 2.5

5.16 $\frac{2}{3}$

5.17 0.24

5.18 $\frac{6}{11}$

5.19 81

5.20 3.85

5.21 960

Try It

5.1 (a) six and seven tenths (b) nineteen and fifty-eight hundredths (c) eighteen thousandths (d) negative two and fifty-three thousandths

5.2 (a) five and eight tenths (b) three and fifty-seven hundredths (c) five thousandths (d) negative thirteen and four hundred sixty-one thousandths

5.3 13.68

5.4 5.894

5.5 0.058

5.6 0.067

5.7 (a) $5\frac{3}{10}$ (b) $6\frac{7}{100}$ (c) $-\frac{117}{500}$

5.8 (a) $8\frac{7}{10}$ (b) $1\frac{3}{100}$ (c) $-\frac{3}{125}$

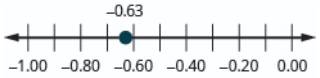
5.9



5.10



5.11



5.12



5.13 (a) > (b) >

5.14 (a) < (b) <

5.15 >

5.16 >

5.17 1.05

5.18 9.17

5.19 (a) 6.58 (b) 6.6 (c) 7

5.20 (a) 15.218 (b) 15.22 (c) 15.2

5.21 17.6

5.22 33.11

5.23 16.49

5.24 23.593

5.25 0.42

5.26 12.58

5.27 −1.53

5.28 −3.58

5.29 27.4815

5.30 87.6148

5.31 −13.427

5.32 −38.122

5.33 0.00348

5.34 0.00603

5.35 (a) 25.8 (b) 258 (c) 2,580

5.36 (a) 142 (b) 1,420 (c) 14,200

5.37 0.07

5.38 0.08

5.39 $0.19

5.40 $0.42

5.41 −0.39

5.42 −0.4

5.43 587.3

5.44 34.25

5.45 200

5.46 350

5.47 $16.52

5.48 $65.97

5.49 $41.28

5.50 $48.75

5.51 $15.47

5.52 $14.25

5.53 $5.97

5.54 $43.00

5.55 0.25

5.56 0.375

5.57 −2.25

5.58 −5.5

5.59 $2.\overline{45}$

5.60 $2.3\overline{18}$

5.61 5.275

5.62 6.35

5.63 >

5.64 <

5.65 <

5.66 >

5.67 $\frac{4}{5}, 0.82, \frac{7}{8}$

5.68 $\frac{3}{4}, \frac{13}{16}, 0.835$

5.69 (a) −183.2 (b) 4.5

5.70 (a) −776.25 (b) 2.2

5.71 11.16

5.72 1.51

5.73 (a) 314 in. (b) 7850 sq. in.

5.74 (a) 628 ft. (b) 31,400 sq. ft.

5.75 (a) 325.304 cm (b) 8425.3736 sq. cm

5.76 (a) 165.792 m (b) 2188.4544 sq. m

5.77 (a) $\frac{220}{147}$ m (b) $\frac{550}{3087}$ sq. m

5.78 (a) $\frac{40}{21}$ in. (b) $\frac{200}{693}$ sq.in.

5.79 (a) no (b) yes (c) no

5.80 (a) yes (b) no (c) no

5.81 $y = -8$

5.82 $y = -8.4$

5.83 $a = 1.07$

5.84 $n = 0.83$

5.85 $b = -12$

5.86 $c = -8$

5.87 $c = 11.7$

5.88 $b = 6.48$

5.89 $y - 4.9 = 2.8$; $y = 7.7$

5.90 $z - 5.7 = 3.4$; $z = 9.1$

5.91 $-4.3x = 12.04$; $x = -2.8$

5.92 $-3.1m = 26.66$; $m = -8.6$

5.93 $\frac{q}{-3.4} = 4.5$; $q = -15.3$

5.94 $\frac{r}{-2.6} = 2.5$; $r = -6.5$

5.95 $j + 3.8 = 2.6$; $j = -1.2$

5.96 $k + 4.7 = 0.3$; $k = -4.4$

5.97 8.5

5.98 9

5.99 21.5 years

5.100 10

5.101 \$7.19

5.102 \$39.62

5.103 43

5.104 30

5.105 8.5

5.106 20.5

5.107 2

5.108 5

5.109 21

5.110 5

5.111 (a) $\frac{1}{10}$ (b) 0.1

5.112 (a) $\frac{1}{20}$ (b) 0.05

5.113 (a) $\frac{5}{8}$ (b) 0.625

5.114 (a) $\frac{2}{8}$ (b) 0.25

5.115 (a) $\frac{3}{8}$ (b) $\frac{3}{2}$

5.116 (a) $\frac{3}{8}$ (b) $\frac{3}{2}$

5.117 (a) $\frac{2}{5}$ (b) $\frac{10}{3}$

5.118 (a) $\frac{2}{9}$ (b) $\frac{5}{1}$

5.119 $\frac{2}{3}$

5.120 $\frac{9}{22}$

5.121 $\frac{37}{8}$

5.122 $\frac{102}{19}$

5.123 $\frac{8}{3}$

5.124 $\frac{2}{9}$

5.125 $\frac{123 \text{ miles}}{2 \text{ hours}}$

5.126 $\frac{121 \text{ miles}}{3 \text{ hours}}$

5.127 \$18.00/hour

5.128 \$19.00/hour

5.129 23.5 mpg

5.130 28 mpg

5.131 \$0.29/box

5.132 \$0.53/bottle

5.133 Brand A costs \$0.11 per bag. Brand B costs \$0.13 per bag. Brand A is the better buy.

5.134 Brand C costs \$0.07 per ounce. Brand D costs \$0.09 per ounce. Brand C is the better buy.

5.135 (a) 689 mi/h hours (b) y parents/22 students (c) \$$d$/9 min

5.136 (a) m mi/9 h (b) x students/8 buses (c) \$$y$/40 h

5.137 (a) 6 (b) 13

5.138 (a) 4 (b) 14

5.139 (a) −2 (b) −15

5.140 (a) −9 (b) −8

5.141 (a) not a real number (b) −9

5.142 (a) not a real number (b) −11

5.143 (a) 7 (b) 5

5.144 (a) 17 (b) 23

5.145 $6 < \sqrt{38} < 7$

5.146 $9 < \sqrt{84} < 10$

5.147 ≈ 3.32

5.148 ≈ 3.61

5.149 y

5.150 m

5.151 $8x$

5.152 $13y$

5.153 $-11y$

5.154 $-10p$

5.155 $10ab$

5.156 $15mn$

5.157 19.2 feet

5.158 52 centimeters

5.159 9 seconds

5.160 3.5 seconds

5.161 42.7 mph

5.162 54.1 mph

Section 5.1 Exercises

1. five and five tenths

3. five and one hundredth

5. eight and seventy-one hundredths

7. two thousandths

9. three hundred eighty-one thousandths

11. negative seventeen and nine tenths

13. 8.03

15. 29.81

17. 0.7

19. 0.001

21. 0.029

23. −11.0009

25. 13.0395

27. $1\frac{99}{100}$

29. $15\frac{7}{10}$

31. $\frac{239}{1000}$

33. $\frac{13}{100}$

35. $\frac{11}{1000}$

37. $-\frac{7}{100000}$

39. $6\frac{2}{5}$

41. $7\frac{1}{20}$

43. $4\frac{3}{500}$

45. $10\frac{1}{4}$

47. $1\frac{81}{250}$

49. $14\frac{1}{8}$

51.



53.



55.



57.



59. >

61. <

63. >

65. >

67. <

69. <

71. 0.7

73. 2.8

75. 0.85

77. 5.79

79. 0.30

81. 4.10

83. (a) 5.78 (b) 5.8 (c) 6

85. (a) 63.48 (b) 63.5 (c) 63

87. (a) $58,966 (b) $59,000 (c) $60,000

89. (a) $142.19 (b) $142

91. Answers will vary.

93. Tim had the faster time. 12.3 is less than 12.32, so Tim had the faster time.

Section 5.2 Exercises

95. 24.48

97. 170.88

99. −9.23

101. 49.73

103. −40.91

105. −7.22

107. −13.5

109. 35.8

111. −27.5

113. 15.73

115. 42.51

117. 102.212

119. 51.31

121. −4.89

123. 0.12

125. 0.144

127. 42.008

129. 26.7528

131. −11.653

133. 337.8914

135. 2.2302

137. 1.305

139. 92.4

141. 55,200

143. 0.03

145. 0.19

147. $0.71

149. $2.44

151. 3

153. −4.8

155. 35

157. 2.08

159. 150

161. 20

163. 19.2

165. 12.09

167. 32.706

169. $48.60

171. 20

173. 2

175. $17.80

177. $24.89

179. $29.06

181. $3.19

183. 181.7 pounds

185. $15.00

187. $296.00

189. $12.75

191. (a) $3 (b) $2 (c) $1.50 (d) $1.20 (e) $1

193. $18.64

195. $259.45

197. (a) $243.57 (b) $79.35

199. The difference: 0.03 seconds. Three hundredths of a second.

Section 5.3 Exercises

201. 0.4

203. −0.375

205. 0.85

207. 2.75

209. −12.4

211. $0.\overline{5}$

213. $1.\overline{36}$

215. $0.\overline{135}$

217. 7

219. 3.025

221. 10.58

223. <

225. >

227. <

229. <

231. >

233. >

235. $0.55, \frac{9}{16}, \frac{3}{5}$

237. $\frac{5}{8}, \frac{13}{20}, 0.702$

239. $-\frac{7}{20}, -\frac{1}{3}, -0.3$

241. $-\frac{7}{9}, -\frac{3}{4}, -0.7$

243. −187

245. 295.12

247. 6.15

249. 20.2

251. 107.11

253. 449

255. 9.14

257. −0.23

259. −3.25

261. 16.29

263. 632.045

265. −5.742

267. (a) 31.4 in (b) 78.5 sq.in.

269. (a) 56.52.ft.
(b) 254.34 sq.ft.

271. (a) 288.88 cm
(b) 6644.24 sq.cm

273. (a) 116.808 m
(b) 1086.3144 sq.m

275. (a) $\frac{22}{5}$ mile
(b) $\frac{77}{50}$ sq.mile

277. (a) $\frac{33}{14}$ yard
(b) $\frac{99}{224}$ sq.yard

279. (a) $\frac{55}{21}$ m (b) $\frac{275}{504}$ sq.m

281. $56.66

283. Answers will vary.

Section 5.4 Exercises

285. (a) no (b) no (c) yes

287. (a) no (b) yes (c) no

289. $y = 2.8$

291. $f = -0.85$

293. $a = -7.9$

295. $c = -4.65$

297. $n = 4.4$

299. $x = -3.5$

301. $j = -4.68$

303. $m = -1.42$

305. $x = 7$

307. $c = -5$

309. $p = 3$

311. $q = -80$

313. $x = 20$

315. $z = 2.7$

317. $a = -8$

319. $x = -0.28$

321. $p = 8.25$

323. $r = 7.2$

325. $x = -6$

327. $p = -10$

329. $m = 8$

331. $q = -\frac{3}{4}$

333. $m = \frac{4}{35}$

335. $y = -\frac{25}{24}$

337. $f = -\frac{11}{20}$

339. $a = -0.8$

341. $r = -1.45$

343. $h = 24$

345. $n - 1.9 = 3.4; 5.3$

347. $-6.2x = -4.96; 0.8$

349. $\frac{y}{-1.7} = -5;\ 8.5$

351. $n + (-7.3) = 2.4; 9.7$

353. $104

355. Answers will vary.

Section 5.5 Exercises

357. 4

359. 35

361. 245.5

363. 11.65

365. $18.84

367. 19.5 minutes

369. 0.329

371. 21

373. 45

375. 4.5

377. 99.65

379. $6.50

381. 40.5 months

383. 2

385. 22

387. 2 children

389. 11 units

391. $\frac{1}{20}, 0.05$

393. $\frac{1}{24}, 0.041\overline{6} \approx 0.042$

395. $\frac{2}{5}, 0.4$

397. $\frac{5}{7}, 0.\overline{714285} \approx 0.714$

399. (a) $285.47 (b) $275.63 (c) $236.25

401. Answers will vary.

Section 5.6 Exercises

403. $\frac{5}{9}$

405. $\frac{7}{8}$

407. $\frac{7}{3}$

409. $\frac{7}{3}$

411. $\frac{1}{5}$

413. $\frac{10}{17}$

415. $\frac{5}{4}$

417. $\frac{2}{7}$

419. $\frac{11}{4}$

421. $\frac{1}{3}$

423. $\frac{6}{23}$

425. $\frac{82}{15}$

427. $\frac{35}{9}$

429. $\frac{9}{4}$

431. $\frac{35 \text{ calories}}{3 \text{ ounces}}$

433. $\frac{41 \text{ lbs}}{15 \text{ sq. in.}}$

435. $\frac{488 \text{ miles}}{7 \text{ hours}}$

437. $\frac{\$119}{8 \text{ hours}}$

439. 11.67 calories/ounce

441. 2.73 lbs./sq. in.

443. 69.71 mph

445. $14.88/hour

447. 32 mpg

449. 2.69 lbs./week

451. 92 beats/minute

453. 8,000

455. $1.09/bar

457. $1.33/pair

459. $0.48/pack

461. $0.60/disc

463. $1.29/box

465. The 50.7-ounce size costs $0.138 per ounce. The 33.8-ounce size costs $0.142 per ounce. The 50.7-ounce size is the better buy.

467. The 18-ounce size costs $0.222 per ounce. The 14-ounce size costs $0.235 per ounce. The 18-ounce size is a better buy.

469. The regular bottle costs $0.075 per ounce. The squeeze bottle costs $0.069 per ounce. The squeeze bottle is a better buy.

471. The half-pound block costs $6.78/lb, so the 1-lb. block is a better buy.

473. $\frac{793 \text{ miles}}{p \text{ hours}}$

475. $\frac{\$3}{0.5 \text{ lbs.}}$

477. $\frac{105 \text{ calories}}{x \text{ ounces}}$

479. $\frac{y}{5x}$

481. 15.2 students per teacher

483. (a) 72 calories/ounce (b) 3.87 grams of fat/ounce (c) 5.73 grams carbs/ounce (d) 3.33 grams protein/ounce

485. Answers will vary.

487. Kathryn should swim for approximately 16.35 minutes. Explanations will vary.

Section 5.7 Exercises

489. 6

491. 8

493. −2

495. −1

497. not a real number

499. not a real number

501. 5

503. 7

505. $8 < \sqrt{70} < 9$

507. $14 < \sqrt{200} < 15$

509. 4.36

511. 7.28

513. y

515. $7x$

517. $-8a$

519. $12xy$

521. 8.7 feet

523. 8 seconds

525. 15.8 seconds

527. 72 mph

529. 53.0 mph

531. 45 inches

533. Answers will vary. 9^2 reads: "nine squared" and means nine times itself. The expression $\sqrt{9}$ reads: "the square root of nine" which gives us the number such that if it were multiplied by itself would give you the number inside of the square root.

Review Exercises

535. three hundred seventy-five thousandths

537. five and twenty-four hundredths

539. negative four and nine hundredths

541. 0.09

543. 10.035

545. −0.05

547. $\frac{33}{40}$

549. $3\frac{16}{25}$

551. <

553. <

555. (a) 12.53 (b) 12.5 (c) 13

557. (a) 5.90 (b) 5.9 (c) 6

559. 24.67

561. 24.831

563. −2.37

565. −1.6

567. 15,400

569. 0.18

571. 4

573. 200

575. $28.22

577. $1.79

579. 0.875

581. −5.25

583. $0.\overline{54}$

585. >

587. >

589. >

591. $\frac{11}{15}, 0.75, \frac{7}{9}$

593. 6.03

595. 1.975

597. −0.22

599. (a) 21.98 ft.
(b) 38.465 sq.ft.

601. (a) 34.54 cm
(b) 94.985 sq.cm

603. (a) no (b) yes

605. (a) no (b) yes

607. $h = -3.51$

609. $p = 2.65$

611. $j = 3.72$

613. $x = -4$

615. $a = -7.2$

617. $s = 25$

619. $-5.9x = -3.54$; $x = 0.6$

621. $m + (-4.03) = 6.8$; $m = 10.83$

623. $269.10

625. $40.94

627. 24.5

629. 16 clients

631. 2

633. $\frac{1}{5}; 0.2$

635. $\frac{7}{4}$

637. $\frac{2}{3}$

639. $\frac{4}{9}$

641. $\frac{7}{9}$

643. $\frac{12 \text{ pounds}}{1 \text{ square inch}}$

645. $\frac{\$35}{2 \text{ hours}}$

647. 12 pounds/sq.in.

649. $17.50/hour

651. $0.42

653. $1.65

655. $0.11, $0.12; 60 tablets for $6.49

657. $\frac{a \text{ adults}}{45 \text{ children}}$

659. $\frac{19}{3+n}$

661. 12

663. −9

665. not a real number

667. 17

669. $12 < \sqrt{155} < 13$

671. 7.55

673. $8b$

675. $15mn$

677. $7y$

679. $11cd$

681. 5.5 feet

683. 72 mph

Practice Test

685. $1\frac{73}{100}$

687. (a) 16.7 (b) 16.75
(c) 17

689. 18.42

691. 0.192

693. −0.08

695. 2

697. 200

699. 1.975

701. −1.2

703. 8.8

705. $26.45

707. (a) 52.1 (b) 51.5 (c) 55

709. The unit prices are \$0.172 per ounce for 64 ounces, and \$0.177 per ounce for 48 ounces; 64 ounces is the better buy.

711. $12n$

713. 15 feet

Chapter 6

Be Prepared

6.1 $\frac{33}{5}$

6.2 0.6

6.3 $\frac{31}{50}$

6.4 32

6.5 10.71

6.6 5

6.7 63

6.8 324

6.9 75

6.10 6.67

6.11 $\frac{1}{12}$

6.12 80

6.13 $\frac{24 \text{ miles}}{2 \text{ hours}}$

Try It

6.1 $\frac{89}{100}$

6.2 $\frac{72}{100}$

6.3 $\frac{62}{100}, 62\%$

6.4 $\frac{41}{100}, 41\%$

6.5 (a) $\frac{12}{25}$ (b) $\frac{11}{10}$

6.6 (a) $\frac{16}{25}$ (b) $\frac{3}{2}$

6.7 (a) $\frac{161}{250}$ (b) $\frac{2}{3}$

6.8 (a) $\frac{17}{40}$ (b) $\frac{7}{80}$

6.9 (a) 0.09 (b) 0.87

6.10 (a) 0.03 (b) 0.91

6.11 (a) 1.15 (b) 0.235

6.12 (a) 1.23 (b) 0.168

6.13 (a) $\frac{6}{25}$ (b) 0.24

6.14 (a) $\frac{11}{25}$ (b) 0.44

6.15 (a) $\frac{3}{10}$ (b) 0.3

6.16 (a) $\frac{1}{8}$ (b) 0.125

6.17 (a) 1% (b) 17%

6.18 (a) 4% (b) 41%

6.19 (a) 175% (b) 8.25%

6.20 (a) 225% (b) 9.25%

6.21 (a) 62.5% (b) 275% (c) 340%

6.22 (a) 87.5% (b) 225% (c) 160%

6.23 42.9%

6.24 57.1%

6.25 $11.\overline{1}\%$, or $11\frac{1}{9}\%$

6.26 $16.\overline{6}\%$, or $16\frac{2}{3}\%$

6.27 36

6.28 33

6.29 117

6.30 126

6.31 68

6.32 64

6.33 \$26

6.34 \$36

6.35 75%

6.36 80%

6.37 125%

6.38 175%

6.39 \$14.67

6.40 \$2.16

6.41 24.1 grams

6.42 2,375 mg

6.43 26%

6.44 37%

6.45 8.8%

6.46 50%

6.47 6.3%

6.48 10%

6.49 ⓐ $45.25 ⓑ $769.25

6.50 ⓐ $20.50 ⓑ $270.50

6.51 9%

6.52 7.5%

6.53 $273

6.54 $394.20

6.55 4%

6.56 5.5%

6.57 $450

6.58 $82

6.59 ⓐ $11.60 ⓑ $17.40

6.60 ⓐ $256.75 ⓑ $138.25

6.61 ⓐ $184.80 ⓑ 33%

6.62 ⓐ $60 ⓑ 15%

6.63 ⓐ $600 ⓑ $1,800

6.64 ⓐ $2,975 ⓑ $11,475

6.65 $160

6.66 $56

6.67 $2,750

6.68 $3,560

6.69 4.5%

6.70 6.5%

6.71 $142.50

6.72 $7,020

6.73 6%

6.74 4%

6.75 $11,450

6.76 $9,600

6.77 $57.00

6.78 $143.50

6.79 ⓐ $\frac{5}{9} = \frac{20}{36}$
ⓑ $\frac{7}{11} = \frac{28}{44}$
ⓒ $\frac{2.50}{8} = \frac{3.75}{12}$

6.80 ⓐ $\frac{6}{7} = \frac{36}{42}$
ⓑ $\frac{8}{36} = \frac{12}{54}$
ⓒ $\frac{3.75}{6} = \frac{2.50}{4}$

6.81 ⓐ no ⓑ yes

6.82 ⓐ no ⓑ no

6.83 77

6.84 104

6.85 65

6.86 24

6.87 −7

6.88 −9

6.89 12 ml

6.90 180 mg

6.91 300

6.92 5 pieces

6.93 590 Euros

6.94 56,460 yen

6.95 $\frac{n}{105} = \frac{60}{100}$

6.96 $\frac{n}{85} = \frac{40}{100}$

6.97 $\frac{36}{n} = \frac{25}{100}$

6.98 $\frac{27}{n} = \frac{36}{100}$

6.99 $\frac{n}{100} = \frac{39}{52}$

6.100 $\frac{n}{100} = \frac{23}{92}$

6.101 $\frac{n}{40} = \frac{65}{100}$; $n = 26$

6.102 $\frac{n}{40} = \frac{85}{100}$; $n = 34$

6.103 $\frac{n}{64} = \frac{125}{100}$; $n = 80$

6.104 $\frac{n}{84} = \frac{175}{100}$; $n = 147$

6.105 $\frac{3.23}{n} = \frac{8.5}{100}$; $n = 38$

6.106 $\frac{4.64}{n} = \frac{7.25}{100}$; $n = 64$

6.107 $\frac{27}{72} = \frac{n}{100}$; $n = 37.5\%$

6.108 $\frac{23}{92} = \frac{n}{100}$; $n = 25\%$

Section 6.1 Exercises

1. $\frac{6}{100}$

3. $\frac{32}{1000}$

5. ⓐ $\frac{57}{100}$ ⓑ 57%

7. ⓐ $\frac{42}{100}$ ⓑ 42%

9. $\frac{1}{25}$

11. $\frac{17}{100}$

13. $\frac{13}{25}$

15. $\frac{5}{4}$

17. $\frac{3}{8}$

19. $\frac{23}{125}$

21. $\frac{19}{200}$

23. $\frac{4}{75}$

25. 0.05

27. 0.01

29. 0.63

31. 0.4

33. 1.15

35. 1.5

37. 0.214

39. 0.078

41. (a) $\frac{3}{200}$ (b) 0.015

43. (a) $\frac{7023}{10{,}000}$ (b) 0.7023

45. (a) $\frac{1}{4}$ (b) 0.25

47. (a) $\frac{3}{5}$ (b) 0.6

49. 1%

51. 18%

53. 135%

55. 300%

57. 0.9%

59. 8.75%

61. 150%

63. 225.4%

65. 25%

67. 37.5%

69. 175%

71. 680%

73. $41\frac{2}{3}\%$ *or* $41.\overline{6}\%$

75. $266.\overline{6}\%$

77. 42.9%

79. $55.\overline{5}\%$

81. 25%

83. 35%

85.

Fraction	Decimal	Percent
$\frac{1}{2}$	0.5	50%
$\frac{9}{20}$	0.45	45%
$\frac{9}{50}$	0.18	18%
$\frac{1}{3}$	0.3	33.3%
$\frac{1}{125}$	0.008	0.8%
2	2.0	200%

87. (a) $\frac{1}{10}$
(b) approximately \$9.50

89. $\frac{17}{20}$; 0.85

91. 80%; 0.8

93. $\frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1.$

95. The Szetos sold their home for five times what they paid 30 years ago.

Section 6.2 Exercises

97. 54

99. 26.88

101. 162.5

103. 18,000

105. 112

107. 108

109. \$35

111. \$940

113. 30%

115. 36%

117. 150%

119. 175%

121. \$11.88

123. \$259.80

125. 24.2 grams

127. 2,407 mg

129. 45%

131. 25%

133. 13.2%

135. 125%

137. 72.7%

139. 2.5%

141. 11%

143. 5.5%

145. 21.2%

147. The original number should be greater than 44.80% is less than 100%, so when 80% is converted to a decimal and multiplied to the base in the percent equation, the resulting amount of 44 is less. 44 is only the larger number in cases where the percent is greater than 100%.

149. Alex should have packed half as many shorts and twice as many shirts.

Section 6.3 Exercises

151. (a) $4.20 (b) $$88.20

153. (a) $9.68 (b) $138.68

155. (a) $17.13 (b) $267.13

157. (a) $61.45 (b) $1,260.45

159. 6.5%

161. 6.85%

163. $20.25

165. $975

167. $859.25

169. 3%

171. 16%

173. 15.5%

175. $139

176. $81

177. $125

179. (a) $26.97 (b) $17.98

181. (a) $128.37 (b) $260.63

183. (a) $332.48 (b) $617.47

185. (a) $576 (b) 30%

187. (a) $53.25 (b) 15%

189. (a) $370 (b) 43.5%

191. (a) $7.20 (b) $23.20

193. (a) $0.20 (b) $0.80

195. (a) $258.75 (b) $373.75

197. (a) $131.25 (b) $126.25 (c) 25% off first, then $20 off

199. (a) Priam is correct. The original price is 100%. Since the discount rate was 40%, the sale price was 60% of the original price.
(b) Yes.

Section 6.4 Exercises

201. $180

203. $14,000

205. 6.3%

207. $90

209. $579.96

211. $14,167

213. $3,280

215. $860

217. $24,679.91

219. 4%

221. 5.5%

223. $116

225. $4,836

227. 3%

229. 3.75%

231. $35,000

233. $3,345

235. $332.10

237. $195.00

239. Answers will vary.

241. Answers will vary.

Section 6.5 Exercises

243. $\frac{4}{15} = \frac{36}{135}$

245. $\frac{12}{5} = \frac{96}{40}$

247. $\frac{5}{7} = \frac{115}{161}$

249. $\frac{8}{1} = \frac{48}{6}$

251. $\frac{9.36}{18} = \frac{2.60}{5}$

253. $\frac{18.04}{11} = \frac{4.92}{3}$

255. yes

257. no

259. no

261. yes

263. $x = 49$

265. $z = 7$

267. $a = 9$

269. $p = -11$

271. $a = 7$

273. $c = 2$

275. $j = 0.6$

277. $m = 4$

279. 9 ml

281. 114 beats/minute. Carol has not met her target heart rate.

283. 159 cal

285. $\frac{3}{4}$ cup

287. $252.50

289. 0.8 Euros

291. 48 quarters

293. 19 gallons, $58.71

295. 12.8 hours

297. 4 bags

299. $\frac{n}{250} = \frac{35}{100}$

301. $\frac{n}{47} = \frac{110}{100}$

303. $\frac{45}{n} = \frac{30}{100}$

305. $\frac{90}{n} = \frac{150}{100}$

307. $\frac{17}{85} = \frac{p}{100}$

309. $\frac{340}{260} = \frac{p}{100}$

311. $\frac{n}{180} = \frac{65}{100}$; 117

313. $\frac{n}{92} = \frac{18}{100}$; 16.56

315. $\frac{n}{26} = \frac{175}{100}$; 45.5

317. $\frac{n}{488} = \frac{300}{100}$; 1464

319. $\frac{7.65}{n} = \frac{17}{100}$; 45

321. $\frac{13.53}{n} = \frac{8.25}{100}$; 164

323. $\frac{14}{56} = \frac{p}{100}$; 25%

325. $\frac{12}{96} = \frac{p}{100}$; 12.5%

327. He must add 20 oz of water to obtain a final solution of 32 oz.

329. Answers will vary.

Review Exercises

331. $\frac{32}{100}$

333. $\frac{13}{100}$, 13%

335. $\frac{12}{25}$

337. $\frac{641}{1000}$

339. 0.06

341. 1.28

343. ⓐ $\frac{27}{200}$ ⓑ 0.135

345. ⓐ $\frac{1}{2}$ ⓑ 0.5

347. 4%

349. 282%

351. 0.3%

353. 75%

355. 362.5%

357. 40%

359. 161

361. 240

363. 25

365. 68%

367. $16.70

369. 28.4%

371. 4%

373. ⓐ $45 ⓑ $795

375. 7.25%

377. $11,400

379. 15%

381. $45

383. (a) $25.47 (b) $59.43

385. (a) $13 (b) 26%

387. (a) $0.48 (b) $1.28

389. $450

391. $4400

393. $900

395. 2.5%

397. $\frac{3}{8} = \frac{12}{32}$

399. $\frac{1}{18} = \frac{23}{414}$

401. yes

403. no

405. 20

407. 4

409. 12 ml

411. 340 calories

413. 13 gallons

415. $\frac{n}{395} = \frac{62}{100}$

417. $\frac{15}{1000} = \frac{p}{100}$

419. $\frac{n}{900} = \frac{85}{100}, 765$

421. $\frac{3.51}{n} = \frac{4.5}{100}, \$78$

Practice Test

423. $0.24, \frac{6}{25}$

425. $3.5, \frac{7}{2}$

427. $33.\overline{3}\%$ *or* $33\frac{1}{3}\%$

429. 25%

431. 40

433. 11.9%

435. 4%

437. $70

439. 6.25%

441. −52

Chapter 7

Be Prepared

7.1 $\frac{319}{100}$

7.2 $0.\overline{45}$

7.3 12

7.4 $8y + 15$

7.5 12

7.6 -15

7.7 0.75

7.8 16

7.9 $12y + 15$

7.10 4

7.11 $\frac{2}{5}$

7.12 $\frac{27}{10}$

7.13 $4,290$

7.14 $\frac{5}{9}$

7.15 $\frac{5}{12}$

Try It

7.1 (a) $\frac{-24}{1}$ (b) $\frac{357}{100}$

7.2 (a) $\frac{-19}{1}$ (b) $\frac{841}{100}$

7.3 (a) rational (b) rational (c) irrational

7.4 (a) rational (b) rational (c) irrational

7.5 (a) rational (b) irrational

7.6 (a) irrational (b) rational

7.7

Number	Whole	Integer	Rational
-3		✓	✓
$-\sqrt{2}$			
0.3			✓
$\frac{9}{5}$			✓
4	✓	✓	✓
$\sqrt{49}$	✓	✓	✓

Number	Irrational	Real
-3		✓
$-\sqrt{2}$	✓	✓
0.3		✓
$\frac{9}{5}$		✓
4		✓
$\sqrt{49}$		✓

7.8

Number	Whole	Integer	Rational
$-\sqrt{25}$		✓	✓
$-\frac{3}{8}$			✓
-1		✓	✓
6	✓	✓	✓
$\sqrt{121}$	✓	✓	✓
2.041975...			

Number	Irrational	Real
$-\sqrt{25}$		✓
$-\frac{3}{8}$		✓
-1		✓
6		✓
$\sqrt{121}$		✓
2.041975...	✓	✓

7.9 (a) $-4 + 7 = 7 + (-4)$
(b) $6 \cdot 12 = 12 \cdot 6$

7.10 (a) $14 + (-2) = -2 + 14$
(b) $3(-5) = (-5)3$

7.11 (a) $(1 + 0.7) + 0.3 = 1 + (0.7 + 0.3)$
(b) $(-9 \cdot 8) \cdot \frac{3}{4} = -9\left(8 \cdot \frac{3}{4}\right)$

7.12 (a) $(4 + 0.6) + 0.4 = 4 + (0.6 + 0.4)$
(b) $(-2 \cdot 12) \cdot \frac{5}{6} = -2\left(12 \cdot \frac{5}{6}\right)$

7.13 $8(4x) = (8 \cdot 4)x = 32x$

7.14 $-9(7y) = (-9 \cdot 7)y = -63y$

7.15 (a) 0.84 (b) 0.84

7.16 (a) 0.975 (b) 0.975

7.17 (a) 24 (b) 24

7.18 (a) 15 (b) 15

7.19 $-48a$

7.20 $-92x$

7.21 $\frac{5}{49}$

7.22 $\frac{11}{25}$

7.23 $\frac{22}{15}$

7.24 $\frac{11}{9}$

7.25 $15.58c$

7.26 $17.79d$

7.27 10.53

7.28 70.4

7.29 $24y$

7.30 $60z$

7.31 $32r - s$

7.32 $41m + 6n$

7.33 $4x + 8$

7.34 $6x + 42$

7.35 $27y + 72$

7.36 $25w + 45$

7.37 $7x - 42$

7.38 $8x - 40$

7.39 $\frac{2}{5}p + 4$

7.40 $\frac{3}{7}u + 9$

7.41 $5y + 3$

7.42 $4n + 9$

7.43 $70 + 15p$

7.44 $4 + 35d$

7.45 $rs - 2r$

7.46 $yz - 8y$

7.47 $xp + 2p$

7.48 $yq + 4q$

7.49 $-18m - 15$

7.50 $-48n - 66$

7.51 $-10 + 15a$

7.52 $-56 + 105y$

7.53 $-z + 11$

7.54 $-x + 4$

7.55 $-3x + 3$

7.56 $2x - 20$

7.57 $5x - 66$

7.58 $7x - 13$

7.59 (a) 120 (b) 120

7.60 (a) 126 (b) 126

7.61 (a) 30 (b) 30

7.62 (a) 3 (b) 3

7.63 (a) -32 (b) -32

7.64 (a) −45 (b) −45

7.65 (a) identity property of addition (b) identity property of multiplication

7.66 (a) identity property of multiplication (b) identity property of addition

7.67 (a) -18 (b) $-\frac{7}{9}$ (c) -1.2

7.68 (a) -47 (b) $-\frac{7}{13}$ (c) -8.4

7.69 (a) $\frac{1}{5}$ (b) -7 (c) $\frac{10}{3}$

7.70 (a) $\frac{1}{18}$ (b) $-\frac{5}{4}$ (c) $\frac{5}{3}$

7.71 (a) 0 (b) 0 (c) 0

7.72 (a) 0 (b) 0 (c) 0

7.73 (a) 0 (b) 0 (c) 0

7.74 (a) 0 (b) 0 (c) 0

7.75 (a) undefined (b) undefined (c) undefined

7.76 (a) undefined (b) undefined (c) undefined

7.77 9

7.78 −18

7.79 p

7.80 r

7.81 0

7.82 0

7.83 undefined

7.84 undefined

7.85 $20y + 50$

7.86 $12z + 16$

7.87 2.5 feet

7.88 54 feet

7.89 8600 pounds

7.90 102,000,000 pounds

7.91 440,000,000 yards

7.92 151,200 minutes

7.93 16 cups

7.94 48 teaspoons

7.95 9 lbs. 8 oz

7.96 21 ft. 6 in.

7.97 4 lbs. 8 oz.

7.98 11 gal. 2 qts.

7.99 5000 m

7.100 250 cm

7.101 2.8 kilograms

7.102 4.5 kilograms

7.103 (a) 0.00725 kL (b) 6300 mL

7.104 (a) 35,000 L (b) 410 cL

7.105 83 cm

7.106 1.04 m

7.107 2 L

7.108 2.4 kg

7.109 2.12 quarts

7.110 3.8 liters

7.111 19,336 ft

7.112 3,470 mi

7.113 15°C

7.114 5°C

7.115 59°F

7.116 50°F

Section 7.1 Exercises

1. (a) $\frac{5}{1}$ (b) $\frac{319}{100}$

3. (a) $\frac{-12}{1}$ (b) $\frac{9279}{1000}$

5. Rational: $0.75, 0.22\overline{3}$.
Irrational: $1.39174\ldots$

7. Rational: $0.\overline{45}, 3.59$.
Irrational: $1.919293\ldots$

9. (a) rational (b) irrational

11. (a) irrational (b) rational

13.

Number	Whole	Integer	Rational
-8		✓	✓
0	✓	✓	✓
1.95286...			
$\frac{12}{5}$			✓
$\sqrt{36}$	✓	✓	✓
9	✓	✓	✓

Number	Irrational	Real
-8		✓
0		✓
1.95286...	✓	✓
$\frac{12}{5}$		✓
$\sqrt{36}$		✓
9		✓

15.

Number	Whole	Integer	Rational
$-\sqrt{100}$		✓	✓
-7		✓	✓
$-\frac{8}{3}$			✓
-1		✓	✓
0.77			✓
$3\frac{1}{4}$			✓

Number	Irrational	Real
$-\sqrt{100}$		✓
-7		✓
$-\frac{8}{3}$		✓
-1		✓
0.77		✓
$3\frac{1}{4}$		✓

17. (a) 4
(b) Teachers cannot be divided
(c) It would result in a lower number.

19. Answers will vary.

Section 7.2 Exercises

21. $7 + 6 = 6 + 7$

23. $7(-13) = (-13)7$

25. $(-12)(-18) = (-18)(-12)$

27. $-15 + 7 = 7 + (-15)$

29. $y + 1 = 1 + y$

31. $-3m = m(-3)$

33. $(21 + 14) + 9 = 21 + (14 + 9)$

35. $(14 \cdot 6) \cdot 9 = 14(6 \cdot 9)$

37. $(-2 + 6) + 7 = -2 + (6 + 7)$

39. $\left(13 \cdot \frac{2}{3}\right) \cdot 18 = 13\left(\frac{2}{3} \cdot 18\right)$

41. $4(7x) = (4 \cdot 7)x$

43. $(17 + y) + 33 = 17 + (y + 33)$

45. (a) 0.97 (b) 0.97

47. (a) 2.375 (b) 2.375

49. (a) 21 (b) 21

51. (a) 8 (b) 8

53. 23

55. $\frac{5}{12}$

57. $\frac{25}{7}$

59. $\frac{65}{23}$

61. -176

63. $\frac{13}{12}$

65. $\frac{23}{15}$

67. $9.89d$

69. 36

71. 29.193

73. $72w$

75. $-46n$

77. $12q$

79. $42u + 30v$

81. $-57p + (-10q)$

83. $a + \frac{6}{5}b$

85. $7.41m + 6.57n$

87. (a) \$975 (b) \$700
(c) \$1675 (d) \$185
(e) \$270 (f) \$1220

89. Answers will vary.

Section 7.3 Exercises

91. $3a + 27$

93. $27w + 63$

95. $7y - 91$

97. $35u - 20$

99. $\frac{1}{3}u + 3$

101. $\frac{4}{5}m + 4$

103. $3x - 4$

105. $2 + 9s$

107. $uv - 10u$

109. $ax + 7x$

111. $-3a - 33$

113. $-81a - 36$

115. $-4q + 28$

117. $-42x + 48$

119. $-q - 11$

121. $-5p + 4$

123. $8u + 4$

125. $-4x + 10$

127. $-42n + 39$

129. $-r + 15$

131. $-c + 6$

133. $3y + 1$

135. $47u + 60$

137. $24x + 4$

139. $6n - 72$

141. $17n + 76$

143. (a) 56 (b) 56

145. (a) $\frac{9}{2}$ (b) $\frac{9}{2}$

147. (a) −89 (b) −89

149. (a) −525 (b) −525

151. (a) 160 (b) 160

153. (a) −1.03 (b) −1.03

155. (a) $3(4 - 0.03) = 11.91$
(b) $1.42

157. Answers will vary.

Section 7.4 Exercises

159. identity property of multiplication

161. identity property of addition

163. $\frac{1}{14}$

165. $-\frac{1}{19}$

167. $\frac{13}{8}$

169. $-\frac{12}{5}$

171. $\frac{5}{2}$

173. −2

175. 0

177. 0

179. undefined

181. 0

183. 0

185. undefined

187. 16

189. $31s$

191. p

193. $2n$

195. 0

197. 0

199. 0

201. undefined

203. undefined

205. undefined

207. $20q - 35$

209. $225h + 360$

211. (a) 8 hours (b) 8
(c) associative property of multiplication

213. Answers will vary.

Section 7.5 Exercises

215. 24 inches

217. 3.75 feet

219. 58 inches

221. 90 feet

223. 0.98 miles

225. 300,000 pounds

227. 110 tons

229. 162 hours

231. 8100 seconds

233. 256 tablespoons

235. 128 ounces

237. 111 ounces

239. 31.25 gallons

241. 4 lbs. 1 oz.

243. 5 weeks and 1 day

245. 9 ft 2 in

247. 8 yards

249. 8000 meters

251. 245 centimeters

253. 10.911 kilometers

255. 3000 milligrams

257. 25,000 milligrams

259. 3800 grams

261. 0.3 liters

263. 65 centimeters

265. 1.82 kilograms

267. 16.8 grams

269. 42,000 milligrams

271. 106.7 centimeters

273. 8.2 meters

275. 40,900 kilograms

277. 5.2 feet

279. 33 pounds

281. 30.2 liters

283. 25°C

285. −10°C

287. −15.6°C

289. 48.9°C

291. 77°F

293. 5°F

295. 46.4°F

297. 60.8°F

299. 110 reflectors

Review Exercises

303. $\frac{-5}{1}$

305. $\frac{18}{10}$

307. $0.\overline{16}, 1.95$

309. (a) irrational (b) rational

311. (a) 4 (b) $-5, -\sqrt{4}, 4$ (c) $-5, -2\frac{1}{4}, -\sqrt{4}, 0.\overline{25}, \frac{13}{5}, 4$ (d) none (e) $-5, -2\frac{1}{4}, -\sqrt{4}, 0.\overline{25}, \frac{13}{5}, 4$

313. −14·5 = 5(−14)

315. $a + 8 = 8 + a$

317. (22 + 7) + 3 = 22 + (7 + 3)

319. $\frac{1}{2}(22y) = \left(\frac{1}{2} \cdot 22\right) y$

321. (a) 5.39 (b) 5.39

323. (a) 13 (b) 13

325. $\frac{11}{15}$

327. −60

329. $5.98\,d$

331. $25\,q$

333. $34\,m + (-25\,n)$

335. $9y - 36$

337. $56a + 96$

339. $yp + 10p$

341. $-4x + 68$

343. (a) 9 (b) 9

345. (a) 36 (b) 36

347. identity property of addition

349. identity property of multiplication

351. −19.4

353. $\frac{7}{15}$

355. $-\frac{1}{5}$

357. $-\frac{9}{4}$

359. 0

361. 0

363. $n + 7$

365. 34

367. $54x - 84$

369. 3.5 feet

371. 15 yards

373. 9000 pounds

375. 64 tablespoons

377. 1.9 gallons

379. 7 hours 10 minutes

381. 3 yards, 12 inches

383. 8.85 kilometers

385. 13,000 milligrams

387. 0.65 liters

389. 855 milliliters

391. 10,000 milligrams

393. 25.6 meters

395. 171.6 pounds

397. 11.4 kilograms

399. −5°C

401. 17.8°C

403. 23°F

405. 75.2°F

Practice Test

407. $\sqrt{144} = 12$ therefore rational.

409. $x \cdot 14 = 14 \cdot x$

411. $(8 \cdot 2) \cdot 5 = 8 \cdot (2 \cdot 5)$

413. ⓐ $-\frac{2}{5}$ ⓑ $\frac{5}{2}$

415. $15y$

417. $\frac{23}{15}$

419. $30y - 4z$

421. $30x - 24$

423. $2a + 3$

425. $66p - 2$

427. 0

429. 0

431. .276 grams

433. 9.317 miles

435. 95°*F*

Chapter 8

Be Prepared

8.1 28

8.2 $x - 5$

8.3 yes

8.4 1

8.5 $-\frac{8}{3}$

8.6 −25

8.7 $4y$

8.8 4

8.9 −21

8.10 3

8.11 12

8.12 478

Try It

8.1 no

8.2 no

8.3 $x = -16$

8.4 $x = -20$

8.5 $n = -1$

8.6 $x = -4$

8.7 $p = \frac{7}{6}$

8.8 $q = \frac{2}{3}$

8.9 $b = 6.4$

8.10 $c = 14$

8.11 $y = 15$

8.12 $z = 2$

8.13 $p = 5$

8.14 $q = -16$

8.15 $h = -1$

8.16 $x = 1$

8.17 $x + 11 = 41$; $x = 30$

8.18 $y - 12 = 51$; $y = 63$

8.19 $4x - 3x = 14$; $x = 14$

8.20 $7a - 6a = -8$; $a = -8$

8.21 $a + 6 = 13$; Athena weighs 7 pounds.

8.22 $26 + h = 68$; Henry has 42 books.

8.23 $19{,}875 = s - 1025$; the sticker price is $20,900.

8.24 $7.75 = n - 3.25$; the price at night is $11.00.

8.25 $y = -16$

8.26 $z = -13$

8.27 $b = 144$

8.28 $c = 128$

8.29 $k = -8$

8.30 $g = -3$

8.31 $n = 35$

8.32 $y = 18$

8.33 $x = 2$

8.34 $n = -5$

8.35 $c = -3$

8.36 $x = -\frac{4}{7}$

8.37 $n = -6$

8.38 $n = -5$

8.39 $x = -4$

8.40 $a = -8$

8.41 $y = 5$

8.42 $m = 9$

8.43 $n = 10$

8.44 $c = 1$

8.45 $p = -7$

8.46 $m = -3$

8.47 $j = 2$

8.48 $h = 1$

8.49 $x = -1$

8.50 $y = 4$

8.51 $q = 1$

8.52 $n = 1$

8.53 $a = -5$

8.54 $k = -6$

8.55 $x = 10$

8.56 $y = -3$

8.57 $x = -5$

8.58 $y = -5$

8.59 $x = 4$

8.60 $y = 1$

8.61 $y = -6$

8.62 $z = 8$

8.63 $a = 2$

8.64 $n = 2$

8.65 $j = \frac{5}{3}$

8.66 $k = \frac{5}{2}$

8.67 $p = -2$

8.68 $q = -8$

8.69 $u = 2$

8.70 $x = 4$

8.71 $n = 1$

8.72 $m = -1$

8.73 $x = \frac{1}{2}$

8.74 $y = 3$

8.75 $v = 40$

8.76 $u = -12$

8.77 $a = -2$

8.78 $c = -2$

8.79 $p = -4$

8.80 $q = 2$

8.81 $n = 2$

8.82 $m = -1$

8.83 $x = 20$

8.84 $x = 10$

8.85 $h = 12$

8.86 $k = -1$

8.87 $n = 9$

8.88 $d = 16$

Section 8.1 Exercises

1. yes

3. no

5. $x = 5$

7. $b = \frac{1}{2}$

9. $p = -11.7$

11. $a = 10$

13. $x = \frac{7}{3}$

15. $y = 13.8$

17. $x = -27$

19. $q = -\frac{1}{4}$

21. $y = \frac{27}{20}$

23. $m = 17$

25. $x = 8$

27. $n = -20$

29. $y = 2$

31. $w = -1.7$

33. $x = -2$

35. $m = -4$

37. $k = 6$

39. $c = -41$

41. $y = 28$

43. $x + (-5) = 33$; $x = 38$

45. $y - 3 = -19$; $y = -16$

47. $p + 8 = 52$; $p = 44$

49. $5c - 4c = 60$; $c = 60$

51. $f - \frac{1}{3} = \frac{1}{12}$; $f = \frac{5}{12}$

53. $-9m + 10m = -25$; $m = -25$

55. Let p equal the number of pages read in the Psychology book. $41 + p = 54$. Jeff read 13 pages in his Psychology book.

57. Let d equal the daughter's age. d = 12 − 5. Eva's daughter's age is 7 years old.

59. 21 pounds

61. 100.5 degrees

63. $121.19

65. $d = \frac{13}{24}$

67. Answers will vary.

Section 8.2 Exercises

69. $p = 9$

71. $x = 3$

73. $y = -6$

75. $m = 7$

77. $a = 15$

79. $x = 0$

81. $z = 28$

83. $c = 36$

85. $q = -48$

87. $p = 80$

89. $r = 25$

91. $x = -32$

93. $q = \frac{5}{2}$

95. $y = -1$

97. $m = -5$

99. $t = \frac{5}{2}$

101. $q = 24$

103. $p = 56$

105. 6 children

107. $1.08

109. 42 yards

111. Answer will vary.

Section 8.3 Exercises

113. $x = 6$

115. $y = 6$

117. $m = -8$

119. $b = -8$

121. $x = -4$

123. $q = -2$

125. $k = -11$

127. $x = 9$

129. $b = -3$

131. $z = 3$

133. $x = -\frac{3}{4}$

135. $r = -2$

137. $x = 19$

139. $f = 7$

141. $q = -5$

143. $c = -4$

145. $x = 2$

147. $y = 4$

149. $m = -6$

151. $a = 7$

153. $a = -40$

155. $p = 15$

157. $z = 3.46$

159. $w = 60$

161. $x = 23$

163. $y = 9$

165. $x = 6$

167. $y = 3$

169. $n = -2$

171. $p = -1$

173. $x = 5$

175. $v = 1$

177. $m = 0.25$

179. $t = -9$

181. $b = 2$

183. $m = 6$

185. $k = \frac{3}{2}$

187. $r = 3$

189. $y = -4$

191. $n = 2$

193. $x = 34$

195. $s = 10$

197. $x = \frac{1}{2}$

199. 30 feet

201. 8 nickels

203. Answers will vary.

205. Answers will vary.

207. Answers will vary.

Section 8.4 Exercises

209. $x = -1$

211. $y = -1$

213. $a = \frac{3}{4}$

215. $x = 4$

217. $m = 20$

219. $x = -3$

221. $w = \frac{9}{4}$

223. $x = 1$

225. $b = 12$

227. $x = 1$

229. $p = -41$

231. $x = -\frac{5}{2}$

233. $y = 10$

235. $j = 2$

237. $x = 18$

239. $x = 18$

241. $x = 20$

243. $n = 9$

245. $d = 8$

247. $q = 11$

249. $d = 18$

251. Answers will vary.

253. Answers will vary.

Review Exercises

255. yes

257. no

259. 12

261. $a = \frac{4}{3}$

263. $u = 17$

265. $c = \frac{12}{11}$

267. $n = 44$

269. $f = \frac{10}{3}$

271. $y = 4$

273. $n = -8$

275. $-6 + m = 25$; $m = 31$

277. $s = 11 - 3$; 8 years old

279. $c - 46.25 = 9.75$; \$56.00

281. $x = 9$

283. $p = 21$

285. $n = 108$

287. $x = 48$

289. $m = 4$

291. $x = 15$

293. $r = 3$

295. $p = 5$

297. $x = -22$

299. $y = -13$

301. $k = -5$

303. $x = 6$

305. $u = -7$

307. $x = -2$

309. $s = -22$

311. $y = 12$

313. $r = 38$

315. $y = 26$

317. $n = 2$

319. $a = \frac{14}{3}$

321. $x = 5$

323. $p = -20$

Practice Test

325. (a) yes (b) no

327. $c = 16$

329. $x = -5$

331. $x = 9$

333. $y = 4$

335. $m = 6$

337. $d = -32$

339. $x = -2$

341. $p = \frac{2}{3}$

343. $2x - 4 = 16$; $x = 10$

Chapter 9

Be Prepared

9.1 $2x - 6$

9.2 36

9.3 2

9.4 3.5

9.5 $20 + 5n$

9.6 6

9.7 $x = 2$

9.8 60

9.9 10

9.10 $w - 3$

9.11 $3h$

9.12 6

9.13 25

9.14 (a) 50.24in.; (b) 200.96sq. in.

9.15 0.196

9.16 125

9.17 32

9.18 $\frac{77}{2}$

9.19 $\frac{35 \text{ miles}}{1 \text{ gallon}}$

9.20 12

9.21 \$200

Try It

9.1 \$180

9.2 40

9.3 2

9.4 7

9.5 \$950

9.6 \$4,200

9.7 25

9.8 4

9.9 3

9.10 6

9.11 9, 15

9.12 27, 31

9.13 −8, −15

9.14 −29, 11

9.15 −4, 0

9.16 −2, −3

9.17 47, 48

9.18 −15, −16

9.19 31, 32, 33

9.20 −11, −12, −13

9.21 9 nickels, 16 dimes

9.22 17 nickels, 5 quarters

9.23 42 nickels, 21 dimes

9.24 51 dimes, 17 quarters

9.25 41 nickels, 18 quarters

9.26 22 nickels, 59 dimes

9.27 330 day passes, 367 tournament passes

9.28 112 adult tickets, 199 senior/child tickets

9.29 32 at 49 cents, 12 at 8 cents

9.30 26 at 49 cents, 10 at 21 cents

9.31 (a) 155° (b) 65°

9.32 (a) 103° (b) 13°

9.33 40°, 140°

9.34 25°, 65°

9.35 21°

9.36 56°

9.37 34°

9.38 45°

9.39 20°, 70°, 90°

9.40 30°, 60°, 90°

9.41 8

9.42 22.5

9.43 10

9.44 17

9.45 8

9.46 12

9.47 12 feet

9.48 8 feet

9.49 (a) cubic (b) linear
(c) square (d) linear
(e) square (f) cubic

9.50 (a) cubic (b) square
(c) cubic (d) linear
(e) square (f) linear

9.51 (a) 8 inches
(b) 3 sq. inches

9.52 (a) 8 centimeters
(b) 4 sq. centimeters

9.53 (a) 340 yd
(b) 6000 sq. yd

9.54 (a) 220 ft (b) 2976 sq. ft

9.55 15 in.

9.56 9 yd

9.57 18 m, 11 m

9.58 11 ft , 19 ft

9.59 8 ft, 24 ft

9.60 5 cm, 4 cm

9.61 26 ft

9.62 29 m

9.63 30 ft, 70 ft

9.64 60 yd, 90 yd

9.65 13 sq. in.

9.66 49 sq. in.

9.67 8 ft

9.68 6 ft

9.69 14 in.

9.70 6 ft

9.71 13 in.

9.72 17 cm

9.73 14 ft

9.74 7 m

9.75 161 sq. yd

9.76 225 sq. cm

9.77 42 sq. cm

9.78 63 sq. m

9.79 40.25 sq. yd

9.80 240 sq. ft

9.81 (a) 31.4 in.
(b) 78.5 sq. in.

9.82 (a) 28.26 ft
(b) 63.585 sq. ft

9.83 17.27 ft

9.84 37.68 ft

9.85 30 cm

9.86 110 ft

9.87 28 sq. units

9.88 110 sq. units

9.89 36.5 sq. units

9.90 70 sq. units

9.91 103.2 sq. units

9.92 38.24 sq. units

9.93 (a) 792 cu. ft
(b) 518 sq. ft

9.94 (a) 1,440 cu. ft
(b) 792 sq. ft

9.95 (a) 216 cu. ft
(b) 228 sq. ft

9.96 (a) 2,772 cu. in.
(b) 1,264 sq. in.

9.97 (a) 91.125 cu. m
(b) 121.5 sq. m

9.98 (a) 389.017 cu. yd.
(b) 319.74 sq. yd.

9.99 (a) 64 cu. ft (b) 96 sq. ft

9.100 (a) 4,096 cu. in. (b) 1536 sq. in.

9.101 (a) 113.04 cu. cm (b) 113.04 sq. cm

9.102 (a) 4.19 cu. ft (b) 12.56 sq. ft

9.103 (a) 3052.08 cu. in. (b) 1017.36 sq. in.

9.104 (a) 14.13 cu. ft (b) 28.26 sq. ft

9.105 (a) 351.68 cu. cm (b) 276.32 sq. cm

9.106 (a) 100.48 cu. ft (b) 125.6 sq. ft

9.107 (a) 3,818.24 cu. cm (b) 1,356.48 sq. cm

9.108 (a) 91.5624 cu. ft (b) 113.6052 sq. ft

9.109 65.94 cu. in.

9.110 235.5 cu. cm

9.111 678.24 cu. in.

9.112 128.2 cu. in.

9.113 330 mi

9.114 7 mi

9.115 11 hours

9.116 56 mph

9.117 (a) $r = 45$ (b) $r = \frac{d}{t}$

9.118 (a) $r = 65$ (b) $r = \frac{d}{t}$

9.119 (a) $h = 20$ (b) $h = \frac{2A}{b}$

9.120 (a) $b = 4$ (b) $b = \frac{2A}{h}$

9.121 (a) $t = 3$ years (b) $t = \frac{I}{Pr}$

9.122 (a) $r = 0.12 = 12\%$ (b) $r = \frac{I}{Pt}$

9.123 (a) $y = 1$ (b) $y = \frac{10-3x}{4}$

9.124 (a) $y = -1$ (b) $y = \frac{18-5x}{2}$

9.125 $b = P - a - c$

9.126 $c = P - a - b$

9.127 $y = 11 - 7x$

9.128 $y = 8 - 11x$

9.129 $y = \frac{9-4x}{7}$

9.130 $y = \frac{1-5x}{8}$

Section 9.1 Exercises

1. There are 30 children in the class.

3. Zachary has 125 CDs.

5. There are 6 boys in the club.

7. There are 17 glasses.

9. Lisa's original weight was 175 pounds.

11. 18%

13. The original price was $120.

15. 4

17. 15

19. 5

21. 12

23. −5

25. 18, 24

27. 8, 12

29. −2, −3

31. 4, 10

33. 32, 46

35. 38, 39

37. −11, −12

39. 25, 26, 27

41. −11, −12, −13

43. The original price was $45.

45. Each sticker book cost $1.25.

47. The price of the refrigerator before tax was $1,080.

49. Answers will vary.

Section 9.2 Exercises

51. 8 nickels, 22 dimes

53. 15 dimes, 8 quarters

55. 12 dimes and 27 nickels

57. 63 dimes, 20 quarters

59. 10 of the $1 bills, 7 of the $5 bills

61. 10 of the $10 bills, 5 of the $5 bills

63. 16 nickels, 12 dimes, 7 quarters

65. 30 child tickets, 50 adult tickets

67. 110 child tickets, 50 adult tickets

69. 40 postcards, 100 stamps

71. 30 at 49 cents, 10 at 21 cents

73. 15 at $10 shares, 5 at $12 shares

75. 9 girls, 3 adults

77. Answers will vary.

79. Answers will vary.

Section 9.3 Exercises

81. (a) 127° (b) 37°

83. (a) 151° (b) 61°

85. 45°

87. 62.5°

89. 62°, 118°

91. 62°, 28°

93. 56°

95. 44°

97. 57°

99. 67.5°

101. 45°, 45°, 90°

103. 30°, 60°, 90°

105. 12

107. 351 miles

109. 15

111. 25

113. 8

115. 12

117. 10.2

119. 16.2

121. 5 feet

123. 14.1 feet

125. 2.9 feet

127. Answers will vary.

Section 9.4 Exercises

129. cubic

131. square

133. linear

135. (a) 10 cm (b) 4 sq. cm

137. (a) 8 cm (b) 3 sq. cm

139. (a) 10 cm (b) 5 sq. cm

141. (a) 260 ft (b) 3825 sq. ft

143. (a) 58 ft (b) 210 sq. ft

145. 24 inches

147. 27 meters

149. 23 m

151. 7 in., 16 in.

153. 17 m, 12 m

155. 13.5 m, 12.8 m

157. 25 ft, 50 ft

159. $l = 11$ m, $w = 7$ m

161. 26 in.

163. 55 m

165. 35 ft, 45 ft

167. 76 in., 36 in.

169. 30 sq. in.

171. 25.315 sq. m

173. 0.75 sq. ft

175. 8 ft

177. 23 in.

179. 11 yd

181. 28 cm

183. 17 yd

185. 6 m

187. 15 ft

189. 24 in.

191. 27.5 in.

193. 12 ft, 13 ft, 14 ft

195. 3 ft, 6 ft, 8 ft

197. 144 sq. ft

199. 2805 sq. m

201. 231 sq. cm

203. 28.56 sq. m

205. 13.5 sq. ft

207. 1036 sq. in.

209. 15 ft

211. $240

213. Answers will vary.

215. Answers will vary.

Section 9.5 Exercises

217. (a) 43.96 in. (b) 153.86 sq. in.

219. (a) 53.38 ft (b) 226.865 sq. ft

221. 62.8 ft

223. 37.68 in.

225. 6.908 ft

227. 52 in.

229. 5.5 m

231. 24 ft

233. 6.5 mi

235. 16 sq. units

237. 30 sq. units

239. 57.5 sq. units

241. 12 sq. units

243. 67.5 sq. units

245. 89 sq. units

247. 44.81 sq. units

249. 41.12 sq. units

251. 35.13 sq. units

253. 95.625 sq. units

255. 187,500 sq. ft

257. 9400 sq. ft

259. (a) 6.5325 sq. ft (b) 10.065 sq. ft

261. Answers will vary.

Section 9.6 Exercises

263. (a) 9 cu. m (b) 27 sq. m

265. (a) 17.64 cu. yd. (b) 41.58 sq. yd.

267. (a) 1,024 cu. ft (b) 640 sq. ft

269. (a) 3,350.49 cu. cm (b) 1,622.42 sq. cm

271. (a) 125 cu. cm (b) 150 sq. cm

273. (a) 1124.864 cu. ft. (b) 648.96 sq. ft

275. (a) 262,144 cu. ft (b) 24,576 sq. ft

277. (a) 21.952 cu. m (b) 47.04 sq. m

279. (a) 113.04 cu. cm (b) 113.04 sq. cm

281. (a) 1,766.25 cu. ft (b) 706.5 sq. ft

283. (a) 14,130 cu. in. (b) 2,826 sq. in.

285. (a) 381.51 cu. cm (b) 254.34 sq. cm

287. (a) 254.34 cu. ft (b) 226.08 sq. ft

289. (a) 29.673 cu. m (b) 53.694 sq. m

291. (a) 1,020.5 cu. cm (b) 565.2 sq. cm

293. (a) 678.24 cu. in. (b) 508.68 sq. in.

295. 37.68 cu. ft

297. 324.47 cu. cm

299. 261.67 cu. ft

301. 64,108.33 cu. ft

303. (a) 31.4 cu. ft (b) 2.6 cu. ft (c) 28.8 cu. ft

305. Answers will vary.

Section 9.7 Exercises

307. 612 mi

309. 7 mi

311. 6.5 hours

313. 3.6 hours

315. 60 mph

317. 80 mph

319. (a) $t = 5$ (b) $t = \frac{d}{r}$

321. (a) $t = 8.5$ (b) $t = \frac{d}{r}$

323. (a) $r = 68$ (b) $r = \frac{d}{t}$

325. (a) $r = 64$ (b) $r = \frac{d}{t}$

327. (a) $b = 14$ (b) $b = \frac{2A}{h}$

329. (a) $h = 30$ (b) $h = \frac{2A}{b}$

331. (a) $P = \$19{,}571.43$
(b) $P = \frac{I}{rt}$

333. (a) $t = 6$ years
(b) $t = \frac{I}{Pr}$

335. (a) $y = 2$
(b) $y = \frac{12-2x}{3}$

337. (a) $y = 13$ (b) $y = 7 - 3x$

339. $b = 90 - a$

341. $a = 180 - b - c$

343. $y = 15 - 8x$

345. $y = -6 + 4x$

347. $y = \frac{7-4x}{3}$

349. $y = 4 + x$

351. $L = \frac{P-2W}{2}$

353. $d = \frac{C}{\pi}$

355. $L = \frac{V}{WH}$

357. 104° F

359. Answers will vary

Review Exercises

361. Answers will vary.

363. There are 116 people at the concert.

365. His original weight was 180 pounds.

367. 38

369. 18, 9

371. 16 dimes, 11 quarters

373. 6 of $5 bills, 11 of $10 bills

375. 35 adults, 82 children

377. 3 of 26 -cent stamps, 8 of 41 -cent stamps

379. 132°

381. 33°, 57°

383. 73°

385. 30°, 60°, 90°

387. 15

389. 26

391. 8

393. 8.1

395. 6 feet

397. cubic

399. square

401. (a) 8 units (b) 3 sq. units

403. (a) 140 m
(b) 1176 sq. m

405. (a) 98 ft. (b) 180 sq. ft.

407. 25 cm

409. 62 m

411. 24.5 cm., 12.5 cm.

413. 135 sq. in.

415. 600 sq. in.

417. 7 in., 7 in.

419. 17 ft., 20 ft., 22 ft.

421. 100 sq. ft.

423. 675 sq. m

425. (a) 18.84 m
(b) 28.26 sq. m

427. 48 in.

429. 30 sq. units

431. 300 sq. units

433. 199.25 sq. units

435. (a) 630 cu. cm
(b) 496 sq. cm

437. (a) 15.625 cu. in.
(b) 37.5 sq. in.

439. (a) 267.95 cu. yd. (b) 200.96 sq. yd.

441. (a) 12.76 cu. in. (b) 26.41 sq. in.

443. (a) 75.36 cu. yd. (b) 100.48 sq. yd.

445. (a) 753.6 cu. cm (b) 477.28 sq. cm

447. 5.233 cu. m

449. 4.599 cu. in.

451. 1520 miles

453. 1.6 hours

455. (a) $t = 6.2$ (b) $t = \frac{d}{r}$

457. (a) $b = 26$ (b) $b = \frac{2A}{h}$

459. (a) $P = \$6000$ (b) $P = \frac{I}{(r \cdot t)}$

461. (a) $y = 4$ (b) $y = \frac{20-6x}{5}$

463. $a = 90 - b$

465. $y = 17 - 4x$

467. $W = \frac{P-2L}{2}$

Practice Test

471. −16

473. 7 quarters, 12 dimes

475. 38°

477. 48.3

479. 10

481. 127.3 ft

483. 2200 square centimeters

485. 282.6 inches

487. 1440

489. 31,400 cubic inches

491. 14.7 miles per hour

493. (a) height $= 52$ (b) $h = \frac{2A}{b}$

Chapter 10

Be Prepared

10.1 $11x$

10.2 $3n + 9$

10.3 100

10.4 $\frac{9}{16}$

10.5 -8

10.6 $2x + 6$

10.7 $-44 + 33a$

10.8 $x^2 + 16x + 63$

10.9 $\frac{1}{3}$

10.10 $32m^{15}$

10.11 $\frac{x}{y}$

10.12 ten thousand

10.13 twelve ten-thousandths

10.14 8

10.15 $2 \cdot 2 \cdot 2 \cdot 7$

10.16 $-18a - 33$

10.17 $4x^4 + 12x^3 - 4x^2$

Try It

10.1 (a) monomial (b) polynomial (c) trinomial (d) binomial (e) monomial

10.2 (a) binomial (b) trinomial (c) polynomial (d) monomial (c) monomial

10.3 (a) 1 (b) 1 (c) 4 (d) 2 (e) 0

10.4 (a) 0 (b) 2 (c) 4 (d) 5 (e) 3

10.5 $17x^2$

10.6 $-3y^2$

10.7 $14n$

10.8 $-2a^3$

10.9 $-2x^2 + 3y^2$

10.10 $-2a^2 + b^2$

10.11 $4x^2 - 8x + 10$

10.12 $11y^2 + 9y - 5$

10.13 $3y^2 + 3y + 3$

10.14 $3u^2 - u + 2$

10.15 $4n^2 + 12n$

10.16 $5a^2 + 8a + 8$

10.17 (a) 13 (b) 3

10.18 (a) 114 (b) −2

10.19 4 feet

10.20 20 feet

10.21 (a) 64 (b) 11

10.22 (a) 81 (a) 21

10.23 (a) $\frac{25}{64}$ (b) 0.4489

10.24 (a) $\frac{8}{125}$ (b) 0.016129

10.25 (a) 16 (b) −16

10.26 (a) 64 (b) −64

10.27 x^{15}

10.28 x^{16}

10.29 p^{10}

10.30 m^8

10.31 6^{10}

10.32 9^{15}

10.33 y^{43}

10.34 z^{39}

10.35 x^{21}

10.36 y^{15}

10.37 (a) x^{28} (b) 7^{32}

10.38 (a) x^{54} (b) 8^{42}

10.39 $196x^2$

10.40 $144a^2$

10.41 $256x^4y^4$

10.42 $216x^3y^3$

10.43 x^{40}

10.44 y^{42}

10.45 $-512x^{12}y^{21}$

10.46 $81a^{20}b^{24}$

10.47 $98n^{14}$

10.48 $48m^5$

10.49 $64u^{18}v^{22}$

10.50 $675x^7y^{18}$

10.51 $-56x^{11}$

10.52 $54y^9$

10.53 $12m^5n^6$

10.54 $12p^{11}q^8$

10.55 $6x + 48$

10.56 $2y + 24$

10.57 $y^2 - 9y$

10.58 $p^2 - 13p$

10.59 $8x^2 + 24xy$

10.60 $18r^2 + 3rs$

10.61 $-32y^3 - 20y^2 + 36y$

10.62 $-54x^3 - 6x^2 + 6x$

10.63 $12x^4 - 9x^3 + 27x^2$

10.64 $24y^4 - 16y^3 - 32y^2$

10.65 $xp + 8p$

10.66 $ap + 4p$

10.67 $x^2 + 17x + 72$

10.68 $a^2 + 9a + 20$

10.69 $20x^2 + 51x + 27$

10.70 $80m^2 + 142m + 63$

10.71 $56y^2 - 13y - 3$

10.72 $15x^2 - 14x - 16$

10.73 $x^2 - xy + 5x - 5y$

10.74 $x^2 - x + 2xy - 2y$

10.75 $x^2 + 15x + 56$

10.76 $y^2 + 16y + 28$

10.77 $y^2 + 5y - 24$

10.78 $q^2 + q - 20$

10.79 $20a^2 + 37a - 18$

10.80 $49x^2 - 28x - 32$

10.81 $12x^2 - 60x - xy + 5y$

10.82 $12a^2 - 54a - 2ab + 9b$

10.83 $12m^2 - 55m + 63$

10.84 $42n^2 - 47n + 10$

10.85 $y^3 - 8y^2 + 9y - 2$

10.86 $3x^3 + 2x^2 - 3x + 10$

10.87 $y^3 - 8y^2 + 9y - 2$

10.88 $3x^3 + 2x^2 - 3x + 10$

10.89 (a) x^3 (b) 7^9

10.90 (a) y^6 (b) 8^8

10.91 (a) $\frac{1}{x^7}$ (b) $\frac{1}{12^{10}}$

10.92 (a) $\frac{1}{m^9}$ (b) $\frac{1}{7^6}$

10.93 (a) b^8 (b) $\frac{1}{z^6}$

10.94 (a) $\frac{1}{p^8}$ (b) w^4

10.95 (a) 1 (b) 1

10.96 (a) 1 (b) 1

10.97 1

10.98 1

10.99 (a) 1 (b) $7x^2$

10.100 (a) $-23x^2$ (b) 1

10.101 (a) $\frac{49}{81}$ (b) $\frac{y^3}{512}$ (c) $\frac{p^6}{q^6}$

10.102 (a) $\frac{1}{64}$ (b) $-\frac{125}{m^3}$ (c) $\frac{r^4}{s^4}$

10.103 a^{11}

10.104 b^{19}

10.105 k^2

10.106 $\frac{1}{d}$

10.107 f^{12}

10.108 $\frac{1}{b^{10}}$

10.109 $\frac{m^{15}}{n^{40}}$

10.110 $\frac{t^{20}}{u^{14}}$

10.111 $\frac{25b^2}{81c^6}$

10.112 $\frac{64p^{12}}{343q^{15}}$

10.113 $\frac{1}{y^{11}}$

10.114 $9x^5$

10.115 $7x^4$

10.116 $16y^3$

10.117 $-\frac{12y}{x^2}$

10.118 $\frac{9}{a^5}$

10.119 $\frac{2a^6}{3b^2}$

10.120 $-\frac{3q^6}{5p^8}$

10.121 $\frac{4y^2}{7x^4}$

10.122 $\frac{5}{8m^5n^3}$

10.123 $2xy^2$

10.124 $-4ab^5$

10.125 (a) $\frac{1}{8}$ (b) $\frac{1}{100}$

10.126 (a) $\frac{1}{9}$ (b) $\frac{1}{10{,}000}$

10.127 (a) $\frac{1}{25}$ (b) $-\frac{1}{25}$

10.128 (a) $\frac{1}{4}$ (b) $-\frac{1}{4}$

10.129 (a) 2 (b) $\frac{1}{18}$

10.130 (a) 2 (b) $\frac{1}{256}$

10.131 $\frac{1}{y^7}$

10.132 $\frac{1}{z^8}$

10.133 (a) $\frac{8}{p}$ (b) $\frac{1}{8p}$ (c) $-\frac{1}{8p}$

10.134 (a) $\frac{11}{q}$ (b) $\frac{1}{11q}$ (c) $-\frac{1}{11q}$

10.135 (a) x^4 (b) $\frac{1}{y^5}$ (c) $\frac{1}{z^9}$

10.136 (a) a^5 (b) $\frac{1}{b^4}$ (c) $\frac{1}{c^{15}}$

10.137 $\frac{1}{p^3q^3}$

10.138 $\frac{1}{r^2s^8}$

10.139 $-\frac{12v^5}{u}$

10.140 $\frac{30d^3}{c^8}$

10.141 $\frac{1}{x^4}$

10.142 $\frac{1}{y^4}$

10.143 $\frac{64}{a^8}$

10.144 $\frac{8}{c^{12}}$

10.145 x^{11}

10.146 y^{13}

10.147 9.6×10^4

10.148 4.83×10^4

10.149 7.8×10^{-3}

10.150 1.29×10^{-2}

10.151 1,300

10.152 92,500

10.153 0.00012

10.154 0.075

10.155 0.06

10.156 0.009

10.157 400,000

10.158 20,000

10.159 18

10.160 16

10.161 7

10.162 11

10.163 $8x^2$

10.164 $9y^3$

10.165 $3x$

10.166 $5m^2$

10.167 $4(x + 3)$

10.168 $6(a + 4)$

10.169 $9(a + 1)$

10.170 $11(x + 1)$

10.171 $11(x - 4)$

10.172 $13(y - 4)$

10.173 $4(y^2 + 2y + 3)$

10.174 $6(x^2 + 7x - 2)$

10.175 $x(9x + 7)$

10.176 $a(5a - 12)$

10.177 $2x^2(x + 6)$

10.178 $3y^2(2y - 5)$

10.179 $9y(2y + 7)$

10.180 $8k(4k + 7)$

10.181 $6y(3y^2 - y - 4)$

10.182 $4x(4x^2 + 2x - 3)$

10.183 $-5(y + 7)$

10.184 $-8(2z + 7)$

10.185 $-7a(a - 3)$

10.186 $-x(6x - 1)$

Section 10.1 Exercises

1. binomial

3. trinomial

5. polynomial

7. monomial

9. 5

11. 1

13. 0

15. $15x^2$

17. $-8u$

19. $5a + 7b$

21. $-4a - 3b$

23. $16x$

25. $-17x^6$

27. $12y^2 + 4y + 8$

29. $-3x^2 + 17x - 1$

31. $4a^2 - 7a - 11$

33. $4m^2 - 10m + 2$

35. $11z + 8$

37. $12s^2 - 16s + 9$

39. $2p^3 + p^2 + 9p + 10$

41. $x^2 + 3x + 4$

43. $11w - 66$

45. (a) 187 (b) 40 (c) 2

47. (a) −104 (b) 4 (c) 40

49. 19 feet

51. 10 mpg

53. Answers will vary.

Section 10.2 Exercises

55. 1,024

57. $\frac{1}{4}$

59. 0.008

61. 625

63. −625

65. −10,000

67. $-\frac{8}{27}$

69. −0.25

71. x^9

73. a^5

75. 3^{14}

77. z^6

79. x^{a+2}

81. y^{a+b}

83. u^8

85. y^{20}

87. 10^{12}

89. x^{90}

91. x^{2y}

93. 5^{xy}

95. $25a^2$

97. $-216m^3$

99. $16r^2s^2$

101. $256x^4y^4z^4$

103. x^{14}

105. a^{36}

107. $45x^3$

109. $200a^5$

111. $8m^{18}$

113. $1{,}000x^6y^3$

115. $16a^{12}b^8$

117. $\frac{8}{27}x^6y^3$

119. $1{,}024a^{10}$

121. $25{,}000p^{24}$

123. $x^{18}y^{18}$

125. $144m^8n^{22}$

127. $-60x^6$

129. $72u^7$

131. $4r^{11}$

133. $36a^5b^7$

135. $8x^2y^5$

137. $\frac{1}{2}x^3y^3$

139. 1,679,616

141. Answers will vary.

143. Answers will vary.

Section 10.3 Exercises

145. $4x + 40$

147. $15r - 360$

149. $-3m - 33$

151. $-8z + 40$

153. $u^2 + 5u$

155. $n^3 - 3n^2$

157. $12x^2 - 120x$

159. $-27a^2 - 45a$

161. $24x^2 + 6xy$

163. $55p^2 - 25pq$

165. $3v^2 + 30v + 75$

167. $8n^3 - 8n^2 + 2n$

169. $-8y^3 - 16y^2 + 120y$

171. $5q^5 - 10q^4 + 30q^3$

173. $-12z^4 - 48z^3 + 4z^2$

175. $2y^2 - 9y$

177. $8w - 48$

179. $x^2 + 10x + 24$

181. $n^2 + 9n - 36$

183. $y^2 + 11y + 24$

185. $a^2 + 22a + 96$

187. $u^2 - 14u + 45$

189. $z^2 - 32z + 220$

191. $x^2 + 3x - 28$

193. $v^2 + 7v - 60$

195. $6n^2 + 11n + 5$

197. $20m^2 - 88m - 9$

199. $16c^2 - 1$

201. $15u^2 - 82u + 112$

203. $2a^2 + 5ab + 3b^2$

205. $5x^2 - 20x - xy + 4y$

207. $u^3 + 7u^2 + 14u + 8$

209. $3a^3 + 31a^2 + 5a - 50$

211. $y^3 - 16y^2 + 69y - 54$

213. $2x^3 - 9x^2 - 17x - 6$

215. (a) 195 (b) 195
(c) Answers will vary.

217. Answers will vary.

Section 10.4 Exercises

219. 4^6

221. x^9

223. r^4

225. $\frac{1}{y^{16}}$

227. $\frac{1}{10^{12}}$

229. $\frac{1}{a^8}$

231. 1

233. 1

235. -1

237. (a) 1 (b) 10

239. (a) 1 (b) $-27x^5$

241. (a) 1 (b) 15

243. 7

245. $\frac{243}{32}$

247. $\frac{m^3}{216}$

249. $\frac{x^{10}}{y^{10}}$ **251.** $\frac{a^2}{9b^2}$ **253.** x^3

255. u^2 **257.** $\frac{1}{y^2}$ **259.** 1

261. $\frac{1}{x^{18}}$ **263.** a^{14} **265.** y^3

267. $\frac{1}{x^{10}}$ **269.** $\frac{16r^{12}}{625s^4}$ **271.** 1

273. $\frac{1}{r^3}$ **275.** $3x^8$ **277.** $8b^6$

279. $\frac{-18}{x^6}$ **281.** $2x$ **283.** $\frac{5}{6x^6}$

285. $\frac{6r^2}{s^8}$ **287.** $\frac{n^6}{8}$ **289.** $-\frac{4y^6}{5x^2}$

291. $\frac{4z}{3x^2y^3}$ **293.** $\frac{10n^2}{m^4}$ **295.** $5u^4v^3$

297. (a) $18n^{10}$ (b) $12n^{10}$ (c) $45n^{20}$ (d) 5

299. (a) q^8 (b) q^{15} **301.** (a) z (b) $\frac{1}{z}$

303. $6y^6$ **305.** $15c^6$ **307.** $3x^2$

309. $-yz^2$ **311.** 1,000,000 **313.** Answers will vary.

315. Answers will vary.

Section 10.5 Exercises

317. $\frac{1}{64}$ **319.** $\frac{1}{32}$ **321.** $\frac{1}{10}$

323. $\frac{4}{9}$ **325.** $\frac{3}{5}$ **327.** $\frac{3}{4}$

329. (a) $\frac{1}{64}$ (b) $-\frac{1}{64}$ **331.** (a) $\frac{1}{4096}$ (b) $-\frac{1}{4096}$ **333.** (a) $\frac{10}{3}$ (b) $\frac{1}{30}$

335. (a) $\frac{3}{25}$ (b) $\frac{1}{225}$ **337.** $\frac{1}{p^3}$ **339.** $\frac{1}{m^5}$

341. (a) $\frac{3}{q}$ (b) $\frac{1}{3q}$ (c) $-\frac{1}{3q}$ **343.** (a) $\frac{10}{k}$ (b) $\frac{1}{10k}$ (c) $-\frac{1}{10k}$ **345.** r^3

347. $\frac{1}{q^5}$ **349.** $\frac{1}{z^8}$ **351.** $\frac{1}{m}$

353. $\frac{1}{x}$ **355.** $\frac{u^2}{v^2}$ **357.** $\frac{1}{a^2b^4}$

359. $\frac{1}{p^5q^7}$ **361.** $-\frac{21q^5}{p^3}$ **363.** $\frac{32}{a^3b}$

365. $\frac{1}{q^{100}}$ **367.** $\frac{1}{x^4}$ **369.** $\frac{1}{p^6}$

371. m^6

373. $\frac{9}{q^{10}}$

375. $\frac{n^{18}}{64}$

377. b^8

379. m^7

381. $\frac{1}{r^3}$

383. p^3

385. 2.8×10^5

387. 1.29×10^6

389. 4.1×10^{-2}

391. 1.03×10^{-5}

393. 6.85×10^9

395. 5.7×10^{-9}

397. 830

399. 16,000,000,000

401. 0.028

403. 0.0000000615

405. $15,000,000,000,000

407. 0.00001

409. 0.003

411. 0.00000735

413. 20,000,000

415. 50,000,000

417. (a) 1.25×10^{-4} (a) 8,000

419. 11,441,304,000

421. Answers will vary.

Section 10.6 Exercises

423. 15

425. 25

427. 4

429. 5

431. $3x$

433. $12p^3$

435. $2a$

437. $10y$

439. $5x^3$

441. $7b^2$

443. $5(y + 3)$

445. $4(b - 5)$

447. $7(x - 1)$

449. $3(n^2 + 7n + 4)$

451. $6(q^2 + 5q + 7)$

453. $c(9c + 22)$

455. $x(17x + 7)$

457. $q(4q + 7)$

459. $3r(r + 9)$

461. $10u(3u - 1)$

463. $b(a + 8)$

465. $11y(5 - y^3)$

467. $15c^2(3c - 1)$

469. $6c(c^2 - d^2)$

471. $24x^2(2x + 3)$

473. $18a^3(8a^3 + 5)$

475. $10(y^2 + 5y + 4)$

477. $12(u^2 - 3u - 9)$

479. $5p^2(p^2 - 4p - 3)$

481. $8c^3(c^2 + 5c - 7)$

483. $-7(p + 12)$

485. $-6b(3b + 11)$

487. $-8a^2(a - 4)$

489. $-9b^3(b^2 - 7)$

491. $-4(4t^2 - 20t - 1)$

493. Answers will vary.

Review Exercises

494. trinomial

496. binomial

498. 2

500. 0

502. $15p$

504. $-3n^5$

506. $10a^2 + 4a - 1$

508. $6y^2 - 3y + 3$

510. $8q^3 + q^2 + 6q - 29$

512. 995

514. 2,955

516. −163

518. 64 feet

520. 216

522. 0.25

524. p^{13} **526.** a^{6} **528.** y^{12}

530. 3^{10} **532.** $64n^2$ **534.** $256a^8b^8$

536. $27a^{15}$ **538.** x^{21} **540.** $-54p^5$

542. $56x^3y^{11}$ **544.** $70 - 7x$ **546.** $-625y^4 + 5y$

548. $a^2 + 7a + 10$ **550.** $6x^2 - 19x - 7$ **552.** $n^2 + 9n + 8$

554. $5u^2 + 37u - 24$ **556.** $p^2 + 11p + 28$ **558.** $27c^2 - 3c - 4$

560. $x^3 - 2x^2 - 24x - 21$ **562.** $m^3 - m^2 - 72m - 180$ **564.** 2^6 *or* 64

566. $\frac{1}{n^9}$ **568.** 1 **570.** 1

572. $\frac{9}{25}$ **574.** $\frac{125m^3}{n^3}$ **576.** a^{2}

578. $\frac{1}{x^{40}}$ **580.** $\frac{1}{n}$ **582.** $9p^9$

584. $-\frac{3}{y^4}$ **586.** $\frac{4a^5}{b^2}$ **588.** $\frac{4m^9}{n^4}$

590. $\frac{1}{36}$ **592.** $\frac{5}{16}$ **594.** x^{6}

596. $\frac{1}{u^3v^5}$ **598.** k^{6} **600.** b^{10}

602. 5.3×10^6 **604.** 9.7×10^{-2} millimeter **606.** 29,000

608. 0.375 **610.** 6,000 **612.** 30,000,000,000

614. 5 **616.** $4x^2$ **618.** $8(2u - 3)$

620. $6p(p + 1)$ **622.** $-9a^3(a^2 + 1)$ **624.** $5(y^2 - 11y + 9)$

Practice Test

626. (a) trinomial (b) 4 **628.** $6x^2 - 3x + 11$ **630.** n^{5}

632. $-48x^5y^9$ **634.** $s^2 + 17s + 72$ **636.** $55a^2 - 41a + 6$

638. $24a^2 + 34ab - 45b^2$ **640.** x^{14} **642.** $\frac{4r}{s^6}$

644. $3y^2 - 7x$ **646.** $\frac{1}{8y^3}$ **648.** x^{9}

650. $-6x(x + 5)$ **652.** 0.000525 **654.** 300,000

Chapter 11

Be Prepared

11.1 2 **11.2** 16 **11.3** 5

11.4 -1

11.5 $y = \frac{20-5x}{2}$

11.6 -9

11.7 -4

11.8 y-axis

11.9 b

11.10 $-\frac{1}{2}$

11.11 0, undefined

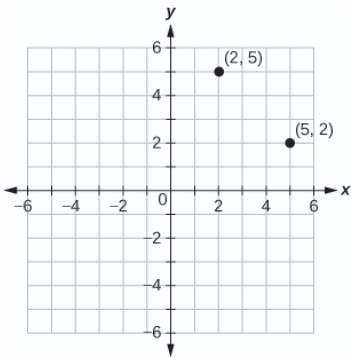
11.12 $-5, -5, 5$

Try It

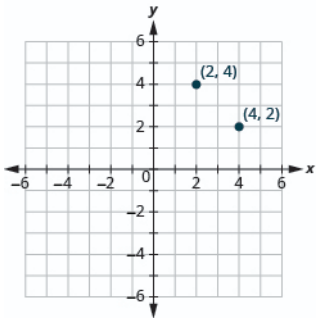
11.1 (a) 1C
(b) Engineering Building

11.2 (a) 1A (b) Library

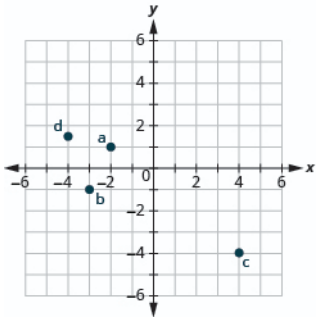
11.3



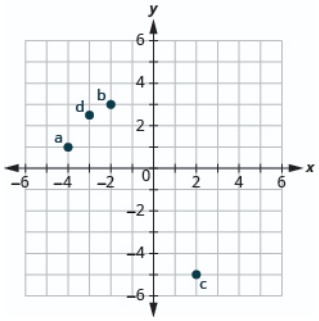
11.4



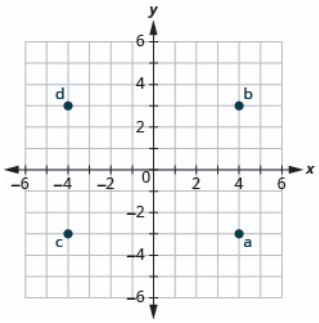
11.5 (a) Quadrant II, (b) Quadrant III, (c) Quadrant IV, (d) Quadrant II



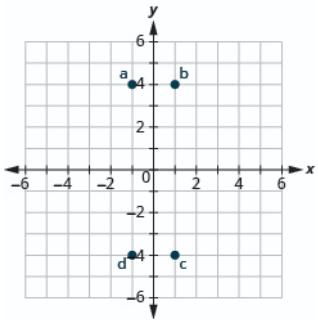
11.6 (a) Quadrant II, (b) Quadrant II, (c) Quadrant IV, (d) Quadrant II



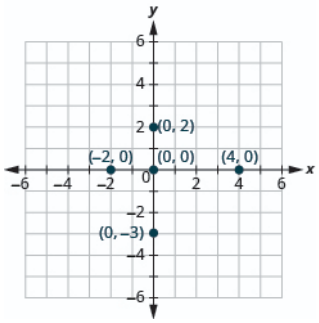
11.7



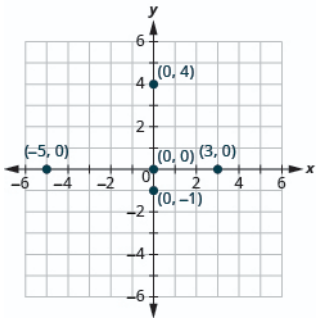
11.8



11.9



11.10



11.11
1. A: (5,1)
2. B: (−2,4)
3. C: (−5,−1)
4. D: (3,−2)

11.12
1. A: (4,2)
2. B: (−2,3)
3. C: (−4,−4)
4. D: (3,−5)

11.13
1. A: (4,0)
2. B: (0,3)
3. C: (−3,0)
4. D: (0,−5)

11.14
1. A: (−3,0)
2. B: (0,−3)
3. C: (5,0)
4. D: (0,2)

11.15 ⓐ , ⓒ

11.16 ⓑ , ⓒ

11.17 ⓑ

11.18 ⓐ , ⓑ

11.19

$y = 3x - 1$		
x	y	(x, y)
0	−1	(0, −1)
−1	−4	(−1, −4)
2	5	(2, 5)

11.20

$y = 6x + 1$		
x	y	(x, y)
0	1	(0, 1)
1	7	(1, 7)
−2	−11	(−2, −11)

11.21

$2x - 5y = 20$		
x	y	(x, y)
0	−4	(0, −4)
10	0	(10, 0)
−5	−6	(−5, −6)

11.22

$3x - 4y = 12$		
x	y	(x, y)
0	−3	(0, −3)
4	0	(4, 0)
−4	−6	(−4, −6)

11.23 Answers will vary.

11.24 Answers will vary.

11.25 Answers will vary.

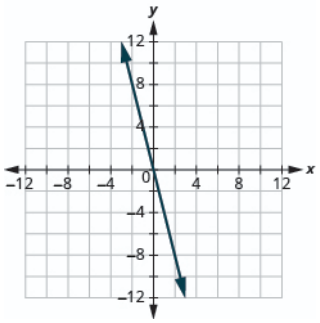
11.26 Answers will vary.

11.27 Answers will vary.

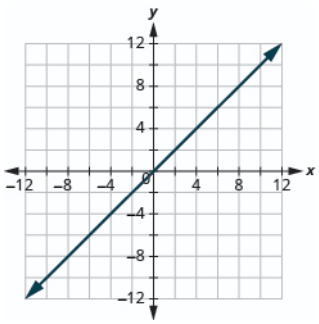
11.28 Answers will vary.

11.29
1. ⓐ yes ⓑ yes
2. ⓐ no ⓑ no
3. ⓐ no ⓑ no
4. ⓐ yes ⓑ yes

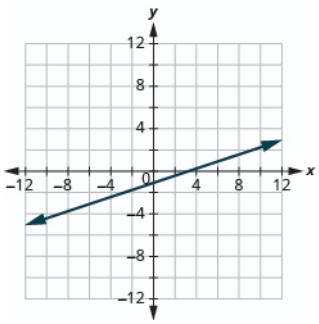
11.30



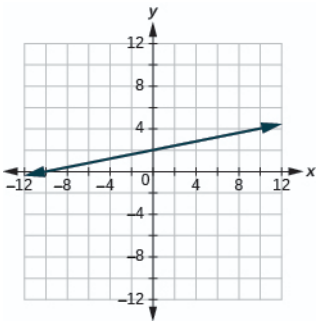
11.31



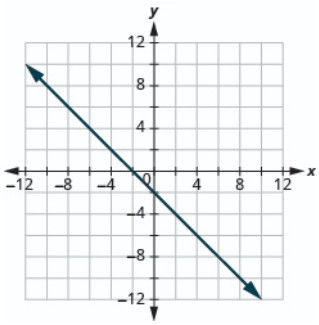
11.32



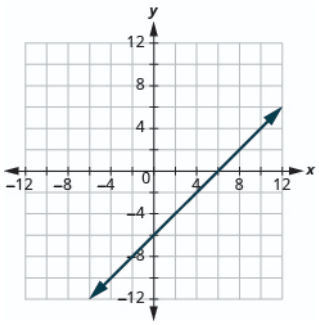
11.33



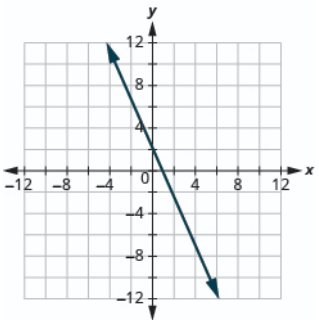
11.34



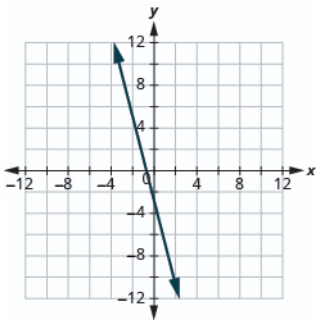
11.35



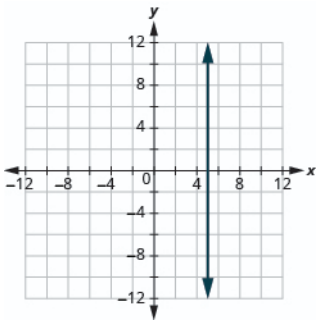
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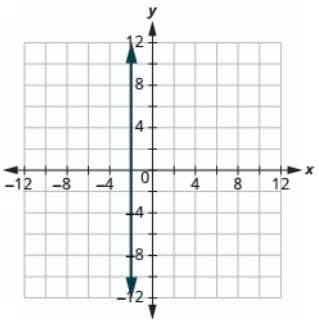
11.37



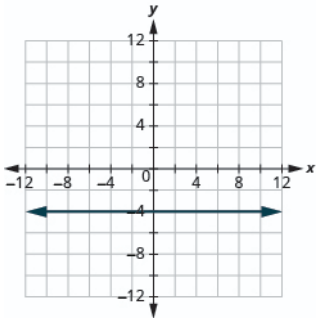
11.38



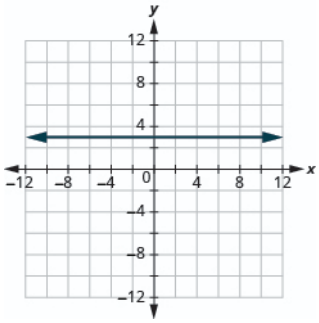
11.39



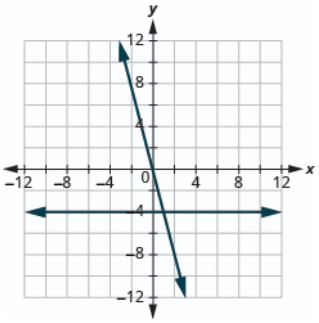
11.40



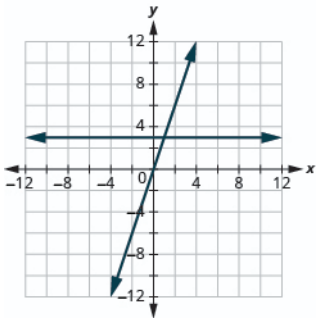
11.41



11.42



11.43



11.44 *x*-intercept (2,0):
y-intercept (0,−2)

11.45 *x*-intercept (3,0);
y-intercept (0,2)

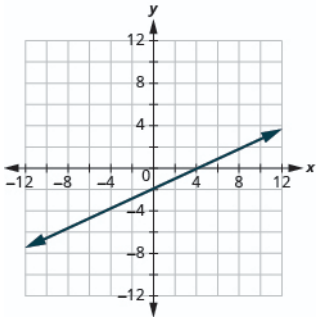
11.46 $(4, 0)$ and $(0, 12)$

11.47 $(8, 0)$ and $(0, 2)$

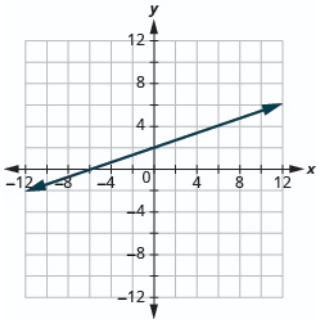
11.48 *x*-intercept (4,0);
y-intercept: (0,−3)

11.49 *x*-intercept (4,0);
y-intercept: (0,−2)

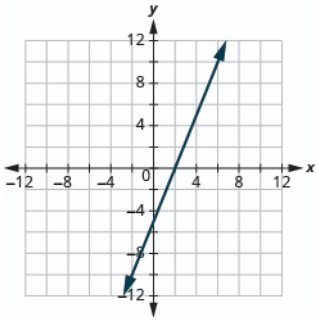
11.50



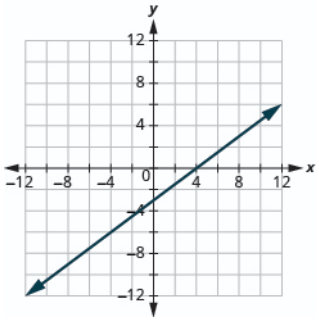
11.51



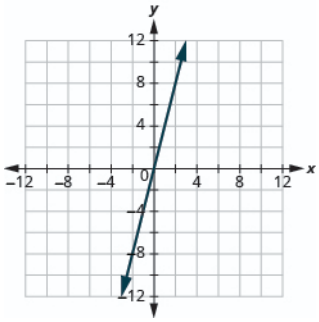
11.52



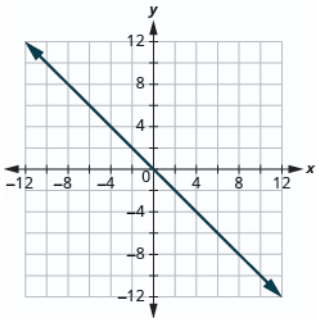
11.53



11.54



11.55



11.56
ⓐ intercepts
ⓑ horizontal line
ⓒ plotting points
ⓓ vertical line

11.57
ⓐ vertical line
ⓑ plotting points
ⓒ horizontal line
ⓓ intercepts

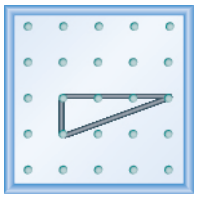
11.58 $\frac{4}{3}$

11.59 $\frac{1}{4}$

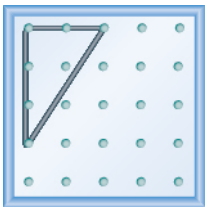
11.60 $-\frac{2}{3}$

11.61 $-\frac{4}{3}$

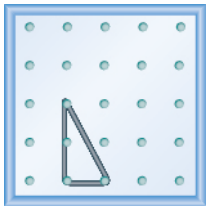
11.62



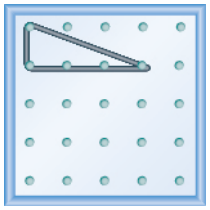
11.63



11.64



11.65



11.66 $\frac{2}{5}$

11.67 $\frac{3}{4}$

11.68 $-\frac{4}{3}$

11.69 $-\frac{3}{5}$

11.70 $\frac{5}{4}$

11.71 $\frac{3}{2}$

11.72 undefined

11.73 0

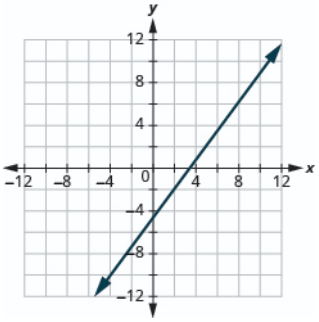
11.74 1

11.75 1

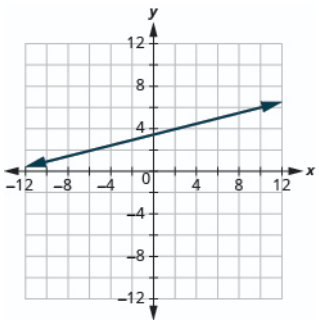
11.76 −1

11.77 10

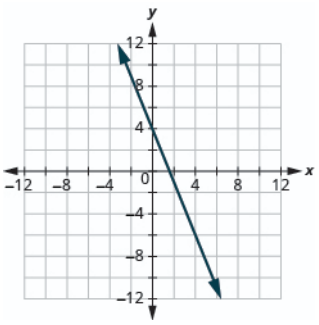
11.78



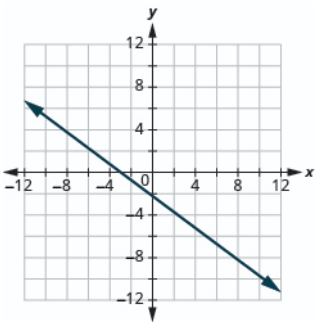
11.79



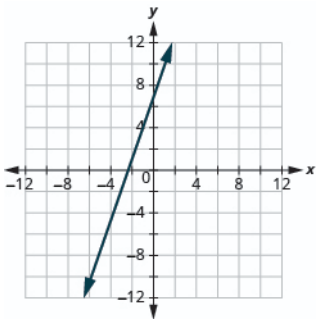
11.80



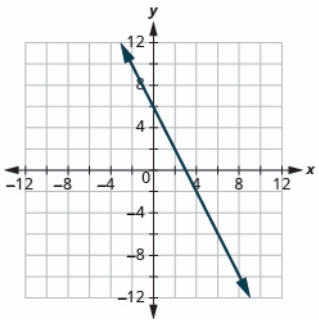
11.81



11.82



11.83



11.84 $\frac{7}{12}$

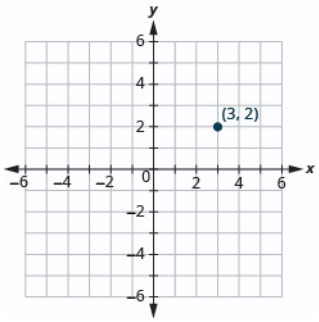
11.85 $\frac{5}{12}$

11.86 $-\frac{1}{36}$

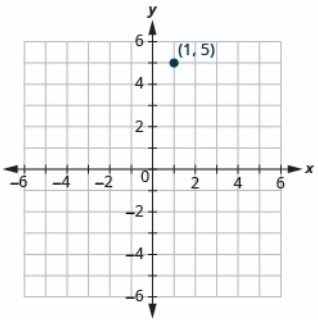
11.87 $-\frac{1}{48}$

Section 11.1 Exercises

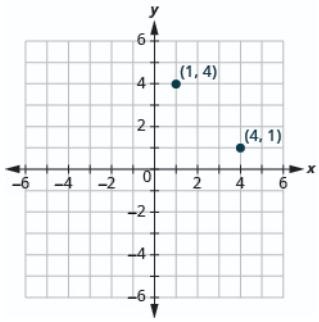
1.



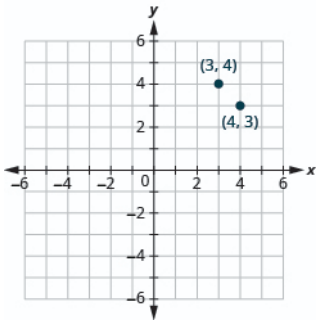
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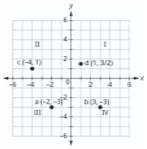
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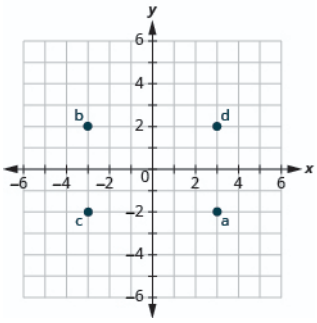
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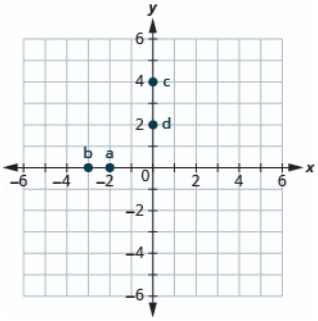
9.



11.



13.



15. C(1, -3) D(4, 3)

17. S(-2, 4) T(-4, -2)

19. C(0, -1) D(-1, 0)

21. ⓐ , ⓑ

23. ⓐ , ⓒ

25. ⓑ , ⓒ

27. ⓐ , ⓑ

29.

x	y	(x, y)
-1	-6	$(-1, -6)$
0	-4	$(0, -4)$
2	0	$(2, 0)$

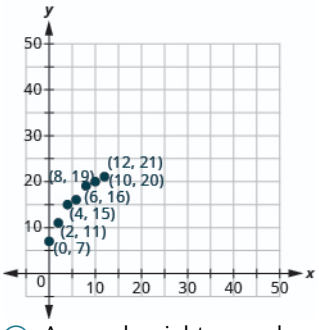
31.

x	y	(x, y)
-2	7	$(-2, 7)$
0	5	$(0, 5)$
3	2	$(3, 2)$

33.

x	y	(x, y)
-2	1	$(-2, 1)$
0	-2	$(0, -2)$
2	-5	$(2, -5)$

35. ⓐ



ⓑ Age and weight are only positive.

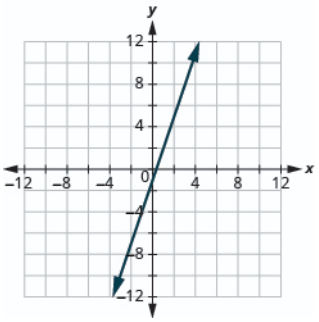
37. Answers may vary.

Section 11.2 Exercises

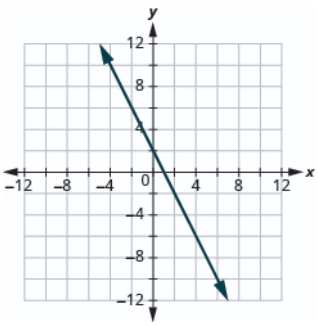
39.
1. ⓐ yes ⓑ yes
2. ⓐ no ⓑ no
3. ⓐ yes ⓑ yes
4. ⓐ no ⓑ no

41.
1. ⓐ yes ⓑ yes
2. ⓐ yes ⓑ yes
3. ⓐ yes ⓑ yes
4. ⓐ no ⓑ no

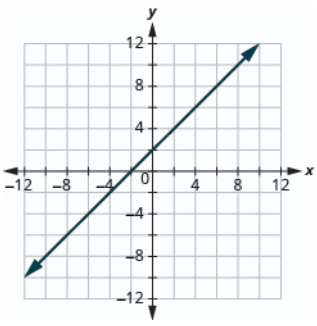
43.



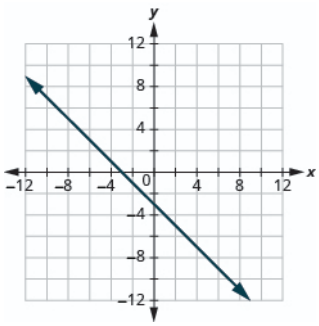
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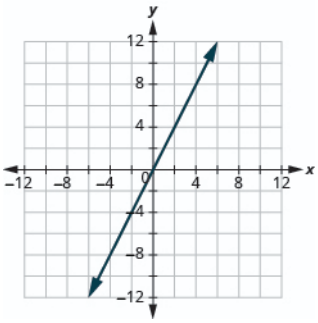
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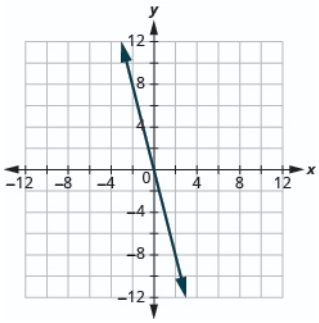
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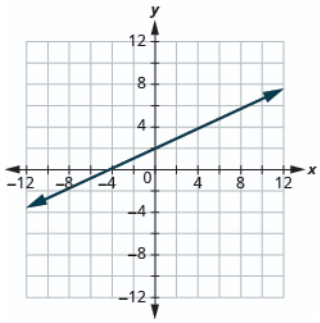
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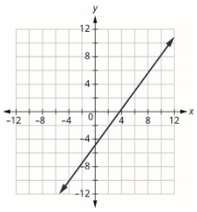
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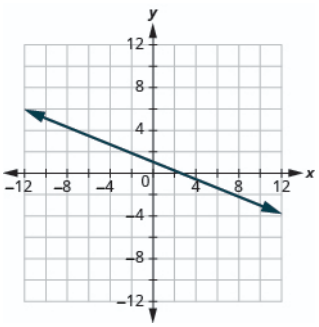
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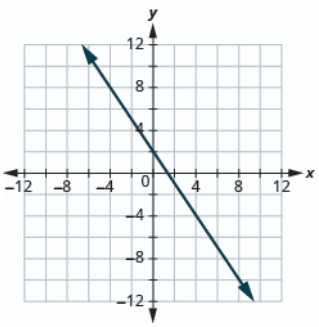
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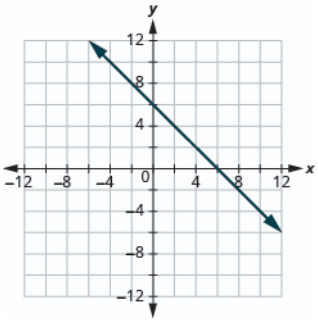
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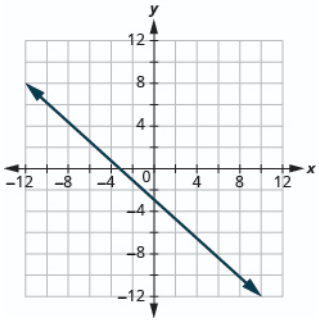
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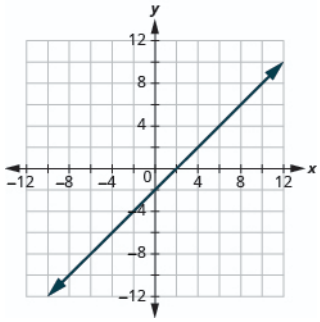
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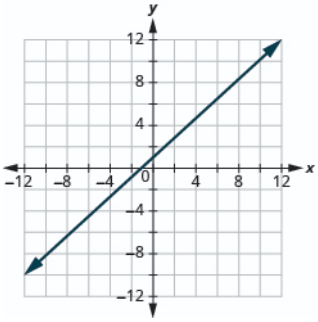
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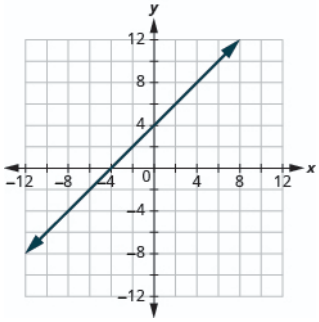
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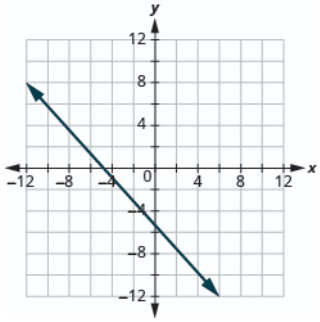
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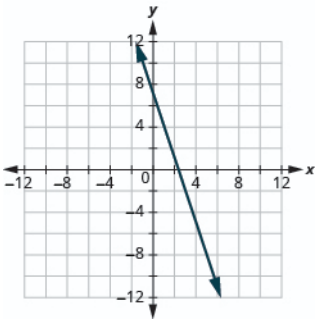
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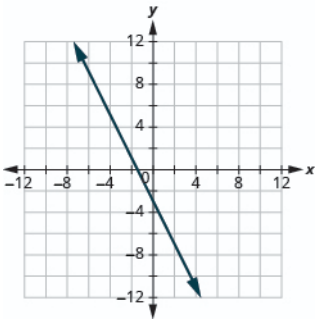
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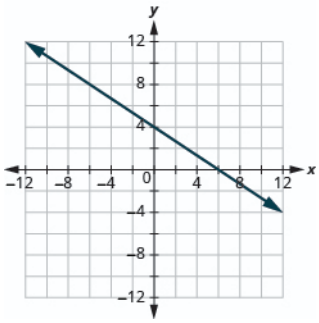
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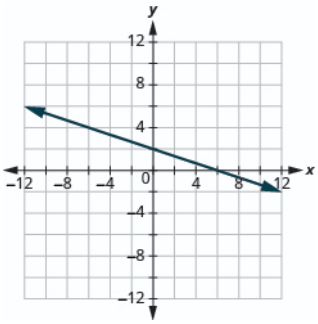
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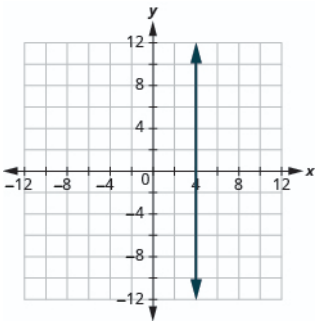
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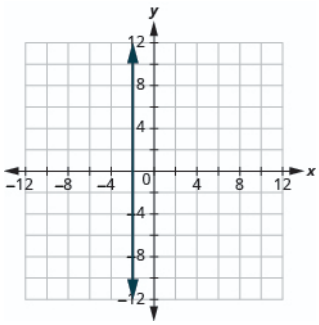
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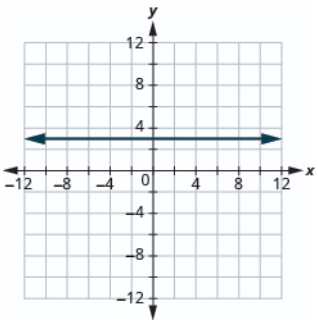
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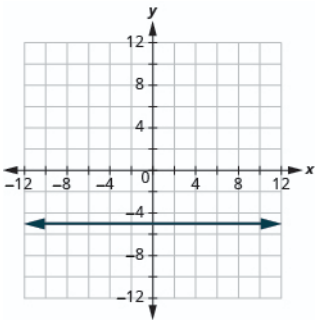
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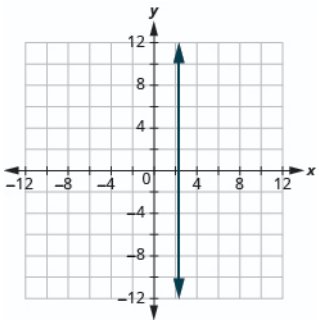
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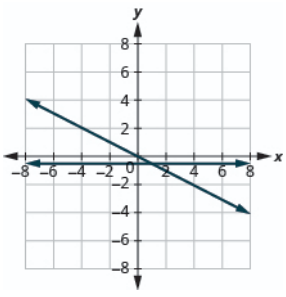
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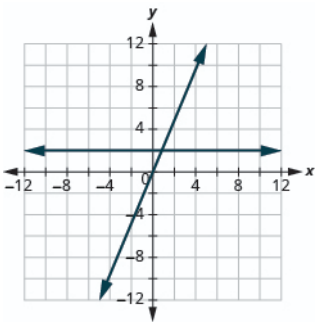
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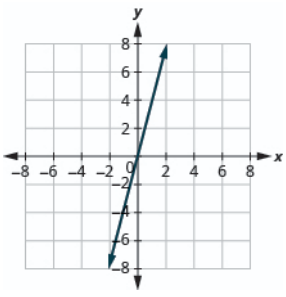
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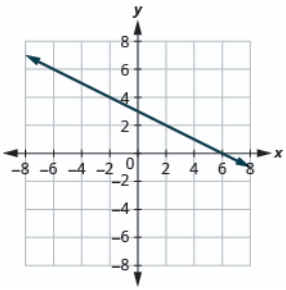
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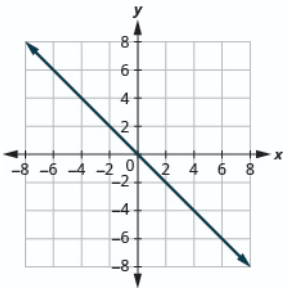
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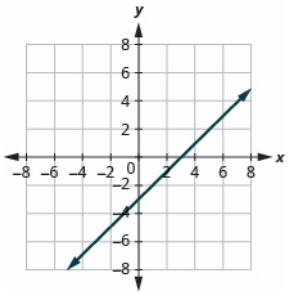
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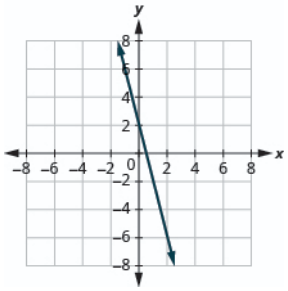
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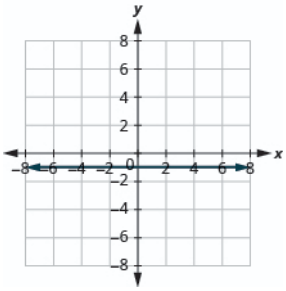
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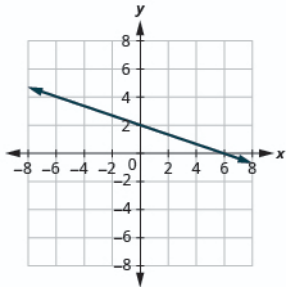
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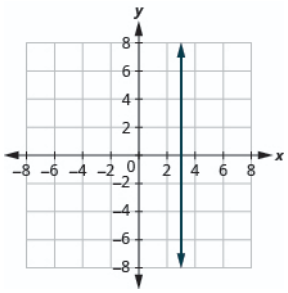
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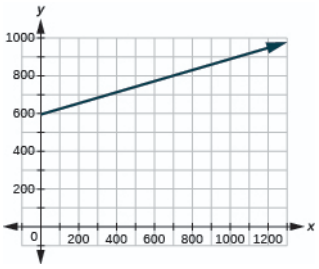
109.



111.



113.



$722, $850, $978

115. Answers will vary.

Section 11.3 Exercises

117. (3,0),(0,3)

119. (5,0),(0,−5)

121. (−2,0),(0,−2)

123. (−1,0),(0,1)

125. (0,0)

127. (4,0),(0,4)

129. (−2,0),(0,−2)

131. (5,0),(0,−5)

133. (−3,0),(0,3)

135. (8,0),(0,4)

137. (2,0),(0,6)

139. (12,0),(0,−4)

141. (2,0),(0,−8)

143. (5,0),(0,2)

145. (4,0),(0,−6)

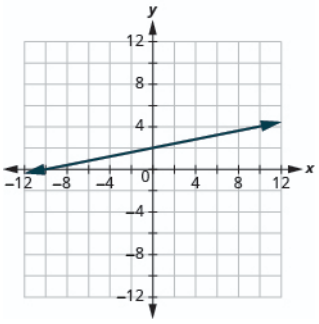
147. (3,0),(0,−1)

149. (−10,0),(0,2)

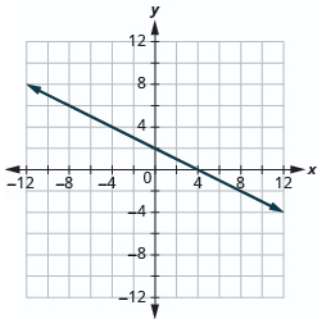
151. (0,0)

153. (0,0)

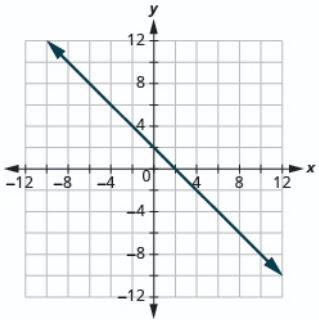
155.



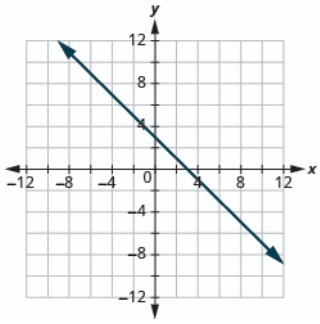
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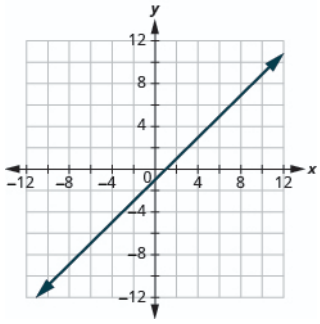
159.



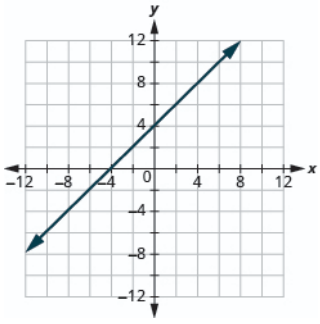
161.



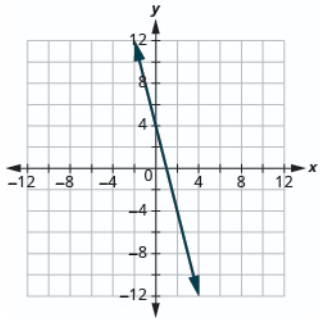
163.



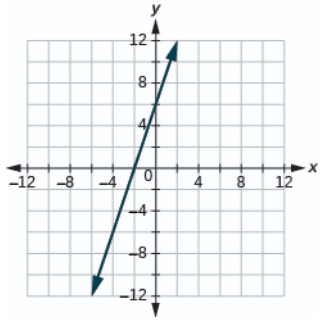
165.



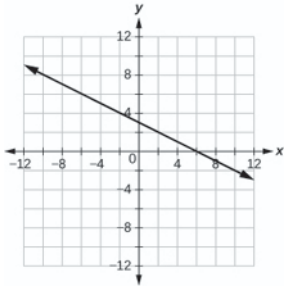
167.



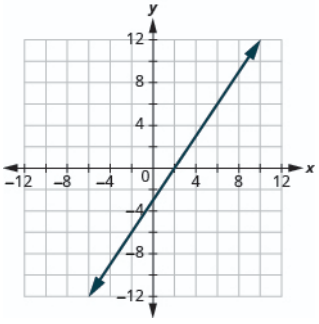
169.



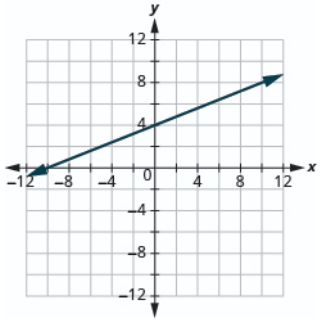
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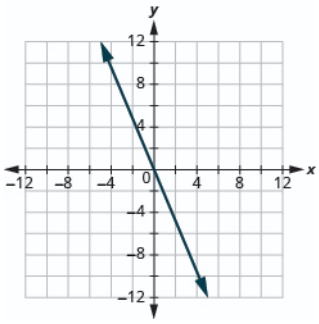
173.



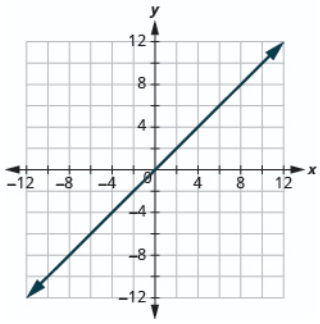
175.



177.



179.



181. vertical line

183. horizontal line

185. plotting points

187. intercepts

189. plotting points

191. horizontal line

193. intercepts

195. plotting points

197. (a) (0,1,000),(15,0). (b) At (0,1,000) he left Chicago 0 hours ago and has 1,000 miles left to drive. At (15,0) he left Chicago 15 hours ago and has 0 miles left to drive.

199. Answers will vary.

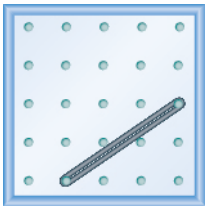
201. Answers will vary.

Section 11.4 Exercises

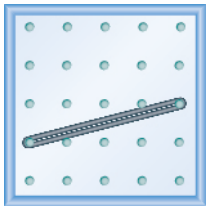
203. $\frac{1}{4}$

205. $-\frac{3}{2}$

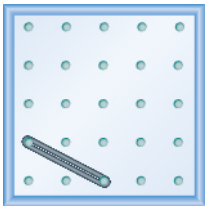
207.



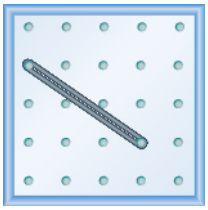
209.



211.



213.



215. $\frac{2}{5}$

217. $\frac{5}{4}$

219. $-\frac{1}{3}$

221. $-\frac{3}{4}$

223. $\frac{3}{4}$

225. $-\frac{5}{2}$

227. $-\frac{2}{3}$

229. $\frac{1}{4}$

231. 0

233. undefined

235. 0

237. undefined

239. $\frac{5}{2}$

241. $\frac{3}{4}$

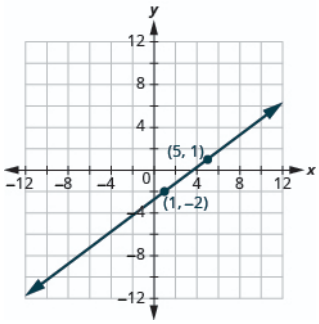
243. $-\frac{5}{2}$

245. $-\frac{8}{5}$

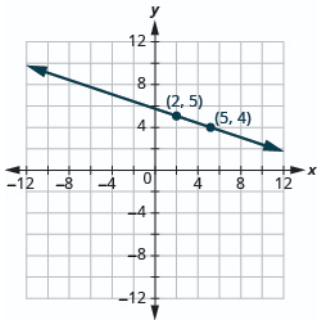
247. $\frac{7}{3}$

249. -1

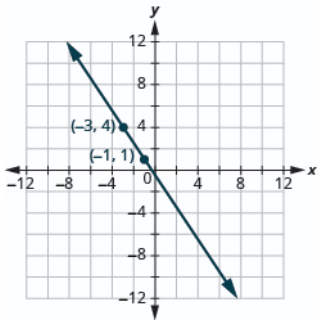
251.



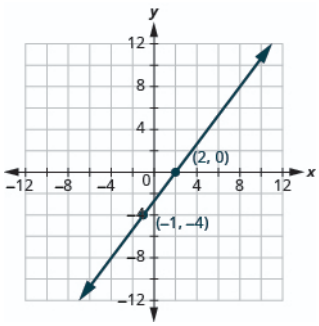
253.



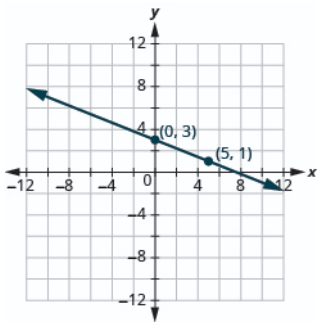
255.



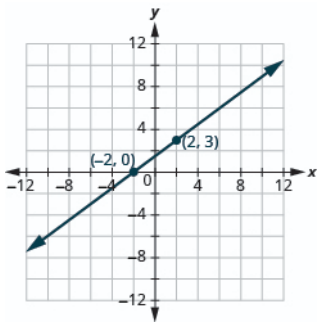
257.



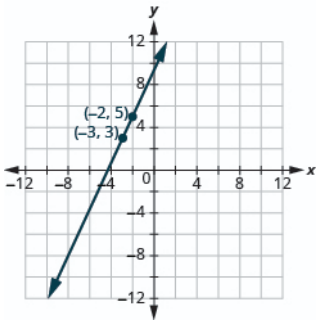
259.



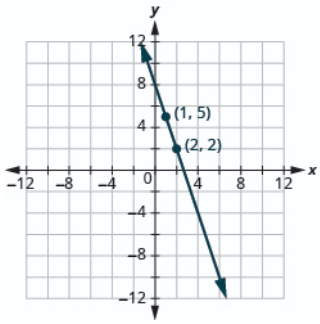
261.



263.



265.



267. $\frac{1}{3}$

269. (a) $\frac{3}{50}$ (b) rise $= 3$; run $= 50$

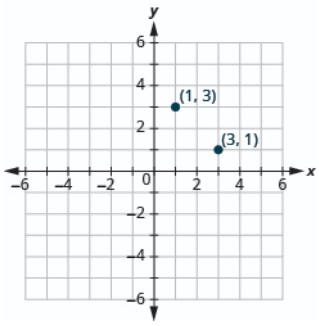
271. (a) 288 inches (24 feet) (b) Models will vary.

273. Answers will vary.

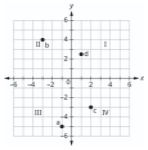
275. Answers will vary.

Review Exercises

277.



279. ⓐ III ⓑ II ⓒ IV ⓐ I



281. ⓐ (5,3) ⓑ (2,−1) ⓒ (−3,−2) ⓓ (−1,4)

283. ⓐ (2,0) ⓑ (0,−5) ⓒ (−4,0) ⓓ (0,3)

285. ⓑ ,ⓒ

287.

x	y	(x, y)
0	−1	$(0, -1)$
1	3	$(1, 3)$
−2	−9	$(-2, -9)$

289.

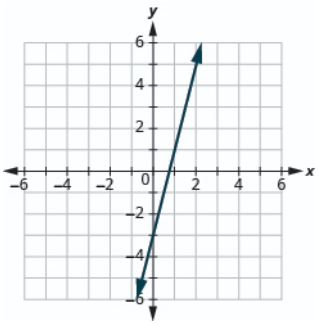
x	y	(x, y)
5	0	$(5, 0)$
1	2	$(1, 2)$
−1	3	$(-1, 3)$

291. Answers will vary.

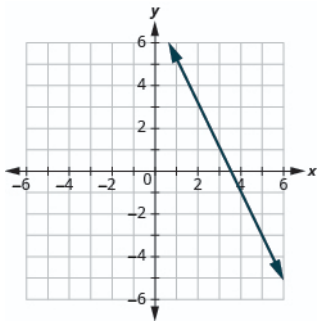
293. Answers will vary.

295.
1. ⓐ yes ⓑ yes
2. ⓐ no ⓑ no
3. ⓐ yes ⓑ yes
4. ⓐ yes ⓑ yes

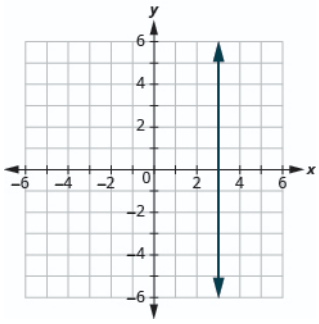
297.



299.



301.

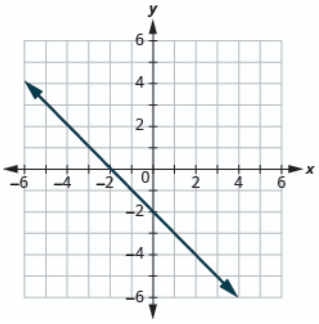


303. (0,3) (3,0)

305. (−1,0) (0,1)

307. (0,0)

309.



311. horizontal line

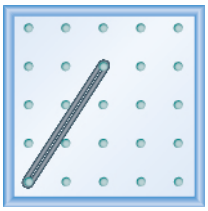
313. intercepts

315. plotting points

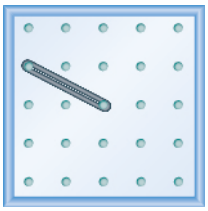
317. $\frac{4}{3}$

319. $-\frac{2}{3}$

321.



323.



325. 1

327. $-\frac{1}{2}$

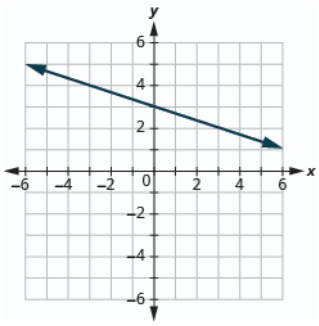
329. undefined

331. 0

333. −4

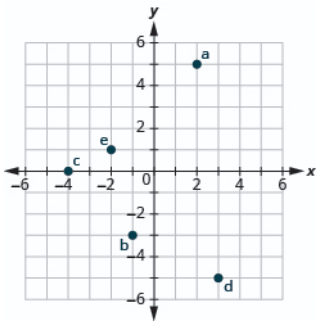
335. $\frac{1}{2}$

337.



Practice Test

339.



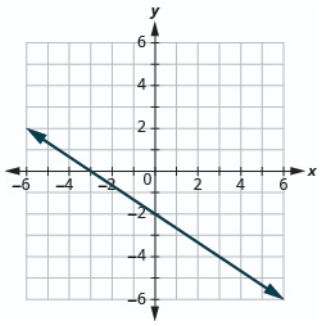
341. (4,0), (0,−2)

343. no; $1 + 4 \cdot 3 \neq 12$

345.

x	y	(x, y)
0	8	$(0, 8)$
2	0	$(2, 0)$
3	−4	$(3, -4)$

347.



349. $-\frac{5}{2}$

351. 0

353. $\frac{1}{50}$